

The Volatility of Marginal Information Rents in Dynamic Mechanism Design

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Abstract

This paper brings martingale methods of continuous-time contracting to Myersonian dynamic mechanism design. In a benchmark setting, I show that implementability is characterized by the volatility of the agent’s *marginal information rent*—how the marginal rent responds to innovations in his type. Once transfers satisfy the usual envelope conditions, truthful reporting after the initial report is locally incentive compatible if and only if the discounted volatility is a nonnegative supermartingale. When the volatility has finite variation, the same condition is necessary and sufficient for global incentive compatibility after the initial report. Using this characterization, I cast the design problem as optimal control, which applies even when the first-order approach fails: one program solves for the optimal mechanism in a regular finite-variation class, and another for the optimal locally incentive-compatible mechanism. The two bracket the designer’s true payoff. I apply the results to financial market design (a monopolistic dealer, then an exchange).

Keywords: dynamic mechanism design, contracts, persistent private information, optimal control, market design, market microstructure

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1 Introduction

Dynamic mechanism design in the Myersonian tradition has developed largely in discrete time (Courty and Li, 2000; Battaglini, 2005; Eső and Szentes, 2007; Pavan et al., 2014).¹ By contrast, dynamic contracting more broadly has benefited from continuous-time martingale methods (DeMarzo and Sannikov, 2006; Sannikov, 2008; Williams, 2011). The two literatures study closely related problems (the first with transferable utility, the second without) but have used surprisingly distinct methods. This paper extends martingale methods to the Myersonian setting and studies how the two literatures' approaches relate. In the process, the paper obtains new results for both literatures.

The main result characterizes implementability in terms of a core object: the volatility of the agent's marginal information rent—the sensitivity of the marginal rent to innovations in his type. Once transfers satisfy the usual envelope conditions, truthful reporting after the initial report is locally incentive compatible if and only if the volatility is nonnegative and, suitably discounted, a supermartingale under truthful reporting. When the volatility has finite variation, the same conditions are necessary and sufficient for global incentive compatibility after the initial report. The characterization also applies without transfers and is thus new to both literatures, for reasons discussed below.² Because the supermartingale condition is local in time, the design problem admits techniques from optimal control. Taken together, the results offer a complete characterization of locally incentive-compatible mechanisms,³ identify a natural class on which the characterization is global, and provide a method for optimizing over each class. I apply the method to a classic problem in financial-market-design where the standard first-order approach fails.

In the benchmark model, a principal chooses an allocation for a single agent, charges transfers, and may bear a convex cost of provision. The agent's marginal value for the allocation, his private type, follows a Markov diffusion. We assume the type follows an Ornstein–Uhlenbeck (OU) process but later extend to the case of a geometric Brownian motion (GBM). The principal

¹An exception is Bergemann and Strack (2015), who solve dynamic revenue-maximization problems in continuous time within the first-order approach, using continuous-time versions of the envelope methods from the discrete theory.

²In brief, the canonical reference for contracting with adverse selection, Williams (2011), restricts the agent to negative drift distortions, whereas I allow distortions of either sign.

³Local incentive compatibility, defined formally in Definition 3, rules out a payoff gain at quadratic order along every feasible deviation direction. Technically, when that effect is zero, a small deviation may be profitable due to higher-order effects on payoffs. Under the conditions of Appendix G.5, the mechanism can be perturbed at arbitrarily small cost to the designer so that the quadratic effect is strictly negative in every nonzero direction.

adjusts the allocation and transfers in response to the agent’s *reported* type path. The agent freely chooses his initial report and then controls the report’s drift thereafter. The agent’s *marginal information rent* is the derivative of his continuation utility with respect to his current type under truthful reporting. By the envelope theorem, it equals the expected discounted allocation stream, weighted by the impulse response of the type process (see (1)).

The marginal rent can be represented as an Itô process with a drift and volatility. I show that implementability after the initial report rests entirely on this volatility (Theorem 1a, Lemma 1). Local incentive compatibility is equivalent to volatility being a nonnegative supermartingale after discounting and moreover, implies global incentive compatibility when the volatility is of finite variation. At the initial report, local incentive compatibility is characterized by a comparison between initial volatility and the slope of the marginal rent schedule (that is, marginal rent as a function of the initial type); under a mild technical condition, global incentive compatibility in the finite-variation class is equivalent to an integral condition comparing initial volatility with the marginal rent schedule (Theorem 1b).

This characterization makes the principal’s problem amenable to optimal control with the current report, the marginal rent, and volatility as state variables. I formulate two programs that bracket the principal’s true payoff: one program optimizes over finite-variation mechanisms satisfying the date-zero integral condition and gives an implementable lower bound; the other optimizes over locally incentive-compatible mechanisms and gives an upper bound. In both programs, I use dynamic programming to characterize the principal’s optimal continuation after time zero. Given the resulting value function, the principal selects the initial marginal rent and volatility schedules to solve a convex program. The finite-variation class is natural for two reasons: 1) it is the largest class for which all higher-order effects of deviations vanish, and 2) it is closely related to “linear mechanisms” in discrete time (in a sense described shortly), in which the marginal rent is linear, at each date, in the agent’s current report.

The two programs are applicable even when the first-order approach, whose validity requires strong assumptions (Pavan et al., 2014; Battaglini and Lamba, 2019), fails. In those cases, both programs preserve the central dynamic tradeoff the designer must then balance between limiting rents due to information today versus creating surplus (to extract) from flexibility to respond to future innovations in the type. In the environments covered by the irrelevance theorem of Eső and Szentes (2017), this tradeoff disappears when the first-order approach is valid: the designer earns as much as if the agent’s future types were observed directly. Analysis of the invalid case is thus economically important, and this

paper provides one route into it (Pavan, 2017).

I also show how the analysis connects to the discrete-time theory. There, incentive compatibility amounts to the envelope condition together with *integral monotonicity* (Pavan et al., 2014): truthful reporting must dominate every one-shot misreport in an integrated sense (Section 2). Integral monotonicity, however, is non-local and well known to be intractable to impose, both computationally and analytically. I consider “linear mechanisms” in which the marginal rent responds linearly to the innovation in the type in each period. The loading on the innovation is predictable: it may depend on all past reports but is fixed before the innovation is realized. These loadings are the discrete-time analog of the marginal rent volatility. I show that integral monotonicity after the initial report collapses by elementary algebra to a pointwise bound on the growth in loadings: the finite-difference form of drift condition in the finite-variation case. Also an exactly analogous integral condition disciplines the initial report. This is the sense in which “linear mechanisms” are the discrete-time counterpart of continuous-time finite-variation mechanisms, giving the finite-variation class a concrete interpretation (Section 3).

The incentive-compatibility characterization does not rely on transfers, so it extends beyond the Myersonian setting (Proposition 6). To my knowledge, this result is also new to dynamic contracting more broadly. The key difference from existing work is the set of deviations. The canonical reference, Williams (2011), restricts the agent to negative drift distortions. This one-sided restriction makes truthful reporting a boundary choice and supports the first-order analysis common in that literature. The restriction prevents the agent from ever correcting a lie, and its economic motivation is perhaps unclear (Bloedel et al., 2023). I allow drift distortions of either sign, so truthful reporting is interior: first-order effects must vanish, and second-order effects determine local incentives.

I apply the machinery to a classic setting in finance: a market maker trading with a single trader whose hedging needs are private and persistent—a dynamic version of the monopolistic dealer’s problem (Glosten, 1989). I show that the first-order approach fails here outright because the trader’s side of the market is endogenous: whether he buys or sells depends on the terms of trade. I solve the two control programs. Instead of the no-trade interval a standard bid–ask spread would induce (and which arises under the first-order approach), the optimal mechanisms fill small trading needs at reduced scale, balancing rent extraction today against surplus extraction tomorrow. I then show that a related design problem—a revenue-maximizing exchange in which many traders participate and the market must clear—reduces to those single-trader problems. The setting belongs to a benchmark class of models in financial

market microstructure, in which traders bear quadratic holding costs (e.g., Du and Zhu, 2017; Malamud and Rostek, 2017; Chen and Duffie, 2021; Antill and Duffie, 2021).

Related literature. The paper sits at the junction of two literatures: Myersonian dynamic mechanism design and dynamic contracting more broadly. On the Myersonian side where the designer seeks to maximize revenue,⁴ the field goes back to Baron and Besanko (1984) and Besanko (1985). Contributions include Courty and Li (2000), Battaglini (2005), Eső and Szentes (2007), Krämer and Strausz (2011), and Boleslavsky and Said (2013). Pavan et al. (2014) give the general treatment. Pavan (2017) and Bergemann and Välimäki (2019) survey this literature, which has primarily developed in discrete time. I analyze this setting in continuous time, arriving at a distinct characterization of implementability, and exploit tools from optimal control. The closest paper is by Bergemann and Strack (2015), who derive revenue-maximizing mechanisms in continuous time within the first-order approach, by the envelope methods of the discrete theory; the present paper instead confronts second-order incentives with martingale methods.

Within the Myersonian tradition, a recent line of work analyzes the failure of the first-order approach directly. Battaglini and Lamba (2019) show that with persistent types the failure is generic. Krasikov and Lamba (2021, 2026) solve for optimal contracts in specific environments where the approach fails; Garrett and Pavan (2015) and Garrett et al. (2018) develop variational methods that deliver optimality conditions and robust predictions without assuming validity of the first-order approach; and Li and Shi (2025) retain the local incentive constraints in a two-period screening problem where the standard relaxation breaks down, showing that stochastic mechanisms can then be essential. The control programs of Section 5 take a similar approach in an infinite-horizon setting: the local program optimizes subject to all local incentive constraints, while the finite-variation program imposes global continuation incentive compatibility together with a sufficient condition for truthful reporting at date zero. In static settings, local first- and second-order conditions are often sufficient for global incentive compatibility (Carroll, 2012), but in dynamic settings no comparable result is available.

On the contracting side, the paper extends the martingale methods of continuous-time contract theory (Sannikov, 2008; Williams, 2011) to screening with transfers and a nondegenerate initial type. The first-order approach,

⁴As opposed to the parallel strand on implementing efficient allocations in dynamic environments; see Bergemann and Välimäki (2019) for a survey of both.

however, is popular in contract theory, including the dynamic insurance and public finance literature (e.g., [Zhang, 2009](#); [Kapička, 2013](#); [Farhi and Werning, 2013](#)). Most closely, [Williams \(2011\)](#) obtains sufficient conditions for incentive compatibility in a model of dynamic insurance, with deviations restricted to negative drift distortions; his conditions constrain the loading of the agent’s promised marginal utility—the counterpart, in his setting, of the rent volatility here. In contrast, I consider drift distortions of either sign and obtain a characterization of local incentive compatibility, which is fundamentally second order: with both signs admitted, truthful reporting is interior rather than a corner of the deviation set.

This paper is partly inspired by [Bloedel et al. \(2023\)](#), who revisit Williams’s model and identify issues in its analysis of incentive compatibility and optimality. They go on to analyze a class of *self-insurance contracts*—contracts implementable as consumption–saving problems for the agent—and show that the optimal one generically strictly dominates the contract Williams had identified as optimal in his leading application. They also show that the analysis has close discrete-time analogues. [Acciaio et al. \(2025\)](#) likewise revisit Williams’s model, characterizing incentive compatibility within parametric families of contracts linear in the report.

Outline. Section 2 reviews the discrete-time characterization of incentive compatibility and the first-order approach. Section 3 offers intuition for the main results in discrete time through linear mechanisms. Section 4 sets out the continuous-time model. Section 5 presents the volatility characterization and the two control programs used to bound the designer’s payoff. Section 6 applies the results to financial market design—first the dealer, then the exchange. Section 7 extends the analysis to geometric Brownian motion, contracts without transfers, discusses contracts beyond finite variation that are globally certifiable. Section 8 concludes.

2 Background

This section informally reviews the characterization of incentive compatibility for Markov environments in [Pavan et al. \(2014\)](#). Time is discrete, $t \in \{0, 1, \dots\}$. The agent privately observes his type θ_t , which follows a Markov process, and reports a type each period. A direct mechanism specifies an allocation process $X = \{X_t\}$ and a transfer process $T = \{T_t\}$, with X_t and T_t measurable with respect to the report history through period t . The agent’s flow utility is $u(\theta_t, X_t) - T_t$, discounted by $\beta \in (0, 1)$ per period.

Using the Markov property, θ_s can be written as a function of θ_t and the shocks arriving between t and s . The impulse response $I_{s,t} = \partial\theta_s/\partial\theta_t$ is the marginal effect of the time- t type on the time- s type, holding those intermediate shocks fixed. Consider a one-shot misreport: the agent's current type is θ_t , he reports $\hat{\theta}_t$, and he reports truthfully thereafter. Let $X^{\hat{\theta}_t}$ be the allocation stream that follows. The marginal information rent following this report is

$$D_t(\hat{\theta}_t, \theta_t) = \mathbb{E} \left[\sum_{s=t}^{\infty} \beta^{s-t} I_{s,t} u_{\theta}(\theta_s, X_s^{\hat{\theta}_t}) \mid \mathcal{F}_t \right]. \quad (1)$$

This is the expected discounted stream of marginal utilities weighted by the impulse response. $D_t(\theta_t, \theta_t)$ is the on-path marginal rent. Write $W_t(\theta_t)$ for the agent's continuation payoff at a truthful history under truthful reporting.

Specializing [Pavan et al. \(2014, Theorems 1–3\)](#) to our setting gives the following informal characterization:

- (i) *Envelope*. Every incentive-compatible mechanism satisfies

$$W'_t(\theta_t) = D_t(\theta_t, \theta_t)$$

at each truthful history.

- (ii) *Integral monotonicity*. Under their regularity conditions, an allocation rule is implementable⁵ if and only if, at every truthful history and for every current type θ_t and alternative report $\hat{\theta}_t$,

$$\int_{\hat{\theta}_t}^{\theta_t} [D_t(l, l) - D_t(\hat{\theta}_t, l)] dl \geq 0. \quad (2)$$

Condition (i) is the envelope condition, the dynamic analogue of the Myersonian payoff formula. Given the allocation, it pins down the truthful continuation payoff, and therefore expected transfers, up to a constant. By itself, it is only necessary and follows from local optimality. Condition (ii) is the global restriction. For every one-shot misreport, it compares the on-path and off-path marginal rents after integrating between the report and the true type. This comparison incorporates the entire continuation allocation rather than imposing period-by-period allocation monotonicity.

The difficulty lies in condition (ii). The off-path marginal rent depends on the full continuation allocation induced by each current misreport. Imposing the

⁵Implementable means that some transfer rule makes truthful reporting optimal at every history, on and off path, which [Pavan et al. \(2014\)](#) call a strongly truthful equilibrium.

condition therefore requires a continuum of constraints at every history, current type, and alternative report, coupled across periods through the continuation allocation. This makes direct analytic and numerical optimization difficult. The first-order approach instead optimizes subject only to condition (i) and then checks condition (ii).

3 Discrete-Time Preview: Linear Mechanisms

This section gives discrete-time intuition for the volatility characterization. For the linear mechanisms defined below, integral monotonicity reduces to a one-step bound after the initial report and an integral condition at date zero.

Fix $r > 0$, $\lambda \geq 0$, $\gamma > 0$, and $\bar{\theta} \in \mathbb{R}$. Let the interval $\Delta t > 0$ between dates satisfy $1 - \lambda\Delta t > 0$, and set

$$\beta := e^{-r\Delta t}, \quad \beta_\lambda := \beta(1 - \lambda\Delta t).$$

The type follows the Euler scheme for an OU process:

$$\theta_{t+1} = \theta_t + \lambda(\bar{\theta} - \theta_t)\Delta t + \gamma\varepsilon_{t+1},$$

where the innovations are independent across periods and of θ_0 , with mean zero and variance Δt . The agent's flow utility gross of transfers is $\theta_t X_t \Delta t$, so $u_\theta = X_t \Delta t$. The impulse response from period t to period s is $(1 - \lambda\Delta t)^{s-t}$.

Let $\hat{\theta}_t$ denote the report in period t . The innovation inferred from two consecutive reports is

$$\hat{\varepsilon}_{t+1} := \frac{\hat{\theta}_{t+1} - \hat{\theta}_t - \lambda(\bar{\theta} - \hat{\theta}_t)\Delta t}{\gamma}. \quad (3)$$

At each report history, let $\mathbb{E}_t$ denote expectation treating the reports received as truthful and assuming truthful continuation. Under $\mathbb{E}_t$, $\hat{\varepsilon}_{t+1} = \varepsilon_{t+1}$ and so has mean zero. The marginal rent is

$$D_t := \mathbb{E}_t \left[\sum_{s=t}^{\infty} \beta_\lambda^{s-t} X_s \Delta t \right] = X_t \Delta t + \beta_\lambda \mathbb{E}_t[D_{t+1}]. \quad (4)$$

At a history that is truthful through period $t-1$ with report $\hat{\theta}_t$, $D_t = D_t(\hat{\theta}_t, \hat{\theta}_t)$ in the notation of Section 2. For an initial report y , write $D(y) := D_0$, so $D(y) = D_0(y, y)$.

In the continuous-time model of the next section, the martingale representation theorem implies

$$dD_t = [(r + \lambda)D_t - X_t]dt + \psi_t dB_t \quad (5)$$

for a predictable process ψ . In general, there is no analogous representation theorem in discrete time.

Definition 1 (Linear mechanism). *A mechanism is linear if there are loadings ψ_t , measurable with respect to the report history through period t , such that*

$$D_{t+1} = \frac{D_t - X_t \Delta t}{\beta_\lambda} + \psi_t \hat{\varepsilon}_{t+1}. \quad (6)$$

Taking $\mathbb{E}_t$ in (6) recovers the recursion in (4). The loading ψ_t is the discrete counterpart of the continuous-time volatility ψ . Moreover, $\beta_\lambda^{-1} = 1 + (r + \lambda)\Delta t + O(\Delta t^2)$, so $\mathbb{E}_t[D_{t+1}] - D_t$ coincides with $[(r + \lambda)D_t - X_t]\Delta t$ up to first order.

Proposition 1. *Let the mechanism be linear, with all marginal rents $D_t(\hat{\theta}_t, \theta_t)$ well defined and finite. Truthful reporting satisfies integral monotonicity at every period $t \geq 1$ if and only if, at every truthful history through period $t - 1$ and for every report in period t ,*

$$\beta(1 - \lambda\Delta t)^2 \psi_t \leq \psi_{t-1}. \quad (7)$$

At period 0, integral monotonicity holds if and only if

$$\int_y^x \left[D(l) - D(y) - \beta(1 - \lambda\Delta t)^2 \frac{\psi_0(y)}{\gamma} (l - y) \right] dl \geq 0 \quad \forall x, y. \quad (8)$$

Proof. Fix a truthful history through $t - 1$, with $t \geq 1$. Condition on this history and $\theta_t = l$, and consider a one-shot deviation to $\hat{\theta}_t$. Write $\psi_t(\hat{\theta}_t)$ for the loading after this report. Iterating (6) forward gives⁶

$$\sum_{s=0}^{\infty} \beta_\lambda^s X_s \Delta t = D(\hat{\theta}_0) + \sum_{s=0}^{\infty} \beta_\lambda^{s+1} \psi_s \hat{\varepsilon}_{s+1}. \quad (9)$$

By (3), the misreport changes only two inferred innovations:

$$\hat{\varepsilon}_t = \varepsilon_t + \frac{\hat{\theta}_t - l}{\gamma}, \quad \hat{\varepsilon}_{t+1} = \varepsilon_{t+1} + (1 - \lambda\Delta t) \frac{l - \hat{\theta}_t}{\gamma}.$$

⁶The terminal term vanishes almost surely by the definition of D_n and martingale convergence.

All later innovation terms have mean zero since reports are truthful. Thus

$$\begin{aligned}
\beta_\lambda^t D_t(\hat{\theta}_t, l) &= \mathbb{E} \left[\sum_{s=0}^{\infty} \beta_\lambda^s X_s \Delta t \right] - \sum_{s=0}^{t-1} \beta_\lambda^s X_s \Delta t \quad \text{by (4)} \\
&= D(\hat{\theta}_0) + \mathbb{E} \left[\sum_{s=0}^{\infty} \beta_\lambda^{s+1} \psi_s \hat{\varepsilon}_{s+1} \right] - \sum_{s=0}^{t-1} \beta_\lambda^s X_s \Delta t \\
&= D(\hat{\theta}_0) + \sum_{s=0}^{t-2} \beta_\lambda^{s+1} \psi_s \varepsilon_{s+1} - \sum_{s=0}^{t-1} \beta_\lambda^s X_s \Delta t + \beta_\lambda^t \psi_{t-1} \hat{\varepsilon}_t + \beta_\lambda^{t+1} \psi_t(\hat{\theta}_t) \mathbb{E}[\hat{\varepsilon}_{t+1}] \\
&= D(\hat{\theta}_0) + \sum_{s=0}^{t-2} \beta_\lambda^{s+1} \psi_s \varepsilon_{s+1} - \sum_{s=0}^{t-1} \beta_\lambda^s X_s \Delta t \\
&\quad + \beta_\lambda^t \psi_{t-1} \left(\varepsilon_t + \frac{\hat{\theta}_t - l}{\gamma} \right) + \beta_\lambda^{t+1} \psi_t(\hat{\theta}_t) (1 - \lambda \Delta t) \frac{l - \hat{\theta}_t}{\gamma}.
\end{aligned}$$

The sum through $t-2$ is empty when $t=1$. Setting $\hat{\theta}_t = l$ gives $\beta_\lambda^t D_t(l, l)$. Using $\beta_\lambda(1 - \lambda \Delta t) = \beta(1 - \lambda \Delta t)^2$,

$$D_t(l, l) - D_t(\hat{\theta}_t, l) = \left[\psi_{t-1} - \beta(1 - \lambda \Delta t)^2 \psi_t(\hat{\theta}_t) \right] \frac{l - \hat{\theta}_t}{\gamma}.$$

Integrating over l from $\hat{\theta}_t$ to θ_t ,

$$\int_{\hat{\theta}_t}^{\theta_t} [D_t(l, l) - D_t(\hat{\theta}_t, l)] dl = \frac{\psi_{t-1} - \beta(1 - \lambda \Delta t)^2 \psi_t(\hat{\theta}_t)}{2\gamma} (\theta_t - \hat{\theta}_t)^2.$$

For a fixed report, the coefficient is independent of θ_t , so nonnegativity for every true type is equivalent to (7).

At date zero, the same argument for true type l and report y yields

$$D_0(y, l) = D(y) + \beta(1 - \lambda \Delta t)^2 \psi_0(y) \frac{l - y}{\gamma}.$$

Since $D_0(l, l) = D(l)$, integral monotonicity is exactly (8). $\square$

As $\Delta t \downarrow 0$, (7) converges to the drift bound in Theorem 1a, and (8) converges to the integral condition in Theorem 1b. The pathwise one-step bound has continuous-time limit $d\psi_t \leq (r + 2\lambda)\psi_t dt$. Because this inequality is pathwise, it rules out a Brownian component in ψ . Any ψ satisfying this condition has finite variation. The discrete-time argument recovers the finite-variation characterization previewed in the introduction.

In continuous time, every mechanism satisfies (5), so this conclusion may appear to make finite variation necessary for incentive compatibility. It is not: an incentive-compatible mechanism may have ψ with a Brownian component. In this sense, restricting to linear mechanisms and then taking the continuous-time limit does not give a complete picture. However, a closely related condition characterizes local incentive compatibility. In the analysis below I show that local incentive compatibility requires $e^{-(r+2\lambda)t}\psi_t$ to be a supermartingale: $\text{drift}(\psi_t) \leq (r + 2\lambda)\psi_t$.

4 Benchmark Model

Fix a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying a standard Brownian motion $B = \{B_t\}_{t \geq 0}$ and an independent random variable $\theta_0 \sim F$. Let $\{\mathcal{F}_t\}_{t \geq 0}$ be the usual augmentation of the filtration generated by θ_0 and B . Since $\theta_0 \perp\!\!\!\perp B$, the process B remains Brownian with respect to this filtration. Write $\mathbb{E}_x[\cdot] := \mathbb{E}[\cdot \mid \theta_0 = x]$.

Types. The principal allocates to an agent whose marginal value is privately known and evolves according to

$$d\theta_t = \lambda(\bar{\theta} - \theta_t) dt + \gamma dB_t, \quad (10)$$

with initial condition θ_0 . Above, $\lambda \geq 0$ is the rate of mean reversion, $\bar{\theta}$ the long-run mean, and $\gamma > 0$ the volatility. The distribution F of initial types has finite variance and a density f that is positive on the interior of its support Θ , an interval in $\mathbb{R}$.

Reports. The principal does not observe the agent's type. She commits to a mechanism that responds to the *reported* type.

A reporting strategy consists of an initial report $y \in \Theta$ and a progressively measurable process a . Define

$$\hat{B}_t := B_t + \int_0^t a_s ds.$$

The reported type satisfies

$$d\hat{\theta}_t = \lambda(\bar{\theta} - \hat{\theta}_t) dt + \gamma d\hat{B}_t, \quad \hat{\theta}_0 = y. \quad (11)$$

Conditional on $\theta_0 = x$, truthful reporting is $(y, a) = (x, 0)$.

Definition 2 (Admissible reports). *A reporting strategy (y, a) is admissible if, for every $x \in \Theta$:*

- (O1) *Conditional on $\theta_0 = x$, on every finite horizon, its induced law of reports is absolutely continuous with respect to the law of θ conditional on $\theta_0 = y$.*
- (O2) *The drift distortion satisfies*

$$\mathbb{E}_x \int_0^\infty e^{-rt} a_t^2 dt < \infty.$$

Condition (O1) prevents a report from generating, on any finite horizon, an event that has zero probability under the truthful law.⁷ In particular, report laws with a different quadratic variation or with jumps are singular and therefore excluded (hence the restriction to drift distortions). Every pair (y, a) with $y \in \Theta$ and bounded a is admissible.

Mechanisms. A mechanism $M = (X, T)$ assigns an allocation and a transfer to every report history. The allocation takes values in a convex set $\mathcal{X} \subseteq \mathbb{R}$, possibly $\mathbb{R}$ itself, and $T_t \in \mathbb{R}$ is the flow transfer from the agent to the principal. Both rules are progressively measurable with respect to the history of reports. Write X and T for the allocation and transfer processes under truthful reporting, and $\hat{X}$ and $\hat{T}$ for those induced by an arbitrary reporting strategy (y, a) . Throughout, mechanisms are required to satisfy

$$\int_0^\infty e^{-(r+\lambda)t} X_t dt \in L^2, \quad \mathbb{E} \int_0^\infty e^{-rt} |T_t| dt < \infty, \quad (12)$$

so that the marginal rent and discounted transfers below are well defined. The conditions defining an *admissible mechanism* are given in Appendix A. They include integrability, transversality, and continuity requirements. The appendix explains the role of each condition.

Agent's problem. The agent's net flow payoff is $\theta_t X_t - T_t$. Conditional on $\theta_0 = x$, his expected discounted payoff from reporting (y, a) is

$$U_x(y, a) := \mathbb{E}_x \int_0^\infty e^{-rt} (\theta_t \hat{X}_t - \hat{T}_t) dt.$$

Write $U(x) := U_x(x, 0)$ for his truthful utility. When the true initial type is clear, I suppress the subscript x .

⁷By Girsanov's theorem, Novikov's condition is sufficient for (O1).

A mechanism is *incentive compatible* if

$$U_x(y, a) \leq U(x) \quad \forall x \in \Theta$$

for every admissible report (y, a) . Thus the agent cannot gain by misreporting his initial type, distorting his subsequent reports, or doing both.

The agent's outside option is normalized to zero. A mechanism is *individually rational* if

$$U(x) \geq 0 \quad \forall x \in \Theta.$$

Following the dynamic mechanism-design literature (Pavan et al., 2014; Bergemann and Strack, 2015), participation is imposed only at date zero, after the agent observes his initial type.

Principal's problem. The principal receives the transfers and bears a convex flow cost of provision $C: \mathcal{X} \rightarrow \mathbb{R}_+$ with $C(0) = 0$ (as in Mussa and Rosen (1978)); the costless case $C \equiv 0$ is permitted. She maximizes

$$\Pi(M) := \mathbb{E} \int_0^\infty e^{-rt} (T_t - C(X_t)) dt$$

over admissible mechanisms satisfying incentive compatibility and individual rationality.

Discussion. The benchmark model is designed to illustrate the main ideas as simply as possible. The analysis extends along several dimensions.

Section 7.1 extends the analysis to geometric Brownian motion.

The analysis of implementability below also extends to flow payoffs of the form

$$\theta g(X) - c(X) + h(\theta) - T,$$

with g continuous and strictly increasing. Setting $q := g(X)$ and $\tilde{T} := T + c(X)$ rewrites the payoff as $\theta q - \tilde{T} + h(\theta)$. Because $h(\theta)$ is independent of reports, it affects participation but not reporting incentives. The implementability conditions therefore apply with q in place of X . The principal receives $T = \tilde{T} - c(X)$, so $c(X)$ becomes part of her allocation cost. The application in Section 6 uses this reduction.

Moreover, the implementability conditions do not depend on the principal's objective. Changing C or adding a flow payoff $S(\theta_t, X_t)$ alters the objective of the control problem in Section 5.3. This allows objectives that combine revenue and surplus.

5 Main Results

This section develops the main results previewed in the introduction. I characterize incentive compatibility in terms of constraints on the marginal rent and its volatility. I then use these constraints to formulate the two optimal control programs that bound the principal's true payoff.

5.1 Marginal rent and its volatility

I first define the key objects. Since the impulse response for an OU process is $I_{s,t} = e^{-\lambda(s-t)}$, the *marginal information rent* from Section 2 is:

$$D_t := \mathbb{E}_t \left[\int_t^\infty e^{-(r+\lambda)(s-t)} X_s \, ds \right], \quad (13)$$

where $\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_t]$.

Under (12),

$$e^{-(r+\lambda)t} D_t + \int_0^t e^{-(r+\lambda)s} X_s \, ds$$

is a square-integrable martingale under truthful reporting. The martingale representation theorem therefore gives

$$dD_t = [(r + \lambda)D_t - X_t]dt + \psi_t dB_t \quad (14)$$

for a predictable process ψ , the marginal rent volatility.

At each reported history, D_t is computed treating the reports received as truthful and assuming truthful continuation. Under a deviation, the dynamics satisfy (14) with B replaced by $\hat{B}$.

For an initial report y , write $D(y) := D_0$ and let $\psi_0(y)$ denote the initial value of ψ .

5.2 Implementability

The marginal rent and its volatility enter incentives through the agent's continuation value. Let

$$W_t := \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} (\hat{\theta}_s X_s - T_s) \, ds \right]$$

denote the continuation value under truthful continuation ($a_s = 0, s > t$).

Suppose the current report differs from the type by $\delta_t := \hat{\theta}_t - \theta_t$, and the agent stops distorting from time t onward. The OU dynamics imply $\delta_s = e^{-\lambda(s-t)}\delta_t$ for $s > t$. His true continuation payoff is therefore

$$W_t + \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} (\theta_s - \hat{\theta}_s) \hat{X}_s ds \right] = W_t - D_t \delta_t.$$

Suppose the agent has not yet lied $\delta_t = 0$ and consider a small distortion a_t . The distortion moves W_t , contributing drift $a_t \text{vol}(W_t)$, while δ_t grows at rate γa_t , at a cost of D_t per unit. The net drift effect is

$$a_t (\text{vol}(W_t) - \gamma D_t).$$

Since a_t may take either sign, truthful reporting requires $\text{vol}(W_t) = \gamma D_t$. If only negative distortions are feasible, as in Williams (2011), $\text{vol}(W_t) \geq \gamma D_t$ suffices and if the inequality is strict, every lie incurs a first-order loss. If the principal's objective is smooth, the strict inequality only costs her marginally and local incentives are determined at first order. With both positive and negative distortions, the first-order effect must vanish, and second-order effects determine local incentives.

Definition 3 (Local incentive compatibility). *Fix $x \in \Theta$. A pair (h, a) is a feasible deviation direction at x if $(x + \varepsilon h, \varepsilon a)$ is an admissible report for every sufficiently small $\varepsilon > 0$. Truthful reporting is locally incentive compatible at x if, for every feasible deviation direction (h, a) ,*

$$\limsup_{\varepsilon \downarrow 0} \frac{U_x(x + \varepsilon h, \varepsilon a) - U_x(x, 0)}{\varepsilon^2} \leq 0.$$

It is locally incentive compatible after the initial report if the same condition holds for every feasible direction with $h = 0$, and globally incentive compatible after the initial report if $U_x(x, a) \leq U_x(x, 0)$ for every admissible distortion a .

A strictly negative limsup rules out profitable small deviations along that direction. Appendix G.5 gives conditions under which mechanisms satisfying the weak inequality can be approximated, at arbitrarily small revenue cost, by mechanisms satisfying it strictly.

The subsection proceeds as follows. The first-order conditions pin down the present value of transfers (Proposition 2) as in the discrete time theory (Pavan et al., 2014). Once they hold, the payoff from any report reduces to a formula in the volatility of the marginal rent (Lemma 1), and second-order effects deliver the incentive conditions after the initial report (Theorem 1a) and at the initial report (Theorem 1b).

In what follows, let

$$W_t^X := \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \hat{\theta}_s X_s \, ds \right]$$

denote continuation utility gross of transfers.

Proposition 2 (Implementing transfers). *Let an allocation X satisfying (12) be given.*

(i) *Local incentive compatibility after the initial report requires*

$$\text{vol}(W_t) = \gamma D_t \quad dt \otimes d\mathbb{P}\text{-a.e.} \quad (15)$$

(ii) *Incentive compatibility at the initial report requires*

$$U(x) - U_c = \int_c^x D(l) \, dl \quad \forall c, x \in \Theta, \quad (16)$$

where $U_c := U(c)$. Equivalently,

$$\mathbb{E}_x \int_0^\infty e^{-rt} T_t \, dt = W_0^X(x) - \int_c^x D(l) \, dl - U_c. \quad (17)$$

Conversely, let

$$H_t := \gamma D_t - \text{vol}(W_t^X).$$

Suppose that H is square-integrable on every finite horizon under every admissible report and that the transfer below forms an admissible mechanism. Then, for any $c \in \Theta$ and $U_c \in \mathbb{R}$,

$$T_t = r \left[W_0^X(\hat{\theta}_0) - \int_c^{\hat{\theta}_0} D(l) \, dl - U_c \right] - r \int_0^t H_s \, d\hat{B}_s$$

satisfies (i)–(ii).

The proof is in Appendix C. The conditions in the proposition remove any first-order effects of a deviation. Once they are met, the payoff from an arbitrary deviation has a simple formula.

Lemma 1 (Master deviation formula). *Let M be admissible with transfers satisfying Proposition 2(i), and let (y, a) be an admissible report. Then*

$$U_x(y, a) = U(y) + (x - y)D(y) - \mathbb{E}_x \int_0^\infty e^{-rt} \psi_t \delta_t a_t \, dt, \quad (18)$$

with ψ evaluated along the report, and where the gap δ obeys

$$d\delta_t = (-\lambda \delta_t + \gamma a_t) \, dt, \quad \delta_0 = y - x.$$

The first two terms on the right-hand side of (18) give the payoff from an initial report y with no subsequent drift distortion. The integral accounts for later distortions and depends on allocations only through ψ .

The proof is in Appendix B. For intuition, consider again the effect of a drift distortion on the agent's continuation payoff $W_t - D_t\delta_t$. The net effect is

$$a_t(\text{vol}(W_t) - \gamma D_t) - \psi_t \delta_t a_t.$$

The first term is as before when we assumed $\delta_t = 0$ and Condition (15) ensures it vanishes. The second arises because the distortion moves D_t with drift $a_t\psi_t$ and arises only when the agent has deviated in the past $|\delta_t| > 0$. The cumulative effect on payoffs appears in the integral in (18).

The integral term implies two restrictions. First, ψ must be nonnegative. If it is negative on a set of positive measure, a sufficiently small constant-sign distortion supported there gives $\delta_t a_t \geq 0$ and a strictly positive gain in (18). Second, ψ cannot grow too fast. Write

$$\rho := r + 2\lambda.$$

Fix a truthful initial report, so $\delta_0 = 0$, and substitute $\gamma a_t dt = d\delta_t + \lambda \delta_t dt$ into the integral. If ψ is an Itô process, integration by parts gives⁸

$$\begin{aligned} U_x(x, a) - U(x) = & -\frac{1}{2\gamma} \mathbb{E}_x \int_0^\infty e^{-rt} \delta_t^2 [\rho \psi_t - \text{drift}(\psi_t)] dt \\ & + \frac{1}{2\gamma} \mathbb{E}_x \int_0^\infty e^{-rt} \delta_t^2 \text{vol}(\psi_t) a_t dt. \end{aligned} \quad (19)$$

The first term is nonpositive for deviations of any size when the drift of ψ is at most $\rho\psi$. The second contains the volatility of ψ . It is of higher order for the scaled deviations in Definition 3, so local incentives depend on the first term alone. These higher-order terms vanish when ψ has finite variation in which case local and global incentives are equivalent.

Theorem 1a (Incentives after the initial report). *Let M be admissible and suppose transfers satisfy Proposition 2(i). Truthful reporting is locally incentive compatible after the initial report if and only if $e^{-\rho t}\psi_t$ admits a nonnegative càdlàg supermartingale representative, written $e^{-\rho t}\overline{\psi}_t$, under truthful continuation from every initial report.*⁹

⁸The general calculation retains a terminal term that is nonpositive under the supermartingale condition below; see Appendix D.

⁹ ψ is identified only up to a $dt \otimes d\mathbb{P}$ -null set; a representative is a predictable process in this equivalence class.

If the finite-variation part of $\bar{\psi}$ is absolutely continuous, the condition is equivalent to $\bar{\psi} \geq 0$ and

$$\text{drift}(\bar{\psi}_t) \leq \rho \bar{\psi}_t \quad dt \otimes d\mathbb{P}\text{-a.e.}$$

If $\bar{\psi}$ has finite variation, local and global incentive compatibility after the initial report are equivalent.

I use this representative below and omit the bar. The condition implies that, for $s < t$,

$$\mathbb{E}_s[\psi_t] \leq e^{\rho(t-s)} \psi_s.$$

Thus the volatility's conditional expected growth rate cannot exceed ρ . The rate $\rho = r + 2\lambda$ combines the $r + \lambda$ discounting in the marginal rent with the additional rate λ at which a reporting gap decays. The proof of Theorem 1a is in Appendix D.

At date zero, incentive compatibility also restricts the marginal rent schedule as a function of the initial type. The binding deviations are ones where any initial lie is corrected arbitrarily quickly. Appendix E uses these deviations together with the master deviation formula to prove the following.

Theorem 1b (Incentives at the initial report). *Let M be admissible. Suppose transfers satisfy the conditions of Proposition 2 and truthful continuation is locally incentive compatible after every truthful initial report.*

Local incentives. *At any $x \in \Theta$ where the relevant one-sided derivatives of D exist, truthful reporting is locally incentive compatible if and only if*

$$\psi_0(x) \leq \gamma D'_+(x) \quad \text{if } x < \sup \Theta, \quad \psi_0(x) \leq \gamma D'_-(x) \quad \text{if } x > \inf \Theta. \quad (20)$$

If D is differentiable at x , these inequalities reduce to

$$\psi_0(x) \leq \gamma D'(x).$$

Global incentives. *If ψ has finite variation after every truthful initial report, truthful reporting is globally incentive compatible whenever*

$$\int_y^x \left[D(l) - D(y) - \frac{\psi_0(y)}{\gamma} (l - y) \right] dl \geq 0 \quad \forall x, y \in \Theta. \quad (21)$$

Remark 1. *The integral condition (21) is also necessary under mild regularity: whenever the volatility cannot change appreciably while a fixed initial lie*

is corrected over a vanishingly short interval—as for smooth Markov finite-variation mechanisms with locally bounded, absolutely continuous volatility dynamics—incentive compatibility implies (21).¹⁰

Condition (21) is the continuous-time date-zero form of integral monotonicity from Section 2. For an initial report y , compare $D(l)$ with the affine function $D(y) + (\psi_0(y)/\gamma)(l - y)$, which plays the role of the off-path marginal rent $D_t(\hat{\theta}_t, l)$. The local condition compares their slopes; the global condition compares their integrals. Figure 1 illustrates the two restrictions.

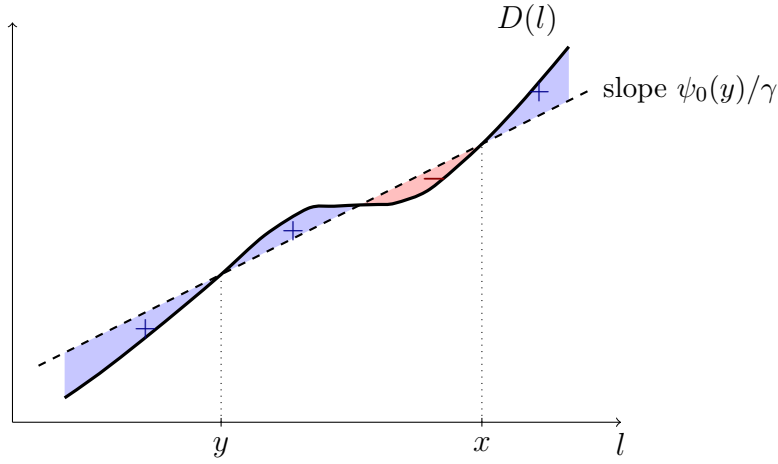


Figure 1: The date-zero conditions. The solid curve is the marginal rent D ; the dashed line through $D(y)$ has slope $\psi_0(y)/\gamma$. The local condition requires this slope not to exceed the slope of D at y (at a kink, the minimum of the one-sided slopes). The global condition requires the signed area between the marginal rent and the line, from y to every possible true type x , to be nonnegative. Blue regions contribute positively to the integral and red regions negatively.

The principal internalizes that reducing the slope of D (for example, to iron the allocation to reduce date-zero information rents) lowers the maximum initial marginal rent volatility and hence limits her ability to expose expected discounted future allocations to type innovations.

¹⁰Consider a report that starts at y and eliminates the gap over $[0, \varepsilon]$. If the volatility changes by $o(1)$ along the correction as $\varepsilon \downarrow 0$, the deviation attains the bound used in the proof of the theorem, and incentive compatibility yields (21).

5.3 Optimal control

The sufficient finite-variation conditions and the necessary local conditions lead to two control programs. Each program first optimizes the continuation from a given report, marginal rent, and volatility, then chooses the date-zero schedules. Under the verification conditions below, the programs give lower and upper bounds on the principal's payoff, respectively.

Eliminating transfers. Fix a base type $c \in \Theta$, let $U_c := U(c)$, and, for $\theta \in \text{int } \Theta$, define

$$h_c(\theta) := \frac{\mathbb{1}\{\theta \geq c\} - F(\theta)}{f(\theta)},$$

which generalizes the inverse hazard rate and reduces to $(1 - F)/f$ when c is the lowest type. Its values at finite endpoints are immaterial. Substituting the expected transfer from Proposition 2(ii) into the principal's objective and integrating over initial types gives

$$\Pi = \mathbb{E} \int_0^\infty e^{-rt} [\theta_t X_t - C(X_t)] dt - \mathbb{E}[h_c(\theta_0)D(\theta_0)] - U_c, \quad (22)$$

and the envelope formula makes individual rationality equivalent to

$$U_c + \int_c^\theta D(l) dl \geq 0 \quad \forall \theta \in \Theta. \quad (23)$$

The principal's payoff is surplus minus date-zero information rents.

For fixed date-zero schedules, the rent terms in (22) are constant, so the continuation problem maximizes surplus,

$$\mathbb{E}_t \int_t^\infty e^{-r(s-t)} [\hat{\theta}_s X_s - C(X_s)] ds,$$

subject to delivering the promised marginal rent and volatility. Continuation utility is *not* a state variable: Proposition 2 shows that transfers can always be constructed to deliver the correct first-order incentives.

States and controls. The continuation state is $(\hat{\theta}, D, \psi)$. Along the report, the marginal rent and volatility satisfy

$$dD_t = [(r + \lambda)D_t - X_t]dt + \psi_t d\hat{B}_t, \quad (24)$$

$$d\psi_t = \rho\psi_t dt + v_t d\hat{B}_t - dK_t, \quad \psi_t \geq 0. \quad (25)$$

The allocation X controls the drift of D ; v is the diffusion coefficient of ψ ; and K lowers ψ . The processes v and K are predictable, and K is càdlàg and nondecreasing with $K_0 = 0$; write K^c for its continuous part. The second equation is the semimartingale form of the requirement that $e^{-\rho t}\psi_t$ be a nonnegative supermartingale (Theorem 1a). The local program permits a choice of v ; the finite-variation program sets $v = 0$. Once ψ reaches zero it remains there—a nonnegative supermartingale is absorbed at zero—so $v = 0$ and K is constant thereafter.

These dynamics prevent the principal from resetting ψ freely at each date.

The variational inequality. For an interior state $(\hat{\theta}, D, \psi) \in \mathbb{R}^2 \times (0, \infty)$, let $V(\hat{\theta}, D, \psi)$ be a candidate continuation value. For an allocation $x \in \mathcal{X}$ and a diffusion coefficient v , let $\mathcal{L}^{x,v}$ denote the generator of the states when K is constant (i.e., the supermartingale constraint is binding):

$$\begin{aligned} \mathcal{L}^{x,v}V &:= \lambda(\bar{\theta} - \hat{\theta})V_{\hat{\theta}} + [(r + \lambda)D - x]V_D + \rho\psi V_{\psi} \\ &\quad + \frac{\gamma^2}{2}V_{\hat{\theta}\hat{\theta}} + \gamma\psi V_{\hat{\theta}D} + \frac{\psi^2}{2}V_{DD} + \frac{v^2}{2}V_{\psi\psi} + v(\gamma V_{\hat{\theta}\psi} + \psi V_{D\psi}). \end{aligned} \quad (26)$$

The two continuation problems differ only in the controls available for the volatility. Write $P \in \{\text{FV}, \text{loc}\}$ for the program, and let

$$\mathcal{A}_{\text{FV}} := \mathcal{X} \times \{0\}, \quad \mathcal{A}_{\text{loc}} := \mathcal{X} \times \mathbb{R}$$

be the corresponding sets of controls (x, v) . Both programs are governed by the variational inequality

$$\min\{\mathfrak{B}_P[V], V_{\psi}\} = 0 \quad \text{on } \mathbb{R}^2 \times (0, \infty), \quad (27)$$

where

$$\mathfrak{B}_P[V] := rV - \sup_{(x,v) \in \mathcal{A}_P} [\hat{\theta}x - C(x) + \mathcal{L}^{x,v}V].$$

Intuitively, since the designer can always lower volatility, $V_{\psi} \geq 0$. Where $V_{\psi} > 0$, K remains constant, the drift of ψ is $\rho\psi$, and the Bellman equation binds: $\mathfrak{B}_P[V] = 0$. Where $V_{\psi} = 0$, an infinitesimal reduction has no first-order value cost, and the Bellman inequality $\mathfrak{B}_P[V] \geq 0$ may be strict.

The date-zero programs. It remains to choose the date-zero schedules. Let V^P denote a candidate value for program P . In both programs, D and ψ_0 are such that the objective below is well defined. The date-zero program is

$$\Pi_P := \sup_{D(\cdot), \psi_0(\cdot), U_c} \int_{\Theta} [V^P(\theta, D(\theta), \psi_0(\theta)) - h_c(\theta)D(\theta)] dF(\theta) - U_c, \quad (28)$$

subject to individual rationality (23) and the date-zero incentive constraints of the program:

FV: $\psi_0 \geq 0$ and the integral condition (21);

loc: D has a nondecreasing representative on $\text{int } \Theta$, with derivative D' , and $0 \leq \psi_0 \leq \gamma D'$ F -almost everywhere.¹¹

When V^P is the true continuation value, both date-zero programs are convex: their feasible sets are convex, and their objectives are concave in the delivered schedules.¹²

From a candidate value to a mechanism. Given a solution of the variational inequality, construct a candidate optimal mechanism as follows.

1. Solve the date-zero program for initial promised marginal rent and volatility schedules.
2. Choose measurable feedback controls $(x, v) \in \mathcal{A}_P$ that attain the supremum in (27). From each initial state, the resulting path must satisfy the Bellman equality $\mathfrak{B}_P[V] = 0$ at almost every time with $\psi > 0$.
3. Increase K^c only where $V_\psi = 0$. When possible, raise K through a jump to lower ψ to the lowest level that leaves $V(\hat{\theta}, D, \psi)$ unchanged.
4. At $\psi = 0$, set $v = 0$, keep K constant, and choose the allocation to satisfy the Bellman equality (57) at the boundary (see Appendix F).

Because $K_0 = 0$, the initial state must satisfy these requirements without an initial jump. The verification theorem also requires an admissible solution of the state equations and delivery of the promised marginal rent. Appendix F states the martingale, transversality, and attainment conditions.

Theorem 2 (Verification). *Fix a program $P \in \{\text{FV}, \text{loc}\}$ and write $V = V^P$. Let $M^* = (X^*, T^*)$ be an admissible mechanism, with initial marginal-rent schedule D^* , initial volatility schedule ψ_0^* , and utility U_c^* for the base type c . Suppose Condition 1 holds and:*

- (i) V solves (27) for program P , together with the boundary condition (56);

¹¹Lemma 3 in the online appendix shows that D is nondecreasing in any incentive-compatible mechanism. If D is absolutely continuous, monotonicity follows directly from the derivative bound.

¹²The state dynamics are linear, the allocation set is convex, and the flow surplus $\theta x - C(x)$ is concave in x . Mixing two feasible mechanisms therefore delivers the mixed marginal rent and volatility and at least the corresponding mixture of surplus.

- (ii) for every initial type $\theta \in \Theta$, the continuation of M^* from $(\theta, D^*(\theta), \psi_0^*(\theta))$ satisfies (24)–(25) and the state and control restrictions for P (stated under “States and controls” in Section 5.3). It satisfies the Bellman equality (57), and reductions through K leave V unchanged as in (58);
- (iii) (D^*, ψ_0^*, U_c^*) maximizes the date-zero program (28); if $P = \text{loc}$, D^* is absolutely continuous, the derivatives required by (20) exist, and (D^*, ψ_0^*) satisfies that condition at every type;
- (iv) the transfers of M^* satisfy Proposition 2(i)–(ii).

Then M^* is individually rational and attains Π_P . If $P = \text{FV}$, it is globally incentive compatible and optimal among admissible, incentive-compatible, individually rational finite-variation mechanisms satisfying (21). If $P = \text{loc}$, it is locally incentive compatible and optimal among admissible locally incentive-compatible, individually rational mechanisms with an absolutely continuous initial marginal rent.

Corollary 2.1 (The bracket). *Under the conditions of Theorem 2 for both programs,*

$$\Pi_{\text{FV}} \leq \sup_M \Pi(M) \leq \Pi_{\text{loc}},$$

where the supremum is over admissible, incentive-compatible, individually rational mechanisms.

Proofs are in Online Appendix G.

6 Application: Financial Market Design

I apply the machinery to a benchmark problem in market microstructure: a dealer extracts revenue from a trader whose hedging needs are private and persistent, a dynamic version of the dealer’s problem in [Glosten \(1989\)](#).¹³ The first-order approach fails because the trader may be either a buyer or a seller, creating countervailing incentives in the sense of [Lewis and Sappington \(1989\)](#): his date-zero incentive constraints can bind in either direction. I then characterize the solutions to the finite-variation and local programs, discuss their dynamic properties, including the pattern of binding incentive constraints, and show in examples, that the programs yield nearly tight bounds. I then extend the analysis to an exchange with many traders and market clearing.

¹³For simplicity, I assume the asset pays no cash flow though the analysis extends straightforwardly if there is no asymmetric information between the dealer and trader on this dimension. [Glosten \(1989\)](#) allows for private information about the asset’s payoff.

6.1 The dealer's problem

A dealer trades an asset with a trader whose ideal position evolves according to

$$d\theta_t = -\lambda\theta_t dt + dB_t, \quad \theta_0 \sim F,$$

where $\lambda \geq 0$ is the mean-reversion rate and the volatility is normalized to one. Throughout, F has finite variance and admits a continuous density f that is symmetric about zero and positive on the interior of its support.

The trader's flow payoff from position X_t and payment T_t is

$$-\frac{1}{2}(X_t - \theta_t)^2 - T_t;$$

the quadratic term penalizes deviations from the trader's ideal position, and the asset pays no cash flow. A trader who does not participate holds $X_t = 0$ and pays nothing. Relative to this outside option, his flow gain from participation, gross of transfers, is

$$u(\theta_t, X_t) = -\frac{1}{2}(X_t - \theta_t)^2 + \frac{1}{2}\theta_t^2 = \theta_t X_t - \frac{1}{2}X_t^2.$$

Setting $\tilde{T}_t := T_t + X_t^2/2$ therefore gives the trader the bilinear flow payoff $\theta_t X_t - \tilde{T}_t$ and the dealer the flow revenue $\tilde{T}_t - X_t^2/2$: this is the benchmark model with allocation cost $C(X) = X^2/2$. The dealer commits to allocation and transfer rules adapted to the trader's reports and maximizes expected discounted revenue, subject to incentive compatibility and individual rationality. Throughout this application, T continues to denote the trader's actual payment; $\tilde{T}$ is used only to apply the benchmark formulas.

Apply the revenue formula (22) using $\tilde{T}$, base type $c = 0$, and base value $u_0 := U(0)$. For nonzero types in the interior of the support, write

$$\Lambda(\theta) := h_0(\theta) = \begin{cases} (1 - F(\theta))/f(\theta), & \theta > 0, \\ -F(\theta)/f(\theta), & \theta < 0, \end{cases}$$

the *two-sided inverse-hazard weight*. The Myerson virtual value of a trader of initial type θ is $\theta - \Lambda(\theta)$. Buyers' marginal rents are weighted by the upper tail of F , and sellers' by the lower tail. The upward jump of $1/f(0)$ at zero is the source of countervailing incentives below. Under truthful reporting, the marginal rent process is (13), and the envelope formula (16) gives $U(\theta) = u_0 + \int_0^\theta D(l) dl$. Since $\theta_t X_t - \frac{1}{2}X_t^2 = \frac{1}{2}\theta_t^2 - \frac{1}{2}(X_t - \theta_t)^2$, the objective (22) becomes

$$\begin{aligned} \Pi = \mathbb{E} \left[\int_0^\infty e^{-rt} \frac{1}{2} \theta_t^2 dt \right] - \frac{1}{2} \mathbb{E} \left[\int_0^\infty e^{-rt} (X_t - \theta_t)^2 dt \right] \\ - \mathbb{E} [\Lambda(\theta_0) D(\theta_0)] - u_0. \end{aligned} \quad (29)$$

The first term is first-best surplus and is independent of the mechanism. The second is the loss from deviations from the trader's ideal position. The final two terms subtract expected information rents, $\mathbb{E}[U(\theta_0)] = \mathbb{E}[\Lambda(\theta_0)D(\theta_0)] + u_0$. The dealer thus trades off tracking the ideal position, which requires a minimal level of volatility ψ , against reducing information rents.

At date zero, implementability is checked by comparing the marginal-rent schedule with its initial volatility. With $\gamma = 1$, the continuation and integral conditions give the report-by-report restriction

$$0 \leq \psi_0(y) \leq \psi_0^{\max}(y) := \inf_{m \in \Theta \setminus \{y\}} \frac{2}{(m-y)^2} \int_y^m [D(l) - D(y)] dl. \quad (30)$$

The upper bound is equivalent to (21). With finite-variation continuations it is sufficient for date-zero incentives; under the regularity in Remark 1, it is also necessary.

6.2 The first-order approach

Here, I adopt the first-order approach and show that it fails. The first-order approach maximizes (29) subject only to the envelope formula and participation, $U(\theta) \geq 0$, ignoring the remaining incentive conditions.

Recall $\rho = r + 2\lambda$. On the positive interior of the support, define the buyer-side virtual value by

$$\theta - \frac{1 - F(\theta)}{f(\theta)}$$

Call F *regular* if this virtual value is nondecreasing and positive somewhere, so that x^* below is interior. By symmetry, the seller side then has the same property. Log-concavity of f is sufficient. Let

$$x^* := \sup \left\{ \theta \in \text{int } \Theta \cap [0, \infty) : \theta - \frac{1 - F(\theta)}{f(\theta)} \leq 0 \right\}.$$

Proposition 3 (The first-order relaxation). *Suppose F is regular. The optimal initial marginal rent is*

$$D^{\text{FOA}}(\theta) = \begin{cases} \frac{\theta - \Lambda(\theta)}{\rho}, & |\theta| > x^*, \\ 0, & |\theta| \leq x^*. \end{cases} \quad (31)$$

Participation binds precisely for $\theta \in [-x^, x^*]$. The corresponding allocation is*

$$X_t^{\text{FOA}} = \begin{cases} \theta_t - \Lambda(\theta_0)e^{-\lambda t}, & |\theta_0| > x^*, \\ \theta_t - \theta_0 e^{-\lambda t}, & |\theta_0| \leq x^*. \end{cases} \quad (32)$$

Under truthful reporting, $D_t = X_t^{\text{FOA}}/\rho$, so $\psi_t \equiv 1/\rho$. The relaxed value of the objective is

$$\Pi^{\text{FOA}} = \frac{1}{2r\rho} + \frac{\rho}{2}\mathbb{E}[D^{\text{FOA}}(\theta_0)^2]. \quad (33)$$

The marginal rent is unique up to F -null sets and the allocation up to $dt \otimes d\mathbb{P}$ -null sets.

The proof in Online Appendix H.1 uses standard Lagrangian methods for static mechanism design with type-dependent outside options (Jullien, 2000). Regularity ensures D^{FOA} nondecreasing. As I soon explain, the first-order solution therefore fails because it violates the *dynamic* integral condition, not ordinary monotonicity (which is a requirement from static mechanism design). This failure occurs even under a monotone hazard rate.

Figure 2 illustrates the solution for F uniform on $[-1, 1]$.

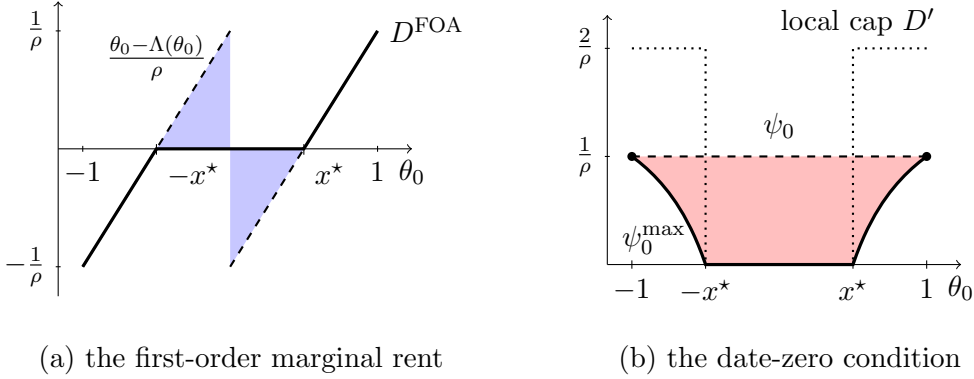


Figure 2: The first-order approach for uniform types. Panel (a) plots the marginal rent (31) (solid) and $(\theta_0 - \Lambda(\theta_0))/\rho$ (dashed). Panel (b) compares the delivered initial marginal rent volatility $1/\rho$ (dashed), the cap $\psi_0^{\max}$ imposed by (30) (solid), and the local cap D' (dotted). The shading in panel (b) marks the violation of the cap.

Panel (a) compares D^{FOA} (solid) with the virtual value divided by ρ (dashed). Holding u_0 fixed, the dealer would choose the dashed marginal rent schedule if participation constraints were ignored.¹⁴ At zero, however, the virtual value jumps downward. Because $U' = D$, setting $u_0 = 0$ under the dashed schedule would give negative utility to types immediately on either side of zero. The first-order solution instead flattens the marginal rent on a central interval:

$$D^{\text{FOA}}(\theta_0) = 0 \quad \text{for } \theta_0 \in [-x^*, x^*],$$

¹⁴Without fixing u_0 , dropping participation would allow the dealer to lower every type's utility by the same amount and obtain unbounded revenue.

where participation constraints bind.

By definition,

$$D(\theta_0) = \mathbb{E} \left[\int_0^\infty e^{-(r+\lambda)t} X_t dt \middle| \theta_0 \right].$$

Thus central types have zero expected trade discounted at rate $r + \lambda$. In fact, for $|\theta_0| \leq x^*$,

$$X_t^{\text{FOA}} = \theta_t - \mathbb{E}[\theta_t | \theta_0],$$

so their expected allocation is zero at every date. At date zero, no central type trades, as if the dealer charged a bid–ask spread. After date zero, however, the allocation responds one for one to innovations in the ideal position. Outside the interval, the dealer partially accommodates the initial ideal position.

Panel (b) shows that the first-order solution is not incentive compatible. The solution delivers initial volatility $\psi_0 = 1/\rho$ at every report. Yet, on $[-x^*, x^*]$, the marginal rent is constant, so $\psi_0^{\max} = 0$. In fact, local incentive constraints are violated on this region, since $\psi_0 > D' = 0$.

Outside the flat $[-x^*, x^*]$, local incentive constraints do not bind. The binding incentive constraints are global. For $x^* < |y| < \sup \Theta$, comparing y with $-y$ gives

$$\psi_0^{\max}(y) \leq \frac{D^{\text{FOA}}(y)}{y} < \frac{1}{\rho}.$$

The integral constraint is therefore violated at *every* interior type. Because $\psi_t \equiv 1/\rho$, Remark 1 applies: violations of the integral constraint correspond to profitable deviations.

6.3 The finite-variation and local programs

Because the first-order approach fails, I now turn to the two programs which show how the dealer trades off information rents against responsiveness. I find that the finite-variation solution, at least for the parameter values in Table 1, is nearly optimal.

6.3.1 Continuation values

Once the initial marginal rent and volatility are chosen, the dealer maximizes continuation surplus. In this application, it turns out that the two programs have the same continuation value, even though the local program permits a Brownian component in ψ . The quadratic structure gives a direct solution for

the continuation value which one can verify solves the variational inequality (27) for both programs.

The states follow (24)–(25), with $v = 0$ in the finite-variation program.

Under the efficient allocation $X_t = \hat{\theta}_t$, the marginal rent is $\hat{\theta}_t/\rho$ and its volatility is $1/\rho$. To measure departures from these values, define

$$\tilde{D}_t := D_t - \frac{\hat{\theta}_t}{\rho}, \quad \zeta_t := X_t - \hat{\theta}_t.$$

Equation (24) then becomes

$$d\tilde{D}_t = [(r + \lambda)\tilde{D}_t - \zeta_t]dt + \left(\psi_t - \frac{1}{\rho}\right)d\hat{B}_t.$$

Since $\hat{\theta}_t X_t - \frac{1}{2}X_t^2 = \frac{1}{2}\hat{\theta}_t^2 - \frac{1}{2}\zeta_t^2$, maximizing continuation surplus is equivalent to minimizing the tracking cost: $\mathbb{E} \int_0^\infty e^{-rt} \zeta_t^2 dt$. Applying Itô's formula to $e^{-rt} \rho \tilde{D}_t^2/2$, taking expectations, and completing the square gives

$$\begin{aligned} \frac{1}{2} \mathbb{E} \int_0^\infty e^{-rt} \zeta_t^2 dt &= \frac{\rho}{2} \tilde{D}_0^2 + \frac{1}{2} \mathbb{E} \int_0^\infty e^{-rt} (\zeta_t - \rho \tilde{D}_t)^2 dt \\ &\quad + \frac{\rho}{2} \mathbb{E} \int_0^\infty e^{-rt} \left(\psi_t - \frac{1}{\rho}\right)^2 dt. \end{aligned} \quad (34)$$

The first term is the cost of the initial promise. The middle integral vanishes when $\zeta_t = \rho \tilde{D}_t$, equivalently $X_t = \rho D_t$. The remaining problem is therefore to keep ψ_t as close as possible to its efficient value $1/\rho$. Since $e^{-\rho t} \psi_t$ must be a nonnegative supermartingale, $\mathbb{E}[\psi_t] \leq e^{\rho t} \psi_0$, and because the loss is quadratic, Jensen's inequality gives

$$\mathbb{E} \left[\left(\psi_t - \frac{1}{\rho}\right)^2 \right] \geq \left[\left(\frac{1}{\rho} - e^{\rho t} \psi_0\right)^+ \right]^2,$$

the bound being trivial once $e^{\rho t} \psi_0 \geq 1/\rho$. For $\psi_0 \leq 1/\rho$, equality holds at every date along the deterministic finite-variation path $\psi_t = \min\{e^{\rho t} \psi_0, 1/\rho\}$, which is therefore optimal even among diffusive volatilities.

Proposition 4 (Continuation value). *From every initial state $(\hat{\theta}, D, \psi) \in \mathbb{R}^2 \times [0, \infty)$, the finite-variation and local continuation problems have the same value:*

$$V(\hat{\theta}, D, \psi) = \frac{\hat{\theta}^2}{2\rho} + \frac{1}{2r\rho} - \frac{\rho}{2} \left(D - \frac{\hat{\theta}}{\rho}\right)^2 - \frac{\rho}{2} P(\psi), \quad (35)$$

where

$$P(\psi) := \int_0^\infty e^{-rs} \left[\left(\frac{1}{\rho} - e^{\rho s} \psi \right)^+ \right]^2 ds.$$

If $0 \leq \psi \leq 1/\rho$, the value is attained in both programs by

$$X_t = \rho D_t, \quad \psi_t = \min \left\{ e^{\rho t} \psi, \frac{1}{\rho} \right\}, \quad v_t = 0.$$

If $\psi > 1/\rho$, the value in (35) is not attained but is approached arbitrarily closely by lowering ψ to $1/\rho$ over a shrinking period of time.

Since P vanishes at and above $1/\rho$, raising the initial volatility past $1/\rho$ leaves continuation value unchanged and only tightens the date-zero incentive constraints. Both date-zero programs may therefore restrict $\psi_0 \leq 1/\rho$ without loss.

Under $X_t = \rho D_t$, the allocation responds to an innovation in the trader's type with coefficient $\rho \psi_t$; efficiency requires this coefficient to equal one, which explains the target $\psi_t = 1/\rho$. When $0 < \psi_0 < 1/\rho$, the optimal path raises ψ at the fastest rate incentives permit until it reaches $1/\rho$, and keeps it there. The dealer's central tradeoff is that lower initial volatility weakens the response to future shocks but relaxes the date-zero constraints on the marginal rent schedule.

6.3.2 The date-zero programs

Because the continuation values coincide, both date-zero programs take the form (28), with the common value V of Proposition 4 in place of V^P and pricing weight $\Lambda = h_0$. The dealer maximizes over measurable D and ψ_0 with $0 \leq \psi_0 \leq 1/\rho$, and over u_0 , subject to participation: $u_0 + \int_0^\theta D(l) dl \geq 0$ for every θ . The finite-variation program additionally imposes the cap (30); the local program instead imposes only the local conditions, D nondecreasing and $\psi_0 \leq D'$ almost everywhere.

For uniform types, both programs admit analytic solutions. Proposition 5 records their main properties; Online Appendix H.3 gives the complete characterizations and proofs. The numerical Gaussian solutions in Table 1 have the same qualitative features.¹⁵

¹⁵There is an interval around zero on which the marginal rent is affine and the initial volatility constant; outside this interval, only constraints comparing θ with $-\theta$ bind. In the local solution, $\psi_0 = D'$ on an inner interval, while outside it the marginal rent coincides with the first-order solution and the volatility is first best.

Proposition 5 (Uniform types). *Suppose F is uniform on $[-1, 1]$. Both date-zero programs have unique solutions up to F -null sets.*

Finite variation. *There are $s \in (0, 1/\rho)$ and $b \in (3/8, 3/4)$ such that, for $|\theta| \leq b$,*

$$D^{\text{FV}}(\theta) = s\theta, \quad \psi_0^{\text{FV}}(\theta) = s.$$

For $b < |\theta| \leq 1$, the equality

$$\psi_0^{\text{FV}}(\theta) = \frac{D^{\text{FV}}(\theta)}{\theta}$$

continues to hold, but the ratio now increases strictly with $|\theta|$ from s to $1/\rho$.

Moreover, every date-zero constraint (21) between two distinct types in $[-b, b]$ binds. Outside this interval, only date-zero constraints between types θ and $-\theta$ bind. Initial volatility is below its first-best level at every interior type and equals $1/\rho$ only at the endpoints.

Local. *There is $a \in (1/2, 1)$ such that, for $|\theta| < a$,*

$$\psi_0^{\text{loc}}(\theta) = (D^{\text{loc}})'(\theta) < \frac{1}{\rho}.$$

For $a \leq |\theta| \leq 1$,

$$D^{\text{loc}}(\theta) = D^{\text{FOA}}(\theta), \quad \psi_0^{\text{loc}}(\theta) = \frac{1}{\rho}.$$

The solution violates (30) at every $\theta \in (-1, 1) \setminus \{0\}$.

At $r = 1$ and $\lambda = 0$, $(s, b, a) = (0.5712, 0.5249, 0.7497)$; Figure 3 plots the two solutions at these parameter values. I discuss their key properties, formalized in the proposition, and offer intuition for them.

Neither program retains the first-order solution's central no-trade interval. Thus, bunching is not robust to dynamics. In contrast, under the finite-variation solution, there is a central region $[-b, b]$ where the marginal rent is affine

$$D^{\text{FV}}(\theta_0) = s\theta_0, \quad X_0^{\text{FV}} = \rho s\theta_0 \quad \text{for } |\theta_0| \leq b,$$

so every nonzero central type trades a fraction $\rho s \in (0, 1)$ of his ideal position. The allocation's response to later type innovations rises from ρs to one as volatility grows at the maximum permitted rate to $1/\rho$. In this central region, local incentive constraints bind.

Outside the central interval, this is not so. For $|\theta_0| > b$, the local condition is slack, but the integral constraint (21) comparing θ_0 with $-\theta_0$ binds. Because the

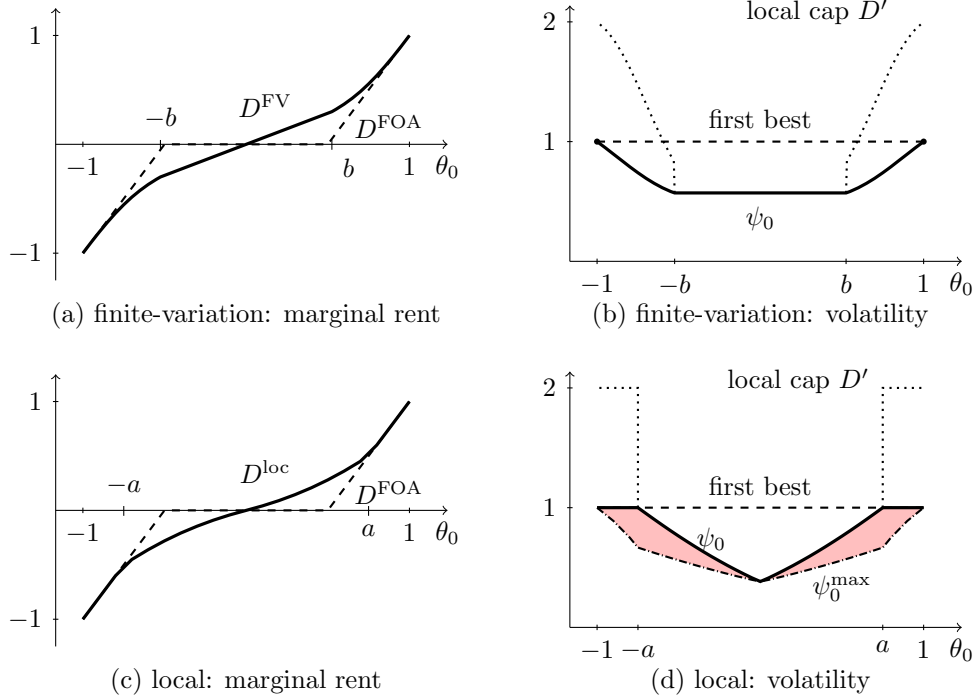


Figure 3: Date-zero solutions for uniform types at $r = 1$ and $\lambda = 0$. Solid curves are the program solutions. In the marginal rent panels, the dashed curve is the first-order marginal rent. In the volatility panels, the dashed line is first-best volatility and the dotted curve is the local cap D' . Panel (d) also shows the global cap $\psi_0^{\max}$ (dash-dotted); the shaded region is the amount by which the local solution violates that cap.

marginal rent is an odd function, this caps the initial volatility at $D^{FV}(\theta_0)/\theta_0$ —the average slope of the marginal rent between these types—rather than at the local slope. Initial volatility is therefore below first best at *every* interior type.

In contrast, the local solution is only distorted on the interval $(-a, a)$. The local solution nevertheless exceeds the cap $\psi_0^{\max}$ implied by its own marginal rent at *every* nonzero interior type as seen from the shaded area in panel (d). Its optimal continuation has deterministic, absolutely continuous volatility, so Remark 1 makes the cap necessary. The local-program solution is therefore not globally incentive compatible.

For different values of λ , Table 1 reports the gap between the finite-variation lower bound and the local upper bound for the uniform and Gaussian distributions. It also reports the values under the static benchmark and first-order solution, whose gap captures the value of responding to later reports. (The static

benchmark fixes the entire allocation path as a function of the initial report; Online Appendix H.6 shows that its optimal allocation is $X_t^{\text{stat}} = \mathbb{E}[X_t^{\text{FOA}} \mid \theta_0]$.) The uniform entries come from the analytic solutions above; the Gaussian entries are computed from discretized versions of the programs (Online Appendix H.7).

	Uniform on $[-1, 1]$		Gaussian	
	$\lambda = 0$	$\lambda = \frac{1}{2}$	$\lambda = 0$	$\lambda = \frac{1}{2}$
Static	15.3%	15.1%	32.3%	31.6%
Finite variation	99.0%	99.0%	98.2%	98.4%
Local program	100%	100%	100%	100%
First-order approach	107.3%	106.0%	110.8%	108.4%
Π_{loc}	0.5435	0.2752	0.6375	0.3258

Table 1: Revenue as a percentage of the local upper bound Π_{loc} . The final row gives its level; $r = 1$ throughout, and the Gaussian columns use the standard normal. The finite-variation value is attained by an implementable mechanism and is a lower bound on unrestricted revenue; the local value is an upper bound; the first-order row is an infeasible relaxation. Uniform entries are analytic; Gaussian entries are computed on a grid.

For uniform types, the finite-variation mechanism attains 99.0% of the local upper bound. The Gaussian computations give gaps of 1.6%–1.8%. The first-order value exceeds the local upper bound by 6–11 percentage points, so most of its apparent gain over finite variation is infeasible. By contrast, the static mechanism earns only 15%–32% of the local value. Responding to later reports is therefore quantitatively important, and the finite-variation mechanism captures nearly all of that value in these examples.

6.4 From dealer to exchange

The analysis extends to an exchange with many traders: under the conditions below, the exchange’s problem reduces to one dealer problem per trader. Consider an exchange that reallocates the asset among N traders, each as in Section 6.1, with independent trading needs. Related models with private trading needs and quadratic costs of deviations from ideal positions typically take the trading protocol as given (e.g., Du and Zhu, 2017; Malamud and Rostek,

2017; Chen and Duffie, 2021; Antill and Duffie, 2021).¹⁶ Here the exchange chooses allocation and transfer rules to maximize total revenue, subject to incentive compatibility, interim participation, and market clearing at every date: $\sum_{i=1}^N X_t^i = 0$.

Each trader's reports may depend on the histories of his own types, his past reports, and the allocations he observes, as in Pavan et al. (2014). Online Appendices H.4–H.5 give the precise definitions of admissible exchange reports and mechanisms and prove the corollary below.

Three classes of dealer mechanisms are relevant: G, the admissible, globally incentive-compatible, individually rational mechanisms; FV, the finite-variation mechanisms feasible in the finite-variation program—those satisfying the conditions of Theorems 1a and 1b and the cap (30); and L, the admissible locally incentive-compatible mechanisms of the local program.

For $\mathcal{C} \in \{\text{G}, \text{FV}, \text{L}\}$, let $\mathcal{D}_\mathcal{C}^i$ denote the admissible, individually rational dealer mechanisms for trader i in class $\mathcal{C}$, and let $\mathcal{E}_\mathcal{C}$ denote the feasible N -trader exchange mechanisms in that class, as defined in Online Appendix H.4. Define

$$\Pi_\mathcal{C}^{\text{ex}} := \sup_{M=\{(X^i, T^i)\}_{i=1}^N \in \mathcal{E}_\mathcal{C}} \sum_{i=1}^N \mathbb{E} \int_0^\infty e^{-rt} T_t^i dt.$$

Corollary 2.2 (Exchange decomposition). *Fix $\mathcal{C} \in \{\text{G}, \text{FV}, \text{L}\}$. Suppose that, for every trader i , the dealer problem*

$$\sup_{(X, T) \in \mathcal{D}_\mathcal{C}^i} \mathbb{E} \int_0^\infty e^{-rt} T_t dt$$

has a solution, and let $X_\mathcal{C}^{i}$ denote its optimal allocation. If $\mathcal{C} = \text{G}$, suppose the solutions satisfy the integrability condition (92) in Online Appendix H.5. Set*

$$\widehat{X}_\mathcal{C}^i := X_\mathcal{C}^{i*} - \frac{1}{N} \sum_{j=1}^N X_\mathcal{C}^{j*}.$$

Then there exist transfers $\{\widehat{T}_\mathcal{C}^i\}_{i=1}^N$ such that

$$\widehat{M}_\mathcal{C} := \{(\widehat{X}_\mathcal{C}^i, \widehat{T}_\mathcal{C}^i)\}_{i=1}^N \in \mathcal{E}_\mathcal{C}$$

and

$$\sum_{i=1}^N \mathbb{E} \int_0^\infty e^{-rt} \widehat{T}_{\mathcal{C},t}^i dt = \Pi_\mathcal{C}^{\text{ex}}.$$

¹⁶An exception is Chen (2024), which studies the static counterpart in which the exchange chooses the mechanism.

The proof first restricts deviations to reports adapted to a trader's own type history. A projection argument shows that the mechanisms in the corollary solve this relaxed problem. The proof then uses additive separability of the allocation in the other traders' reports to show incentive compatibility when reports also respond to observed allocations.

7 Extensions

I summarize three extensions analyzed in Online Appendices I–K.

7.1 Geometric Brownian Motion

Online Appendix I extends the analysis to the case when types follow a geometric Brownian motion, another canonical specification likely to arise in applications:

$$d\theta_t = \mu\theta_t dt + \gamma\theta_t dB_t, \quad \theta_0 \in \Theta \subseteq (0, \infty), \quad r > \mu.$$

Reports follow the same dynamics with B replaced by $\hat{B}$. In contrast with the OU case, the impulse response $I_{s,t} = \theta_s/\theta_t$ is stochastic, so the marginal information rent is

$$D_t = \mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \frac{\theta_s}{\theta_t} X_s ds \right]. \quad (36)$$

Write $\psi_t := \text{vol}(D_t)$ and $A_t := e^{-rt} \hat{\theta}_t \psi_t$. The results for the OU case generalize to this setting. The following theorem is the analog of Theorems 1a and 1b for the GBM case.

Theorem 3 (Implementability under GBM). *Suppose the mechanism satisfies the GBM admissibility conditions of Online Appendix I, and transfers satisfy $\text{vol}(W_t) = \gamma \hat{\theta}_t D_t$ and the date-zero envelope condition (16).*

- (i) *Truthful reporting is locally incentive compatible after the initial report if and only if A admits a nonnegative càdlàg supermartingale representative under truthful continuation from every initial report. If this representative has finite variation, local and global incentive compatibility after the initial report are equivalent.*
- (ii) *Suppose the supermartingale condition in (i) holds. If D is differentiable, truthful reporting is locally incentive compatible if and only if*

$$\psi_0(x) \leq \gamma x D'(x) \quad \forall x \in \Theta. \quad (37)$$

If the representative of A has finite variation after every truthful initial report, truthful reporting is globally incentive compatible whenever

$$\int_y^x \left[D(l) - D(y) - \frac{\psi_0(y)}{\gamma} \frac{l - y}{l} \right] dl \geq 0 \quad \forall x, y \in \Theta. \quad (38)$$

The integral condition is also necessary under the regularity in Remark 1, with A in place of ψ . The finite-variation restriction here is on A , so ψ itself may still have a Brownian component.

7.2 Contracting without Transfers

Online Appendix J extends Theorems 1a and 1b to contracts without transfers showing that it is relevant even outside the Myersonian setting. Call a contract *first-order* if, under truthful continuation from every initial report, the agent's continuation value has the volatility of Proposition 2(i).

Proposition 6 (Incentives without transfers). *Suppose that $T \equiv 0$. Under the maintained admissibility and regularity conditions, the conclusions of Theorem 1a apply as stated to every first-order contract. If, in addition, the contract satisfies the date-zero envelope condition in Proposition 2(ii), then the conclusions of Theorem 1b also apply under the hypotheses stated there.*

Without transfers, however, a recursive formulation generally requires continuation utility as an additional state variable, as in Williams (2011) or Bloedel et al. (2023).

7.3 Beyond Finite Variation

Online Appendix K uses the master formula in Lemma 1 to study global incentive compatibility beyond finite variation. The purpose of the section is to elucidate why global incentive compatibility is difficult to characterize in a way amenable to optimal control, explain why the finite-variation class side-steps this difficulty, and provide intuition for how to construct classes of contracts that can be easily certified as incentive compatible in applications.

Fix a truthful initial report and write $\text{vol}^0(\psi) := \psi$ and $\text{vol}^{j+1}(\psi) :=$

$\text{vol}(\text{vol}^j(\psi))$. Repeating the integration by parts in (19) gives¹⁷

$$U_x(x, a) - U(x) = \sum_{k=2}^{\infty} \frac{(-1)^{k-1}}{k! \gamma^{k-1}} \mathbb{E}_x \int_0^{\infty} e^{-rt} \delta_t^k \times \left[(r + k\lambda) \text{vol}^{k-2}(\psi_t) - \text{drift}(\text{vol}^{k-2}(\psi_t)) \right] dt \quad (39)$$

for the gain from an arbitrary deviation. A simple characterization of incentives is thus difficult to obtain because all higher order terms in δ must be controlled. Finite variation is tractable because it removes every term with $k \geq 3$ since $\text{vol}(\psi) = 0$. The formula also yields intuition for contracts, beyond the finite-variation case, for which incentive compatibility is easy to check: contracts such that successive volatilities follow a simple pattern, so that higher-order terms can be summed. A simple case is when ψ follows a geometric Brownian motion.

Proposition 7 (Proportional volatility). *Let M be an admissible mechanism with transfers satisfying Proposition 2(i). Suppose ψ admits an Itô representative satisfying, under truthful continuation from every initial report,*

$$d\psi_t = m_t dt + \alpha \psi_t dB_t, \quad \psi_t \geq 0, \quad m_t \leq (r + \lambda)\psi_t,$$

where $\alpha \neq 0$ is constant. Then truthful reporting is globally incentive compatible after the initial report.

Here $\text{vol}^j(\psi) = \alpha^j \psi$, so (39) sums to

$$U_x(x, a) - U(x) = \frac{\gamma}{\alpha^2} \mathbb{E}_x \int_0^{\infty} e^{-rt} \left[(m_t - (r + \lambda)\psi_t) h(z_t) - \lambda \psi_t g(z_t) \right] dt,$$

where $z_t := \alpha \delta_t / \gamma$, $h(z) := e^{-z} - 1 + z$, and $g(z) := 1 - (1 + z)e^{-z}$. Both functions are nonnegative. The sufficient bound is stronger than the local bound $m_t \leq (r + 2\lambda)\psi_t$ when $\lambda > 0$; they coincide when $\lambda = 0$.

8 Concluding Discussion

This paper connects Myersonian dynamic mechanism design with the martingale methods of continuous-time contracting and shows that this connection can yield new insights. The main result characterizes incentives in terms of the volatility of the agent's marginal information rent.

¹⁷This formula assumes each volatility has an Itô representation, terminal terms vanish in the integration by parts, and the remainder vanishes as the number of integrations grows.

Under the usual envelope conditions, truthful reporting after the initial report is locally incentive compatible if and only if the discounted volatility is a nonnegative supermartingale. For finite-variation mechanisms, the same condition characterizes global incentive compatibility after the initial report. The associated drift constraint makes the design problem amenable to optimal control with volatility as a state variable. I present two programs that bracket the designer’s payoff even when the first-order approach fails. I apply them to a benchmark model of financial market design, where the first-order approach fails due to countervailing incentives.

Most of the paper carries continuous-time methods into the Myersonian setting. The discrete-time Myersonian methods, however, also yield insights for continuous-time contracting. The integral-monotonicity condition can be used to conjecture, through elementary algebra, the correct conditions for local incentive compatibility. It also reveals finite-variation mechanisms as the counterpart of discrete-time “linear mechanisms.” Further, the paper’s characterization of incentives extends to settings without transfers for the first-order contracts of Section 7. When the agent can distort the drift of his reports in either direction, the supermartingale condition on marginal rent volatility completes the characterization of local incentive compatibility.

Several directions remain open. First, the local analysis appears adaptable to general utility functions. For finite-variation mechanisms, however, the resulting incentive conditions are likely to remain sufficient but may no longer be necessary: concavity in type can introduce slack. A second direction is to consider general Markov diffusions. The two extensions are closely related: when utility is multiplicatively separable in type and allocation, a redefinition of the type variable maps curvature into a change of diffusion, in which case the two extensions are isomorphic.

Finally, the exchange application shows that the results already speak to multi-agent settings in which the optimal allocation is additively separable across agents’ reports. Implementability when the volatility of each agent’s marginal rent depends on the others’ reports is open to investigation.

Appendix

A Admissible Mechanisms

This appendix defines admissible mechanisms. The admissible reports are those of Definition 2. Recall the reporting gap $\delta_t = \hat{\theta}_t - \theta_t$ and the variables W , D , and ψ from Section 5. When the predictable representative of ψ specified below is a semimartingale, let v_t denote its Brownian diffusion coefficient.

Definition 4 (Admissible mechanisms). *A mechanism $M = (X, T)$ is admissible if the following conditions hold.*

(M1) *Under truthful reporting,*

$$\int_0^\infty e^{-(r+\lambda)t} X_t dt \in L^2, \quad \mathbb{E} \int_0^\infty e^{-rt} |T_t| dt < \infty.$$

(M2) *For every $x \in \Theta$ and every admissible report (y, a) ,*

$$\mathbb{E}_x \int_0^\infty e^{-rt} (|\theta_t \hat{X}_t| + |\hat{T}_t|) dt < \infty. \quad (40)$$

Under truthful reporting, the same integrability condition also holds unconditionally.

(M3) *Under truthful continuation from every $y \in \Theta$, ψ has a predictable representative such that, for every $T < \infty$,*

$$\mathbb{E}_y \int_0^T \psi_t^2 dt < \infty, \quad \psi_t \longrightarrow \psi_0(y) \quad \text{in } L^1 \quad (t \downarrow 0),$$

where $\psi_0(y)$ is finite and deterministic. For every $x \in \Theta$ and every admissible report (y, a) ,

$$\int_0^\cdot e^{-rt} (\text{vol}(W_t) - \psi_t \delta_t) dB_t$$

is a true martingale under $\mathbb{P}_x$ on every finite horizon. Whenever ψ admits a semimartingale representative, the process

$$\int_0^\cdot e^{-rt} \delta_t^2 v_t dB_t$$

is also a true martingale under $\mathbb{P}_x$ on every finite horizon.

(M4) For every $x \in \Theta$ and every admissible report (y, a) , $W_T - D_T \delta_T$ is integrable for every $T < \infty$, and

$$\mathbb{E}_x \int_0^\infty e^{-rt} |\psi_t \delta_t a_t| dt < \infty, \quad \mathbb{E}_x [e^{-rT} (W_T - D_T \delta_T)] \rightarrow 0.$$

Whenever ψ admits a semimartingale representative, also require

$$\mathbb{E}_x \int_0^\infty e^{-rt} \delta_t^2 |v_t a_t| dt < \infty.$$

(M5) For $z \in \Theta$, let $\psi^{(z,0)}$ denote ψ when the report starts at z and has no subsequent distortion. Couple these continuations using the same Brownian innovations. For every $x \in \Theta$,

$$\psi_0(y) \rightarrow \psi_0(x) \quad \text{as } y \rightarrow x \text{ within } \Theta.$$

For every $x \in \Theta$ and $T < \infty$,

$$\mathbb{E}_x \int_0^T |\psi_t^{(y,0)} - \psi_t^{(x,0)}|^2 dt \rightarrow 0 \quad \text{as } y \rightarrow x \text{ within } \Theta.$$

Whenever ψ admits a semimartingale representative, every feasible deviation direction (h, a) at x also satisfies

$$\mathbb{E}_x \int_0^\infty e^{-rt} |\delta_t^{(\varepsilon, h, a)}|^2 |v_t^{(\varepsilon, h, a)} \varepsilon a_t| dt = o(\varepsilon^2),$$

where the superscripts on δ and v denote evaluation along the report $(x + \varepsilon h, \varepsilon a)$, and the small- o relation is as $\varepsilon \downarrow 0$.

Condition (M1) provides the square-integrability needed for the martingale representation of the marginal rent and makes expected discounted transfers finite under truthful reporting. Condition (M2) makes the agent's payoff finite under every admissible report.

Condition (M3) gives ψ a well-defined initial value and ensures that the stochastic integrals in the incentive calculations have zero expectation. Condition (M4) makes the deviation terms integrable and the discounted terminal continuation payoff vanish in expectation, justifying the infinite-horizon limit in Lemma 1.

Condition (M5) provides continuity in the initial report and makes the term involving v in (19) negligible at quadratic order for small deviations. If ψ has finite variation, then $v = 0$, so all conditions involving v hold automatically.

B Proof of Lemma 1

In the proofs below, write $g_t := \text{vol}(W_t)$ and $\sigma_t := e^{-(r+\lambda)t}\psi_t$. Thus σ is discounted marginal rent volatility, and $e^{-\lambda t}\sigma_t = e^{-\rho t}\psi_t$.

The law of motion of δ is $d\delta_t = (-\lambda\delta_t + \gamma a_t) dt$. If the agent stops distorting at t , δ decays at rate λ : $\delta_s = e^{-\lambda(s-t)}\delta_t$ for $s \geq t$.

Therefore,

$$\mathbb{E}_t \left[\int_t^\infty e^{-r(s-t)} \delta_s \hat{X}_s ds \right] = D_t \delta_t.$$

The total expected discounted payoff, at time t , from following an arbitrary strategy until time t and switching to no distortion $a_s = 0$ for $s \geq t$ is therefore

$$G_t := \int_0^t e^{-rs} (\theta_s \hat{X}_s - \hat{T}_s) ds + e^{-rt} (W_t - D_t \delta_t).$$

Note that $G_0 = W_0(\hat{\theta}_0) - D(\hat{\theta}_0)\delta_0$, while $\mathbb{E}[G_t] \rightarrow U(\hat{\theta}_0, a)$ by (M2) and (M4).

By Itô's formula,

$$\begin{aligned} dG_t &= e^{-rt} (\theta_t X_t - T_t) dt + d(e^{-rt} W_t) - d(e^{-rt} D_t \delta_t) \\ &= e^{-rt} (\theta_t X_t - T_t) dt - e^{-rt} (\hat{\theta}_t X_t - T_t) dt + e^{-rt} g_t d\hat{B}_t \\ &\quad - \delta_t \left[(-e^{-rt} X_t + \lambda e^{-rt} D_t) dt + e^{\lambda t} \sigma_t d\hat{B}_t \right] - e^{-rt} D_t (-\lambda \delta_t + \gamma a_t) dt. \end{aligned}$$

Canceling terms using $\delta_t = \hat{\theta}_t - \theta_t$ yields

$$\begin{aligned} dG_t &= -e^{-rt} \gamma D_t a_t dt + (e^{-rt} g_t - e^{\lambda t} \sigma_t \delta_t) d\hat{B}_t \\ &= [e^{-rt} (g_t - \gamma D_t) a_t - e^{\lambda t} \sigma_t \delta_t a_t] dt + (e^{-rt} g_t - e^{\lambda t} \sigma_t \delta_t) dB_t. \end{aligned} \quad (41)$$

Proposition 2(i) gives $g_t = \gamma D_t$, so only the second drift term remains and the diffusion term is a $\mathbb{P}$ -martingale by (M3). Taking expectations yields

$$U(\hat{\theta}_0, a) - G_0 = -\mathbb{E} \int_0^\infty e^{\lambda t} \sigma_t \delta_t a_t dt$$

where recall $G_0 = W_0(\hat{\theta}_0) - D(\hat{\theta}_0)\delta_0$, completing the proof.

C Proof of Proposition 2

Fix the allocation X , and write $g_t := \text{vol}(W_t)$ and $g_t^* := \gamma D_t$.

Necessity of (i). By the master deviation formula in Lemma 1, if $g_t \neq \gamma D_t$ on a set of positive measure with positive probability, a small drift distortion of the appropriate sign and produce a first-order gain. I formalize this below.

Fix x and define $C_t = \int_0^t e^{-ru}(g_u - \gamma D_u) du$. As verified below, C_t is integrable for every t . For any $0 \leq s < t$, choose

$$b = \text{sgn}(\mathbb{E}_x[C_t - C_s \mid \mathcal{F}_s]), \quad a_u = b \mathbf{1}_{(s,t]}(u).$$

The agent raises the drift when its conditional expected first-order effect is positive and lowers it when that effect is negative. This direction is chosen at s and held fixed through t . The payoff calculation below gives

$$U_x(x, \varepsilon a) - U(x) = \varepsilon \mathbb{E}_x[b(C_t - C_s)] + o(\varepsilon) = \varepsilon \mathbb{E}_x[\mathbb{E}_x[C_t - C_s \mid \mathcal{F}_s]] + o(\varepsilon).$$

If the coefficient of ε were positive, this would be a profitable local deviation. Hence $\mathbb{E}_x[C_t - C_s \mid \mathcal{F}_s] = 0$ for every $s < t$, making C a martingale. Because C is a time integral, its paths are continuous and have finite variation. Such a martingale must be constant. Since $C_0 = 0$, we obtain $g = \gamma D$, $dt \otimes d\mathbb{P}_x$ -almost everywhere. This holds for every x , proving (i).

Justifying the first-order payoff calculation. For any $\mathcal{F}_s$ -measurable b with $|b| \leq 1$, set $a_u = b \mathbf{1}_{(s,t]}(u)$ and $Q = b(C_t - C_s)$. Superscripts ε denote evaluation along $(x, \varepsilon a)$. From the calculation in the proof of Lemma 1, retaining the term involving $g - \gamma D$, we have

$$U_x(x, \varepsilon a) - U(x) = \varepsilon \mathbb{E}_x Q^\varepsilon - \varepsilon \mathbb{E}_x \int_s^t e^{-ru} \psi_u^\varepsilon \delta_u^\varepsilon a_u du. \quad (42)$$

Integrate (41) along the report $(x, \varepsilon a)$ up to a horizon $T \geq t$. Taking expectations and letting $T \rightarrow \infty$, as in the proof of Lemma 1, gives (42).

We next show that $\mathbb{E}_x Q^\varepsilon \rightarrow \mathbb{E}_x Q$. Because b is chosen before the distortion begins, the density of the distorted report relative to truthful reporting is

$$\xi_\varepsilon = \exp\left\{\varepsilon b(B_t - B_s) - \frac{1}{2}\varepsilon^2 b^2(t - s)\right\}.$$

Integrability under the comparison reports (x, a) and $(x, -a)$ gives $\mathbb{E}_x[|Q|(\xi_1 + \xi_{-1})] < \infty$. For $|\varepsilon| \leq 1$, $\xi_\varepsilon \leq e^{(t-s)/2}(\xi_1 + \xi_{-1})$. Dominated convergence therefore gives $\mathbb{E}_x Q^\varepsilon = \mathbb{E}_x[\xi_\varepsilon Q] \rightarrow \mathbb{E}_x Q$. This also proves $Q \in L^1$; taking $b = 1$ and $s = 0$ establishes the integrability of C_t used above.

Finally, $|\delta_u^\varepsilon| \leq \gamma(t - s)|\varepsilon|$ on $[s, t]$. Condition (M3), Cauchy–Schwarz, and $\mathbb{E}_x \xi_\varepsilon^2 \leq e^{t-s}$ give a uniform bound on $\mathbb{E}_x \int_s^t |\psi_u^\varepsilon| du$ for $|\varepsilon| \leq 1$. The last term in (42) is therefore $O(\varepsilon^2)$, completing the first-order calculation.

Necessity of (ii). Fix a true initial type x and an initial report y , and set $\delta_0 := y - x$. By Lemma 1,

$$U_x(y, 0) = U(y) + (x - y)D(y). \quad (43)$$

Here $U_x(y, 0)$ makes the dependence on the true type explicit.

Incentive compatibility at $t = 0$ and (43) imply, for every $x \in \Theta$,

$$U(x) = \sup_{y \in \Theta} \{U(y) + (x - y)D(y)\}.$$

Hence U is convex and $D(y) \in \partial U(y)$. On every compact subinterval of Θ , U is absolutely continuous and $U' = D$ almost everywhere. Therefore (16) holds.

Using the definition of W_0^X with (16) yields (17).

Constructing transfers. Let

$$H_t := g_t^* - \text{vol}(W_t^X), \quad \kappa_0 := W_0^X(\hat{\theta}_0) - \int_c^{\hat{\theta}_0} D(l)dl - U_c,$$

and define

$$M_t := - \int_0^t H_s d\hat{B}_s, \quad T_t := r(\kappa_0 + M_t).$$

By the conditions in the proposition, M is a martingale on every finite horizon under truthful reporting and all conditional expectations below are well defined.

Because $\mathbb{E}[M_t \mid \theta_0] = 0$, the tower property and Fubini give

$$\mathbb{E} \left[\int_0^\infty e^{-rt} T_t dt \mid \theta_0 \right] = \kappa_0.$$

Along the truthful path, $\hat{\theta}_0 = \theta_0$, and hence

$$U(\theta_0) = W_0^X(\theta_0) - \kappa_0 = U_c + \int_c^{\theta_0} D(l)dl.$$

This verifies (ii).

For $s \geq t$, the martingale property gives $\mathbb{E}_t[T_s] = r(\kappa_0 + M_t)$. Consequently,

$$\mathbb{E}_t \int_t^\infty e^{-r(s-t)} T_s ds = \kappa_0 + M_t,$$

and therefore $W_t = W_t^X - \kappa_0 - M_t = W_t^X - \kappa_0 + \int_0^t H_s d\hat{B}_s$. It follows that $\text{vol}(W_t) = \text{vol}(W_t^X) + H_t = g_t^*$, which is (i).

D Proof of Theorem 1a

As in Appendix B, write $\sigma_t := e^{-(r+\lambda)t}\psi_t$.

Sufficiency Using Lemma 1, I first prove the sufficiency direction of Theorem 1a. I use the following lemma, which applies integration by parts to the formula in Lemma 1.

Lemma 2 (Supermartingale master deviation formula). *Suppose that $e^{-\lambda t}\sigma_t$ is a nonnegative supermartingale under truthful reporting, and write its Doob–Meyer decomposition as $d(e^{-\lambda t}\sigma_t) = -dK_t + \text{vol}(e^{-\lambda t}\sigma_t) d\hat{B}_t$, where K is predictable and increasing and $K_0 = 0$. If transfers satisfy Proposition 2(i) and $(\hat{\theta}_0, a)$ is admissible, then*

$$U(\hat{\theta}_0, a) - W_0(\hat{\theta}_0) + D(\hat{\theta}_0) \delta_0 = \frac{1}{2\gamma} \sigma_0(\hat{\theta}_0) \delta_0^2 - \frac{1}{2\gamma} \mathbb{E} \int_0^\infty e^{2\lambda t} \delta_t^2 dK_t - \frac{L}{2\gamma} + \frac{1}{2\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} \delta_t^2 \text{vol}(\sigma_t) a_t dt \quad (44)$$

where $L := \lim_{T \rightarrow \infty} \mathbb{E}[e^{\lambda T} \sigma_T \delta_T^2] \geq 0$ exists and $\mathbb{E} \int_0^\infty e^{2\lambda t} \delta_t^2 dK_t < \infty$.

Proof. Throughout, let $S_t := e^{-\lambda t}\sigma_t$. We have $d\delta_t = (-\lambda\delta_t + \gamma a_t) dt$, so

$$\gamma \delta_t a_t dt = \frac{1}{2} d(\delta_t^2) + \lambda \delta_t^2 dt \quad \Rightarrow \quad e^{\lambda t} \sigma_t \gamma \delta_t a_t dt = \frac{1}{2} S_t d(e^{2\lambda t} \delta_t^2). \quad (45)$$

For each fixed initial report $\hat{\theta}_0$, $\sigma_0(\hat{\theta}_0) = S_0$ is deterministic.

By (M4), $\mathbb{E} \int_0^\infty e^{-rt} |\psi_t \delta_t a_t| dt < \infty$, so the integral in the master deviation formula (18) is absolutely convergent. On $[0, T]$, substitute (45) and integrate by parts:

$$\int_0^T S_t d(e^{2\lambda t} \delta_t^2) = e^{2\lambda T} S_T \delta_T^2 - S_0 \delta_0^2 - \int_0^T e^{2\lambda t} \delta_t^2 dS_t.$$

Substitute $dS = -dK + \text{vol}(S) d\hat{B}$ and $d\hat{B} = a dt + dB$, and take expectations. The dB -integral has zero expectation by (M3). Using $e^{2\lambda T} S_T = e^{\lambda T} \sigma_T$ gives, for every $T < \infty$,

$$U(\hat{\theta}_0, a) - W_0(\hat{\theta}_0) + D(\hat{\theta}_0) \delta_0 = \frac{1}{2\gamma} \sigma_0(\hat{\theta}_0) \delta_0^2 - \frac{1}{2\gamma} \mathbb{E}[e^{\lambda T} \sigma_T \delta_T^2] - \frac{1}{2\gamma} \mathbb{E} \int_0^T e^{2\lambda t} \delta_t^2 dK_t + \frac{1}{2\gamma} \mathbb{E} \int_0^T e^{\lambda t} \delta_t^2 \text{vol}(\sigma_t) a_t dt - \mathbb{E} \int_T^\infty e^{\lambda t} \sigma_t \delta_t a_t dt. \quad (46)$$

By (M4),

$$\begin{aligned}\mathbb{E} \int_T^\infty e^{-rt} \psi_t \delta_t a_t dt &\longrightarrow 0, \\ \mathbb{E} \int_0^T e^{-rt} \delta_t^2 v_t a_t dt &\longrightarrow \mathbb{E} \int_0^\infty e^{-rt} \delta_t^2 v_t a_t dt.\end{aligned}$$

The two nonnegative quantities $\mathbb{E}[e^{\lambda T} \sigma_T \delta_T^2]$ and $\mathbb{E} \int_0^T e^{2\lambda t} \delta_t^2 dK_t$ must therefore have a finite sum in the limit. The second is increasing in T , so both converge separately. Denoting the first limit by $L \geq 0$ gives (44). $\square$

If the initial report is truthful, $\delta_0 = 0$ and if σ is of finite variation, $\text{vol}(\sigma_t) = 0$. It follows from Lemma 2 that, in this case, a deviation incurs a loss since $dK \geq 0$, proving global incentive compatibility after $t = 0$.

I now show local incentive compatibility after $t = 0$ even when σ may not be of finite variation. Fix a feasible direction a and consider the distortion εa from a truthful initial report. Lemma 2 gives

$$\frac{U(\theta_0, \varepsilon a) - U(\theta_0, 0)}{\varepsilon^2} \leq \frac{\frac{1}{2\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} (\delta_t^{(\varepsilon, a)})^2 |\text{vol}(\sigma_t^{(\varepsilon, a)}) \varepsilon a_t| dt}{\varepsilon^2}.$$

The right-hand side vanishes as $\varepsilon \rightarrow 0$ by (M5).

Necessity. The argument uses two simple deviations. The first deviation pins down the supermartingale property. The deviation creates a small gap between the report and the true type in the interval $[s, s + \eta]$ using a positive drift distortion. In the interval $[s + \eta, t]$ there are no further drift distortions. Finally, in the interval $[t, t + \eta]$ a negative drift distortion is used to close the gap. The master formula shows that the deviation's second-order payoff is governed by the difference between $e^{-\lambda t} \sigma_t$ at the two dates. If this process could rise in conditional expectation, the deviation is profitable. Local incentive compatibility therefore forces $e^{-\lambda t} \sigma_t$ to be a supermartingale.

A second deviation pins down the nonnegativity of the volatility, by creating a small gap but, unlike the first deviation, does not correct it with subsequent drift distortions.

(M3) and the Lebesgue differentiation theorem imply that, outside a set of dates of Lebesgue measure zero,

$$\frac{2}{\eta^2} \int_\tau^{\tau+\eta} (u - \tau) e^{-\lambda u} \sigma_u du \longrightarrow e^{-\lambda \tau} \sigma_\tau, \quad (47)$$

$$\frac{2}{\eta^2} \int_\tau^{\tau+\eta} (\tau + \eta - u) e^{-\lambda u} \sigma_u du \longrightarrow e^{-\lambda \tau} \sigma_\tau \quad (48)$$

in $L^2(\Omega)$ as $\eta \downarrow 0$. These are simply weighted averages of $e^{-\lambda t} \sigma_t$ over an interval that shrinks to τ .

The weights in both integrals are nonnegative, integrate to one, and are bounded by $2/\eta$. Hence, writing $f_u = e^{-\lambda u} \sigma_u$, Jensen's inequality gives

$$\mathbb{E} \left[\left| \frac{2}{\eta^2} \int_{\tau}^{\tau+\eta} (u - \tau) f_u \, du - f_{\tau} \right|^2 \right] \leq \frac{2}{\eta} \int_{\tau}^{\tau+\eta} \mathbb{E} [|f_u - f_{\tau}|^2] \, du.$$

The right side converges to zero for almost every τ by the $L^2(\Omega)$ -valued Lebesgue differentiation theorem. The same argument applies to the weight $2(\tau + \eta - u)/\eta^2$.

Fix $s < t$ among these dates, an event $E \in \mathcal{F}_s$, and $0 < \eta < (t - s)/2$. Consider the reporting distortion

$$a_u^{s,t,\eta} := \frac{e^{-\lambda u}}{\gamma \eta} \mathbf{1}_E \left(\mathbf{1}_{(s, s+\eta]}(u) - \mathbf{1}_{(t, t+\eta]}(u) \right). \quad (49)$$

The distortion is zero before s , so the report is truthful through that date. For each fixed η , this distortion is bounded and has bounded time support. It is therefore admissible and (M5) applies to it.

Under the distortion $\varepsilon a^{s,t,\eta}$,

$$e^{\lambda u} \delta_u^{(\varepsilon)} = \varepsilon \mathbf{1}_E \begin{cases} 0, & u \leq s, \\ (u - s)/\eta, & s < u \leq s + \eta, \\ 1, & s + \eta < u \leq t, \\ (t + \eta - u)/\eta, & t < u \leq t + \eta, \\ 0, & u > t + \eta. \end{cases}$$

Moreover,

$$e^{\lambda u} \sigma_u^{(\varepsilon)} \delta_u^{(\varepsilon)} \varepsilon a_u^{s,t,\eta} \, du = \frac{1}{2\gamma} e^{-\lambda u} \sigma_u^{(\varepsilon)} \, d \left[(e^{\lambda u} \delta_u^{(\varepsilon)})^2 \right].$$

The master formula therefore gives

$$\begin{aligned} & \frac{U(\theta_0, \varepsilon a^{s,t,\eta}) - U(\theta_0, 0)}{\varepsilon^2} \\ &= -\frac{1}{\gamma \eta^2} \mathbb{E} \left[\mathbf{1}_E \left(\int_s^{s+\eta} (u - s) e^{-\lambda u} \sigma_u^{(\varepsilon)} \, du - \int_t^{t+\eta} (t + \eta - u) e^{-\lambda u} \sigma_u^{(\varepsilon)} \, du \right) \right]. \end{aligned} \quad (50)$$

For fixed η , (M3) and Cauchy–Schwarz imply that the two weighted integrals in (50), viewed as functionals of the reported path, satisfy the L^2 condition in Lemma 4. Applying that lemma and taking $\varepsilon \downarrow 0$ yields

$$\begin{aligned} Q_\eta &:= \lim_{\varepsilon \downarrow 0} \frac{U(\theta_0, \varepsilon a^{s,t,\eta}) - U(\theta_0, 0)}{\varepsilon^2} \\ &= -\frac{1}{\gamma\eta^2} \mathbb{E} \left[\mathbf{1}_E \left(\int_s^{s+\eta} (u-s)e^{-\lambda u} \sigma_u \, du - \int_t^{t+\eta} (t+\eta-u)e^{-\lambda u} \sigma_u \, du \right) \right]. \end{aligned}$$

Local incentive compatibility gives $Q_\eta \leq 0$. Letting $\eta \downarrow 0$ and applying (47)–(48) yields

$$\mathbb{E}[\mathbf{1}_E (e^{-\lambda t} \sigma_t - e^{-\lambda s} \sigma_s)] \leq 0. \quad (51)$$

It remains to establish nonnegativity of σ . Fix a date s for which (47) holds and let:

$$b_u^{s,\eta} := \frac{e^{-\lambda u}}{\gamma\eta} \mathbf{1}_E \mathbf{1}_{(s,s+\eta]}(u). \quad (52)$$

Under the scaled distortion $\varepsilon b^{s,\eta}$, $e^{\lambda u} \delta_u^{(\varepsilon)}$ increases linearly from zero to $\varepsilon \mathbf{1}_E$ on $(s, s+\eta]$ and remains at that level thereafter. An analogous calculation as before gives

$$\lim_{\varepsilon \downarrow 0} \frac{U(\theta_0, \varepsilon b^{s,\eta}) - U(\theta_0, 0)}{\varepsilon^2} = -\frac{1}{2\gamma} \mathbb{E} \left[\mathbf{1}_E \frac{2}{\eta^2} \int_s^{s+\eta} (u-s)e^{-\lambda u} \sigma_u \, du \right] \leq 0.$$

Letting $\eta \downarrow 0$ gives

$$\mathbb{E}[\mathbf{1}_E e^{-\lambda s} \sigma_s] \geq 0.$$

Taking

$$E = \{e^{-\lambda s} \sigma_s \leq -c\}, \quad c > 0,$$

shows that $e^{-\lambda s} \sigma_s \geq 0$ almost surely.

We have established nonnegativity and (51) on a set of dates of full Lebesgue measure. Lemma 5 therefore supplies a càdlàg representative $\bar{\sigma}$ such that $e^{-\lambda t} \bar{\sigma}_t$ is a nonnegative supermartingale. The L^1 limit in (M3) supplies its value at $t = 0$.

Finally, apply the local Doob–Meyer decomposition:

$$e^{-\lambda t} \bar{\sigma}_t = \bar{\sigma}_0 + M_t - K_t,$$

where M is a local martingale and K is predictable and nondecreasing. Every local martingale in the initially enlarged Brownian filtration is continuous. Therefore

$$\Delta \bar{\sigma}_t = -e^{\lambda t} \Delta K_t \leq 0,$$

so $\bar{\sigma}$ has no upward jumps, and the continuous finite-variation part of $e^{-\lambda t}\bar{\sigma}_t$ is nonincreasing. Whenever that part is absolutely continuous,

$$\text{drift}(\bar{\sigma}_t) - \lambda \bar{\sigma}_t = -e^{\lambda t} \frac{dK_t^{\text{ac}}}{dt} \leq 0 \quad dt \otimes d\mathbb{P}\text{-a.e.}$$

Equivalently, $\bar{\psi}_t := e^{(r+\lambda)t}\bar{\sigma}_t$ is a representative of ψ , with $e^{-\rho t}\bar{\psi}_t$ a nonnegative supermartingale and, when its finite-variation part is absolutely continuous, $\text{drift}(\bar{\psi}_t) \leq \rho \bar{\psi}_t$. These are the conditions in Theorem 1a.

E Proof of Theorem 1b

As in Appendix B, write $\sigma_t := e^{-(r+\lambda)t}\psi_t$. I begin with the characterization of local IC before turning to global IC in the finite-variation case.

E.1 Local IC at date zero

Fix $\theta_0 \in \Theta$. Consider an ϵ perturbation from truthtelling in a given feasible direction (h, a) . Write $y_\epsilon := \theta_0 + \epsilon h$ for the initial report. By (18), the gain in payoff when the report starts at y_ϵ and $a \equiv 0$ thereafter is

$$\begin{aligned} & U(y_\epsilon) + (\theta_0 - y_\epsilon)D(y_\epsilon) - U(\theta_0) \\ &= - \int_{y_\epsilon}^{\theta_0} [D(l) - D(y_\epsilon)] dl = -\frac{\epsilon^2 h^2}{2} D'(\theta_0) + o(\epsilon^2). \end{aligned} \quad (53)$$

The last equality uses differentiability of D at θ_0 .

Sufficiency. Assume (20), and fix an arbitrary direction (h, a) , and let

$$\Delta_t^{h,a} := \frac{\delta_t^{(\epsilon, h, a)}}{\epsilon}.$$

Apply Lemma 2 from the initial report y_ϵ which gives

$$\begin{aligned} \frac{U(y_\epsilon, \epsilon a) - U(\theta_0)}{\epsilon^2} &\leq \frac{U(y_\epsilon) + (\theta_0 - y_\epsilon)D(y_\epsilon) - U(\theta_0)}{\epsilon^2} + \frac{h^2}{2\gamma} \sigma_0(y_\epsilon) \\ &\quad + \frac{\epsilon}{2\gamma} \mathbb{E} \int_0^\infty e^{\lambda t} (\Delta_t^{h,a})^2 \left| \text{vol}(\sigma_t^{(\epsilon, h, a)}) a_t \right| dt. \end{aligned}$$

By (53) and (M5), the last term converges to zero and $\sigma_0(y_\epsilon) \rightarrow \sigma_0(\theta_0)$ so

$$\limsup_{\epsilon \downarrow 0} \frac{U(y_\epsilon, \epsilon a) - U(\theta_0)}{\epsilon^2} \leq \frac{h^2}{2} \left(\frac{\sigma_0(\theta_0)}{\gamma} - D'(\theta_0) \right) \leq 0.$$

Necessity. Assume local incentive compatibility, and fix a nonzero feasible initial direction h . For each $\tau > 0$, consider

$$a_t^\tau := -\frac{he^{-\lambda t}}{\gamma\tau} \mathbf{1}_{(0,\tau]}(t).$$

This direction is bounded and compactly supported. On $[0, \tau]$, we have

$$e^{\lambda t} \frac{\delta_t^{(\varepsilon, h, a^\tau)}}{\varepsilon} = h \left(1 - \frac{t}{\tau}\right);$$

after τ , the gap is zero. Substitution in the master formula (18) gives

$$\begin{aligned} \frac{U(y_\varepsilon, \varepsilon a^\tau) - U(\theta_0)}{\varepsilon^2} &= \frac{U(y_\varepsilon) + (\theta_0 - y_\varepsilon)D(y_\varepsilon) - U(\theta_0)}{\varepsilon^2} \\ &\quad + \frac{h^2}{\gamma\tau} \mathbb{E} \int_0^\tau S_t^{(\varepsilon, h, a^\tau)} \left(1 - \frac{t}{\tau}\right) dt \end{aligned} \quad (54)$$

where, recall that $S_t = e^{-\lambda t} \sigma_t$.

Lemma 6 and (53) show that the limit as $\varepsilon \downarrow 0$ exists and equals

$$-\frac{h^2}{2} D'(\theta_0) + \frac{h^2}{\gamma\tau} \mathbb{E} \int_0^\tau S_t^{\theta_0} \left(1 - \frac{t}{\tau}\right) dt.$$

By (M3), $S_t^{\theta_0} \rightarrow S_0^{\theta_0}$ in L^1 as $t \downarrow 0$. Local incentive compatibility, followed by $\tau \downarrow 0$, therefore implies

$$0 \geq \frac{h^2}{2} \left(\frac{\sigma_0(\theta_0)}{\gamma} - D'(\theta_0) \right).$$

This is the condition in (20).

E.2 Global IC at date zero in the finite-variation case

For this part of the proof, retain the envelope and truthful-continuation conditions of Theorem 1b, but do not assume that D is differentiable. Suppose that σ is of finite variation and that (21) holds. Then $\text{vol}(\sigma) = 0$, so the feedback term in Lemma 2 vanishes for every admissible deviation. As above, local incentive compatibility after each truthful initial report implies, by Theorem 1a, the nonnegative supermartingale condition required by that lemma.

Applying Lemma 2 with $W_0(\hat{\theta}_0) = U(\hat{\theta}_0)$ and an arbitrary initial gap gives

$$U(\hat{\theta}_0, a) \leq U(\hat{\theta}_0) + (\theta_0 - \hat{\theta}_0) D(\hat{\theta}_0) + \frac{1}{2\gamma} \sigma_0(\hat{\theta}_0) (\theta_0 - \hat{\theta}_0)^2. \quad (55)$$

This bound holds uniformly over continuation strategies. Using $U(\theta_0) - U(\hat{\theta}_0) = \int_{\hat{\theta}_0}^{\theta_0} D(l)dl$, $\frac{1}{2}(\theta_0 - \hat{\theta}_0)^2 = \int_{\hat{\theta}_0}^{\theta_0} (l - \hat{\theta}_0)dl$, and (21),

$$U(\theta_0) - U(\hat{\theta}_0, a) \geq \int_{\hat{\theta}_0}^{\theta_0} \left[D(l) - D(\hat{\theta}_0) - \frac{\sigma_0(\hat{\theta}_0)}{\gamma}(l - \hat{\theta}_0) \right] dl \geq 0.$$

F Verification Conditions

Fix P , V , and M^* as in Theorem 2. For each mechanism, D and ψ are its marginal rent and volatility, defined in (13)–(14). Expectations are under truthful reporting, conditional on the specified initial state.

At $\psi = 0$, apply $\mathcal{L}^{x,0}$ to the restriction $(\hat{\theta}, D) \mapsto V(\hat{\theta}, D, 0)$; no derivative with respect to ψ is needed. The boundary condition is

$$rV(\hat{\theta}, D, 0) - \sup_{x \in \mathcal{X}} \left\{ \hat{\theta}x - C(x) + (\mathcal{L}^{x,0}V)(\hat{\theta}, D, 0) \right\} \geq 0 \quad \forall (\hat{\theta}, D) \in \mathbb{R}^2. \quad (56)$$

The continuation requirements in Theorem 2(ii) have the following meaning. The Bellman equality is

$$rV(\hat{\theta}_t, D_t, \psi_{t-}) = \hat{\theta}_t X_t - C(X_t) + (\mathcal{L}^{X_t, v_t}V)(\hat{\theta}_t, D_t, \psi_{t-}), \quad (57)$$

$dt \otimes d\mathbb{P}$ -almost everywhere, using the boundary convention (56) when $\psi_{t-} = 0$. Reductions through K leave V unchanged if, with probability one,

$$\begin{aligned} \int_{\{0 < t \leq T: \psi_{t-} > 0\}} V_\psi(\hat{\theta}_t, D_t, \psi_{t-}) dK_t^c &= 0 \quad \forall T < \infty, \\ V(\hat{\theta}_t, D_t, \psi_t) &= V(\hat{\theta}_t, D_t, \psi_{t-}) \quad \forall t \text{ with } \Delta K_t > 0. \end{aligned} \quad (58)$$

Condition 1 (Verification conditions). Regularity. V is continuous on $\mathbb{R}^2 \times [0, \infty)$ and has a C^2 restriction to $\psi = 0$. On $\psi > 0$, it is C^2 in $(\hat{\theta}, D)$ and C^1 in ψ if $P = \text{FV}$, and C^2 in $(\hat{\theta}, D, \psi)$ if $P = \text{loc}$. Terms involving v are omitted in the finite-variation program.

Upper bound. Consider any continuation of any admissible mechanism satisfying (24)–(25) and the state and control restrictions for P stated under “States and controls” in Section 5.3. Along every such continuation, from every initial state, $V(\hat{\theta}_T, D_T, \psi_T)$ is integrable for every $T < \infty$. The process

$$M_t^V := \int_0^t e^{-rs} (\gamma V_{\hat{\theta}} + \psi_{s-} V_D + v_s V_\psi)(\hat{\theta}_s, D_s, \psi_{s-}) d\hat{B}_s \quad (59)$$

is a true martingale on every finite horizon. On $\{\psi_{s-} = 0\}$, the integrand is $e^{-rs}\gamma V_{\hat{\theta}}(\hat{\theta}_s, D_s, 0)$. The limit

$$\lim_{T \rightarrow \infty} \mathbb{E} \left[e^{-rT} V(\hat{\theta}_T, D_T, \psi_T) \right] \quad (60)$$

exists and is nonnegative.

Additional requirements for M^* . Discounted surplus is absolutely integrable, both conditional on each initial type and after averaging over types:

$$\begin{aligned} \mathbb{E}_{\theta} \int_0^{\infty} e^{-rt} (|\hat{\theta}_t X_t| + C(X_t)) dt &< \infty \quad (\theta \in \Theta), \\ \int_{\Theta} \mathbb{E}_z \left[\int_0^{\infty} e^{-rt} (|\hat{\theta}_t X_t| + C(X_t)) dt \right] dF(z) &< \infty. \end{aligned} \quad (61)$$

For every initial type, the limit in (60) equals zero.

Online Appendix

G Supplemental Material for Section 5

This appendix provides supplemental results and proofs for Section 5.

G.1 Monotonicity of the Initial Marginal Rent

Lemma 3 (Monotone marginal rents). *In any admissible incentive-compatible mechanism, U is convex and D is nondecreasing on Θ , with $U' = D$ almost everywhere on $\text{int } \Theta$.*

Proof. The admissible report $(y, 0)$ gives

$$U_x(y, 0) = U(y) + (x - y)D(y),$$

because the OU reporting gap is $e^{-\lambda t}(y - x)$. Incentive compatibility in both directions gives

$$(x - y)D(y) \leq U(x) - U(y) \leq (x - y)D(x) \quad \forall x, y \in \Theta.$$

Thus D is nondecreasing. Moreover, $U(x) = \sup_{y \in \Theta} \{U(y) + (x - y)D(y)\}$, so U is convex and $D(y) \in \partial U(y)$. Hence $U' = D$ almost everywhere on $\text{int } \Theta$. $\square$

G.2 Technical Lemmas for Theorem 1a

The first lemma justifies taking limits under small report distortions. The second constructs the supermartingale representative used in Theorem 1a.

Lemma 4 (Continuity under small report distortions). *Fix an initial type x . Let a be bounded with support in $[0, T_0]$, and let $\hat{\theta}^{(\varepsilon)}$ be the report $(x, \varepsilon a)$. For every measurable path functional G with $\mathbb{E}_x[G(\theta)^2] < \infty$,*

$$\mathbb{E}_x[G(\hat{\theta}^{(\varepsilon)})] \longrightarrow \mathbb{E}_x[G(\theta)] \quad (\varepsilon \downarrow 0).$$

Proof. Write $\mathbb{P}_x$ for the conditional law at x . Boundedness of a permits the change of measure

$$\frac{d\mathbb{Q}_\varepsilon}{d\mathbb{P}_x} = L_\varepsilon := \exp \left\{ -\varepsilon \int_0^{T_0} a_t dB_t - \frac{\varepsilon^2}{2} \int_0^{T_0} a_t^2 dt \right\}.$$

Under $\mathbb{Q}_\varepsilon$, the report has the truthful law from x . The density ξ_ε of its law under $\mathbb{P}_x$, relative to the truthful law, is therefore the conditional expectation of L_ε^{-1} given the report under $\mathbb{Q}_\varepsilon$. Conditional Jensen gives

$$\mathbb{E}[\xi_\varepsilon] = 1, \quad \mathbb{E}[\xi_\varepsilon^2] \leq \mathbb{E}_{\mathbb{Q}_\varepsilon}[L_\varepsilon^{-2}] = \mathbb{E}_x[L_\varepsilon^{-1}] \leq e^{\varepsilon^2 \|a\|_\infty^2 T_0},$$

where expectations of ξ_ε are under the truthful path law. The last bound follows by factoring out the exponential martingale with integrand εa . Thus $\|\xi_\varepsilon - 1\|_{L^2} \rightarrow 0$, and

$$\left| \mathbb{E}_x[G(\hat{\theta}^{(\varepsilon)})] - \mathbb{E}_x[G(\theta)] \right| \leq \|G\|_{L^2} \|\xi_\varepsilon - 1\|_{L^2} \rightarrow 0.$$

□

The next lemma extends Doob's regularization argument to inequalities that hold only on a deterministic set of full Lebesgue measure.

Lemma 5 (Supermartingale regularization). *Let R be a predictable process. Suppose there is a set $\mathcal{L} \subset (0, \infty)$ of full Lebesgue measure such that R_t is integrable and nonnegative almost surely for every $t \in \mathcal{L}$, and*

$$\mathbb{E}[R_t \mid \mathcal{F}_s] \leq R_s \quad \text{a.s. whenever } s < t \text{ belong to } \mathcal{L}. \quad (62)$$

Then R has a nonnegative càdlàg supermartingale representative $\bar{R}$ on $(0, \infty)$, equal to $R \, dt \otimes d\mathbb{P}$ -a.e. If, for some $R_0 \in L^1(\mathcal{F}_0)$,

$$R_t \rightarrow R_0 \quad \text{in } L^1 \quad \text{as } t \downarrow 0 \text{ through } t \in \mathcal{L},$$

then $\bar{R}$ extends to a nonnegative càdlàg supermartingale on $[0, \infty)$ by setting $\bar{R}_0 = R_0$.

Proof. Choose a countable dense set $\mathcal{D} \subset \mathcal{L}$. On $\mathcal{D}$, (62) gives the usual supermartingale inequalities. These variables are locally bounded in L^1 : for $0 < a < b$, choose $r \in \mathcal{L}$ with $r < a$. Then

$$\mathbb{E}[R_t] \leq \mathbb{E}[R_r] \quad \text{for } t \in \mathcal{L} \cap [a, b].$$

Doob's upcrossing argument therefore gives finite right and left limits along $\mathcal{D}$ on every compact subinterval of $(0, \infty)$. Define

$$\bar{R}_t := \lim_{\substack{q \downarrow t \\ q \in \mathcal{D}}} R_q, \quad t > 0.$$

The resulting process has càdlàg paths and is adapted because the filtration is right-continuous.

To verify the supermartingale inequality, fix $s < t$, take $p_n \downarrow s$ in $\mathcal{D}$, and first fix $q \in \mathcal{D}$ with $q > t$. Then

$$\mathbb{E}[R_q \mid \mathcal{F}_{p_n}] \leq R_{p_n}.$$

Because $p_n \downarrow s$ and the filtration is right-continuous,

$$\mathcal{F}_{p_n} \downarrow \mathcal{F}_s.$$

Hence, for fixed $q \in \mathcal{D}$,

$$\mathbb{E}[R_q \mid \mathcal{F}_{p_n}] \longrightarrow \mathbb{E}[R_q \mid \mathcal{F}_s] \quad \text{a.s.}$$

Since $R_{p_n} \rightarrow \bar{R}_s$, passing to the limit gives

$$\mathbb{E}[R_q \mid \mathcal{F}_s] \leq \bar{R}_s.$$

Letting $q \downarrow t$ through $\mathcal{D}$ and applying conditional Fatou yields

$$\mathbb{E}[\bar{R}_t \mid \mathcal{F}_s] \leq \bar{R}_s.$$

Thus $\bar{R}$ is a nonnegative càdlàg supermartingale.

It remains to show that $\bar{R}$ represents the original process. The preceding local L^1 bound and predictability allow us to apply Lebesgue differentiation with values in $L^1(\Omega)$. Consequently, for almost every $t > 0$, there are $u_n \downarrow t$, with $u_n \in \mathcal{L}$, such that

$$\mathbb{E}|R_{u_n} - R_t| \longrightarrow 0. \tag{63}$$

Fix such a $t \in \mathcal{L}$. For $q_n \downarrow t$ in $\mathcal{D}$, conditional Fatou and (62) give

$$\bar{R}_t \leq \liminf_{n \rightarrow \infty} \mathbb{E}[R_{q_n} \mid \mathcal{F}_t] \leq R_t.$$

For the reverse inequality, choose $p_n \in \mathcal{D}$ with $t < p_n < u_n$. Then

$$\mathbb{E}[R_{u_n} \mid \mathcal{F}_{p_n}] \leq R_{p_n}$$

and, because R_t is $\mathcal{F}_{p_n}$ -measurable,

$$\|\mathbb{E}[R_{u_n} \mid \mathcal{F}_{p_n}] - R_t\|_{L^1} \leq \mathbb{E}|R_{u_n} - R_t| \longrightarrow 0.$$

After passing to an almost-surely convergent subsequence and using $R_{p_n} \rightarrow \bar{R}_t$, we obtain $R_t \leq \bar{R}_t$. Hence

$$R_t = \bar{R}_t \quad \text{a.s. for almost every } t > 0.$$

Fubini's theorem gives $\bar{R} = R$, $dt \otimes d\mathbb{P}$ -a.e.

Finally, suppose $R_t \rightarrow R_0$ in L^1 as stated. This convergence bounds $\mathbb{E}[R_t]$ near zero, so the same upcrossing argument gives an almost-sure limit of R_q as $q \downarrow 0$ through $\mathcal{D}$. Since $R_q \rightarrow R_0$ in probability, that limit must be R_0 . Setting $\bar{R}_0 = R_0$ makes the paths right-continuous at zero. The preceding conditional-limit argument, now with $p_n \downarrow 0$, proves

$$\mathbb{E}[\bar{R}_t \mid \mathcal{F}_0] \leq \bar{R}_0,$$

and completes the extension. $\square$

G.3 Technical Lemmas for Theorem 1b

The date-zero calculation varies the initial report and the subsequent distortion together. The following lemma justifies that joint limit.

Lemma 6 (Continuity under small deviations). *Fix $x \in \Theta$ and $h \in \mathbb{R}$ such that $y_\varepsilon := x + \varepsilon h \in \Theta$ for all sufficiently small $\varepsilon > 0$. Let a be bounded with support in $[0, T]$, and let $w \in L^2[0, T]$ be deterministic. Write $S_t^{(y, a)} := e^{-\rho t} \psi_t^{(y, a)}$, using the common Brownian innovations of (M5). Then*

$$\mathbb{E}_x \int_0^T w_t S_t^{(y_\varepsilon, \varepsilon a)} dt \longrightarrow \mathbb{E}_x \int_0^T w_t S_t^{(x, 0)} dt.$$

Proof. First hold the initial report at y_ε . The density estimate in Lemma 4 is uniform in that report: the density ξ_ε of the distorted continuation relative to the undistorted one satisfies

$$\|\xi_\varepsilon - 1\|_{L^2}^2 \leq e^{\varepsilon^2 \|a\|_\infty^2 T} - 1 \longrightarrow 0.$$

Set $G_\varepsilon := \int_0^T w_t S_t^{(y_\varepsilon, 0)} dt$. Cauchy–Schwarz gives

$$\|G_\varepsilon\|_{L^2} \leq \|w\|_{L^2[0, T]} \left(\mathbb{E}_x \int_0^T |S_t^{(y_\varepsilon, 0)}|^2 dt \right)^{1/2}.$$

This bound is locally uniform in ε by (M3) and the L^2 -continuity in (M5). Consequently,

$$\left| \mathbb{E}_x \int_0^T w_t [S_t^{(y_\varepsilon, \varepsilon a)} - S_t^{(y_\varepsilon, 0)}] dt \right| \leq \|\xi_\varepsilon - 1\|_{L^2} \|G_\varepsilon\|_{L^2} \longrightarrow 0.$$

Finally, (M5) and Cauchy–Schwarz imply

$$\mathbb{E}_x \int_0^T w_t [S_t^{(y_\varepsilon, 0)} - S_t^{(x, 0)}] dt \longrightarrow 0.$$

Combining the two limits proves the claim. $\square$

G.4 Proof of Theorem 2

For any fixed $c \in \Theta$ and nondecreasing schedule $D : \Theta \rightarrow \mathbb{R}$ in $L^2(F)$, Tonelli's theorem gives

$$\begin{aligned} \int_{\Theta} |h_c(x)D(x)| \, dF(x) &= \int_{\Theta} \int_{\min(c,x)}^{\max(c,x)} |D(l)| \, dl \, dF(x) \\ &\leq \mathbb{E}[|\theta_0 - c|(|D(\theta_0)| + |D(c)|)] < \infty. \end{aligned} \quad (64)$$

The bound follows from monotonicity; finiteness follows from the finite variance of F and Cauchy–Schwarz.

Proof. Fix P , a candidate V satisfying the theorem's hypotheses, and an admissible continuation from a fixed state. Write $\Xi_t = (\hat{\theta}_t, D_t, \psi_t)$. The variational inequality and the boundary inequality give the nonnegative Bellman residual

$$R_t := rV(\Xi_{t-}) - \hat{\theta}_t X_t + C(X_t) - (\mathcal{L}^{X_t, v_t} V)(\Xi_{t-}) \geq 0,$$

with the boundary interpretation of Appendix F. They also give $V_{\psi} \geq 0$ on $\psi > 0$. By continuity, V is nondecreasing in ψ on $[0, \infty)$. Since $\Delta\psi_t = -\Delta K_t \leq 0$,

$$J_t := V(\hat{\theta}_t, D_t, \psi_{t-}) - V(\hat{\theta}_t, D_t, \psi_t) \geq 0.$$

First consider a continuation with finite expected discounted cost. Condition (M2) makes its discounted gross surplus absolutely integrable. Itô's formula gives

$$\begin{aligned} V(\Xi_0) &= \int_0^T e^{-rt} [\hat{\theta}_t X_t - C(X_t)] \, dt + e^{-rT} V(\Xi_T) - M_T^V \\ &\quad + \int_0^T e^{-rt} R_t \, dt + \int_{(0,T]} e^{-rt} \mathbf{1}_{\{\psi_{t-} > 0\}} V_{\psi}(\Xi_{t-}) \, dK_t^c + \sum_{0 < t \leq T} e^{-rt} J_t. \end{aligned} \quad (65)$$

Here M^V is the martingale in (59); in the finite-variation program, no derivative involving a Brownian component of ψ occurs.

To justify the identity at $\psi = 0$, first localize in the interior and pass to the first hitting time of zero. The accumulated terms on the second line are nonnegative. Their monotone limits, continuity of V , and continuity of the martingale give the identity up to that time, including J_t if a jump reaches zero. Thereafter, zero is absorbing and K is constant, so apply Itô's formula to the

C^2 boundary restriction. Localization at compact-state exits is removed in the same way. No derivative in ψ at zero is needed. The finite-horizon integrability and martingale conditions justify taking expectations in (65).

Dropping its nonnegative terms yields

$$\mathbb{E} \int_0^T e^{-rt} [\hat{\theta}_t X_t - C(X_t)] dt + \mathbb{E}[e^{-rT} V(\Xi_T)] \leq V(\Xi_0).$$

Absolute integrability of the flow payoff and the domination transversality condition imply

$$\mathbb{E} \int_0^\infty e^{-rt} [\hat{\theta}_t X_t - C(X_t)] dt \leq V(\Xi_0). \quad (66)$$

A continuation with infinite expected discounted cost has surplus $-\infty$, so the same bound holds. Thus (66) holds for every such continuation.

For each continuation of M^* in hypothesis (ii), (57) makes $R = 0$ almost everywhere, and (58) eliminates both adjustment terms. The integrability and zero terminal limit in Condition 1 therefore give equality in (66) at every initial state $(\theta, D^*(\theta), \psi_0^*(\theta))$. Here D and ψ are the marginal rent and its volatility of M^* , defined in (13)–(14). The state dynamics make $e^{-\rho t} \psi_t$ a nonnegative local supermartingale and hence a supermartingale.

The assembled mechanism is admissible by hypothesis. In the finite-variation program, Theorems 1a and 1b, together with (21), give global incentive compatibility. In the local program, the same theorems and the pointwise date-zero condition in hypothesis (iii) give local incentive compatibility. Constraint (23) gives individual rationality. In both programs, D^* is nondecreasing, so (64) makes the initial rent integrable. Integrating the attained continuation values over initial types and using (22) shows that the mechanism attains Π_P .

Finally, consider a mechanism in the theorem's comparison class. For $P = \text{FV}$, Lemma 3 gives monotonicity of D , and its date-zero schedules satisfy (21) by the restriction on that class. For $P = \text{loc}$, absolute continuity of its marginal rent and Theorems 1a–1b imply $0 \leq \psi_0 \leq \gamma D'$ almost everywhere; hence D is nondecreasing. In either case, (M1) gives $D \in L^2(F)$.

A comparison mechanism with infinite expected discounted cost has payoff $-\infty$. Otherwise, (M2) makes its discounted surplus absolutely integrable. Its conditional expected surplus is therefore F -integrable and, by (66), bounds the candidate value from below. Together with (64), this makes the date-zero objective well defined. The schedules are therefore feasible for (28). Integrating the continuation bound in (22) gives $\Pi(M) \leq \Pi_P$. $\square$

Proof of Corollary 2.1. Mechanisms with infinite expected discounted cost have payoff $-\infty$. For every other mechanism in the comparison class, Theorems 1a–1b and Lemma 3 give the local program’s initial incentive constraints. The integrability argument in the proof of Theorem 2 makes its date-zero objective well defined, so its schedules are feasible for that program. The continuation bound (66), integrated in (22), gives $\Pi(M) \leq \Pi_{\text{loc}}$. The attained finite-variation mechanism belongs to the same comparison class and is globally incentive compatible by Theorem 2, giving the lower bound. $\square$

G.5 Strict Local Incentives and Perturbation

The weak quadratic test in Definition 3 allows equality. This subsection gives conditions under which requiring a strict inequality leaves the supremum of the principal’s payoff unchanged. The argument mixes actual mechanisms, so it preserves delivery of their marginal rents.

Write $U_{M,x}(y, a)$ for the agent’s payoff under mechanism M . Say that M has *strict local incentives* if, at every initial type x ,

$$\limsup_{\varepsilon \downarrow 0} \frac{U_{M,x}(x + \varepsilon h, \varepsilon a) - U_{M,x}(x, 0)}{\varepsilon^2} < 0 \quad (67)$$

for every nonzero feasible direction (h, a) . Here a direction is nonzero if $h \neq 0$ or a is nonzero on a set of positive $dt \otimes d\mathbb{P}_x$ measure. Strictness rules out profitable sufficiently small scalings of each fixed direction; it does not assert a common neighborhood for all directions.

Proposition 8 (Strict approximation). *In the OU model, fix a class of admissible, individually rational mechanisms with finite principal payoffs, closed under pointwise deterministic convex mixtures. Suppose the class contains a benchmark $\bar{M}$ satisfying the transfer conditions of Proposition 2, with a differentiable initial marginal rent $\bar{D}$ satisfying (20) at every type. At included endpoints, require the corresponding finite inward derivative. Suppose also that, after every initial report, the benchmark’s discounted marginal rent volatility is the same deterministic process*

$$\bar{R}_t := e^{-\rho t} \bar{\psi}_t = \int_t^\infty k_s ds, \quad k_s > 0 \text{ for almost every } s, \quad \int_0^\infty k_s ds < \infty. \quad (68)$$

For any locally incentive-compatible $M = (X, T)$ in the class and $\eta \in (0, 1)$, the mechanism

$$M^\eta = (1 - \eta)M + \eta\bar{M}$$

has strict local incentives and satisfies

$$\Pi(M^\eta) \geq (1 - \eta)\Pi(M) + \eta\Pi(\bar{M}). \quad (69)$$

Consequently, every locally incentive-compatible mechanism in the class has strict approximations at arbitrarily small payoff loss, and imposing strict local incentives leaves the supremum unchanged.

Proof. First verify strictness of the benchmark. Fix an initial type x and a feasible direction (h, a) , and put

$$\Delta_t = he^{-\lambda t} + \gamma \int_0^t e^{-\lambda(t-s)} a_s \, ds.$$

The report $(x + \varepsilon h, \varepsilon a)$ has gap $\delta_t = \varepsilon \Delta_t$. By the absolute continuity of report laws in Definition 2, the deterministic identity (68) holds along this report as well. Since

$$d(e^{2\lambda t} \Delta_t^2) = 2\gamma e^{2\lambda t} \Delta_t a_t \, dt,$$

integration by parts on $[0, H]$ gives

$$\begin{aligned} -\mathbb{E}_x \int_0^H e^{-rt} \bar{\psi}_t \Delta_t a_t \, dt &= \frac{\bar{R}_0 h^2}{2\gamma} - \frac{\bar{R}_H}{2\gamma} \mathbb{E}_x [e^{2\lambda H} \Delta_H^2] \\ &\quad - \frac{1}{2\gamma} \mathbb{E}_x \int_0^H k_t e^{2\lambda t} \Delta_t^2 \, dt. \end{aligned}$$

The report energy condition makes these terms finite on each finite horizon. Fix $T < \infty$, take $H \geq T$, and discard the nonpositive terminal term and the integral over $(T, H]$. Condition (M4) permits passage to $H \rightarrow \infty$ in the integral on the left. The master formula (18) and the date-zero envelope expansion therefore imply

$$\begin{aligned} \limsup_{\varepsilon \downarrow 0} \frac{U_{\bar{M},x}(x + \varepsilon h, \varepsilon a) - U_{\bar{M},x}(x, 0)}{\varepsilon^2} \\ \leq \frac{h^2}{2} \left[\frac{\bar{\psi}_0}{\gamma} - \bar{D}'(x) \right] - \frac{1}{2\gamma} \mathbb{E}_x \int_0^T k_t e^{2\lambda t} \Delta_t^2 \, dt. \end{aligned} \quad (70)$$

Use the appropriate inward derivative at an endpoint. The first term is nonpositive. For a nonzero direction, the last integral is strictly positive for some finite T : otherwise positivity of k and continuity of Δ imply $\Delta \equiv 0$ almost surely, and $d\Delta_t = (-\lambda\Delta_t + \gamma a_t)dt$ then gives $h = 0$ and $a = 0$ almost everywhere. Thus $\bar{M}$ has strict local incentives.

For every fixed report, the agent's payoff is linear in allocation and transfer. Hence its deviation gain under M^η is exactly $(1 - \eta)$ times the gain under M , plus η times the gain under $\bar{M}$. The report class is fixed independently of the mechanism. Local incentive compatibility of M and the strict bound for $\bar{M}$ therefore imply (67) for M^η . The assumed closure keeps the mixture admissible and individually rational. Finally, linearity of transfers and convexity of C give (69). Letting $\eta \downarrow 0$ proves the payoff approximation. Since strict local incentives imply the weak test, the two suprema coincide. $\square$

A benchmark allocation. When $\mathcal{X} = \mathbb{R}$, one candidate is

$$\bar{X}_t = q_t \hat{\theta}_t + m_t \hat{\theta}_0,$$

with deterministic $q_t, m_t > 0$. Suppose its implementing transfers give an admissible mechanism with finite principal payoff, and the following integrals are finite. Direct calculation gives

$$\bar{D}'(x) = \int_0^\infty q_t e^{-\rho t} dt + \int_0^\infty m_t e^{-(r+\lambda)t} dt, \quad \bar{R}_t = \gamma \int_t^\infty q_s e^{-\rho s} ds.$$

Thus $k_t = \gamma q_t e^{-\rho t} > 0$ and the date-zero inequality is strict. The envelope utility is a quadratic with positive leading coefficient, so its free constant can be chosen to ensure individual rationality. With constrained allocations, the proposition applies whenever an admissible benchmark satisfying (68) belongs to the class.

G.6 On Numerics for the Control Programs

A monotone semi-Lagrangian scheme approximates the continuation problem (27) in the states $z = (\vartheta, d, \psi)$, where $\rho = r + 2\lambda$. Choose a time step $h > 0$, a finite state grid, bounded control sets, and fixed boundary values. Let $\mathcal{I}_\Delta W$ denote multilinear interpolation with nonnegative weights.

At a decision state, choose an allocation x and a volatility $\bar{\psi} \in [0, \psi]$ after a downward adjustment. For the finite-variation program, define

$$z_\pm(x, \bar{\psi}) = \begin{pmatrix} \vartheta + \lambda(\bar{\theta} - \vartheta)h \pm \gamma\sqrt{h} \\ d + [(r + \lambda)d - x]h \pm \bar{\psi}\sqrt{h} \\ e^{\rho h} \bar{\psi} \end{pmatrix}, \quad \Delta_r(h) = \frac{1 - e^{-rh}}{r}.$$

The Bellman operator is

$$(\mathcal{T}_{h,\Delta}^{\text{FV}}W)(z) = \sup_{\substack{x \in \mathcal{X} \\ 0 \leq \bar{\psi} \leq \psi}} \left\{ \Delta_r(h)[\vartheta x - C(x)] + \frac{e^{-rh}}{2} \sum_{\epsilon \in \{-, +\}} \mathcal{I}_\Delta W(z_\epsilon(x, \bar{\psi})) \right\}, \quad (71)$$

where the controls and transitions use the selected finite grid and boundary treatment. The common sign in the first two coordinates reproduces their covariance, including the mixed term $\gamma\psi V_{\hat{\theta}D}$ in (26). Interpolation, averaging, and maximization preserve order, so the operator is monotone.

For the local program, also choose v and replace the third coordinate by

$$e^{\rho h} \bar{\psi} \pm v\sqrt{h}, \quad e^{\rho h} \bar{\psi} - |v|\sqrt{h} \geq 0.$$

Use the same sign in all three coordinates. The resulting operator $\mathcal{T}_{h,\Delta}^{\text{loc}}$ includes all covariance terms in (26). At $\psi = 0$, both schemes impose $\bar{\psi} = 0$, and the local scheme also imposes $v = 0$. Thus zero volatility is absorbing. State and control domains can be expanded as the grid and time step are refined; interpolation and boundary evaluation retain nonnegative weights.

Fix either program and abbreviate its operator by $\mathcal{T}$. On the finite grid, with fixed boundary values, it satisfies

$$\|\mathcal{T}W - \mathcal{T}Z\|_\infty \leq e^{-rh} \|W - Z\|_\infty.$$

Hence it has a unique fixed point $V_{h,\Delta}$. Choose U^0 with $\mathcal{T}U^0 \leq U^0$; a constant dominating the boundary values and the maximal flow reward divided by r is one such choice. Iterating $U^{n+1} = \mathcal{T}U^n$ decreases to $V_{h,\Delta}$. A maximizing policy at iteration n has value L^n obtained by solving its linear policy-evaluation system. Monotonicity gives

$$\max_{k \leq n} L^k \leq V_{h,\Delta} \leq U^n \quad \text{at every grid point.}$$

The bracket provides a stopping criterion. Equivalently, any computed vector W satisfies the residual bound

$$\|W - V_{h,\Delta}\|_\infty \leq \frac{\|\mathcal{T}W - W\|_\infty}{1 - e^{-rh}}.$$

A recovered finite-variation policy selects an allocation at date zero and at positive decision dates $h, 2h, \dots$, holds it between decisions, and evolves the state under the actual diffusion. At a positive decision date it may reduce ψ to the selected $\bar{\psi}$; between decisions, $\dot{\psi} = \rho\psi$. This produces finitely many

downward jumps on each finite horizon. Because $K_0 = 0$, initial volatility is chosen directly in the date-zero schedule of (28); an initial reduction is incorporated in that choice.

Proposition 2 supplies transfers for the recovered allocation. Theorem 2 then verifies a proposed mechanism using its delivered promises, admissibility, terminal conditions, and date-zero constraints. For a local upper bound, the domination argument uses a candidate satisfying the Bellman and gradient inequalities, (56), and the domination conditions in Appendix F. Comparing this bound with the evaluated payoff of an admissible finite-variation policy gives the revenue bracket of Corollary 2.1.

H Supplemental Material for Section 6

This appendix proves the dealer and exchange results in Section 6 and gives the static benchmark and numerical details. Throughout, $\rho = r + 2\lambda$, and transfers denoted by T are actual payments. Write

$$\mathcal{B}(m, y) := \int_y^m [D(l) - D(y)] dl$$

for the date-zero integral associated with a marginal-rent schedule.

H.1 Proof of Proposition 3

Proof. Write $\rho = r + 2\lambda$ and define

$$\bar{\Lambda}(\theta) := \begin{cases} \Lambda(\theta), & |\theta| > x^*, \\ \theta, & |\theta| \leq x^*. \end{cases}$$

Thus $\rho D^{\text{FOA}}(\theta) = \theta - \bar{\Lambda}(\theta)$. Regularity gives $|\bar{\Lambda}(\theta)| \leq |\theta|$ and $|\theta - \bar{\Lambda}(\theta)| \leq |\theta|$.

Step 1: Simplify the objective. Conditional on θ_0 , first-best surplus is

$$\frac{1}{2} \mathbb{E} \left[\int_0^\infty e^{-rs} \theta_s^2 ds \mid \theta_0 \right] = \frac{\theta_0^2}{2\rho} + \frac{1}{2r\rho}. \quad (72)$$

The tracking identity (34) also gives

$$\frac{1}{2} \mathbb{E} \left[\int_0^\infty e^{-rs} \zeta_s^2 ds \mid \theta_0 \right] \geq \frac{\rho}{2} \tilde{D}_0^2. \quad (73)$$

Equality holds exactly when $\psi = 1/\rho$ and $\zeta_s = \rho \tilde{D}_s$ almost everywhere. In this case, $\tilde{D}_s = \tilde{D}_0 e^{-\lambda s}$ and $\zeta_s = \rho \tilde{D}_0 e^{-\lambda s}$.

Using $\theta X - X^2/2 = \theta^2/2 - \zeta^2/2$, (72), and (73), the objective is at most

$$\frac{\mathbb{E}[\theta_0^2]}{2\rho} + \frac{1}{2r\rho} + \mathcal{S}(D, u), \quad \mathcal{S}(D, u) := -\frac{\rho}{2}\mathbb{E}\left[\left(D(\theta_0) - \frac{\theta_0}{\rho}\right)^2\right] - \mathbb{E}[U(\theta_0)]. \quad (74)$$

It suffices to maximize (74) with respect to D and u subject to $U(\theta) = u + \int_{x^*}^{\theta} D(l) dl$ and $U \geq 0$. We may without loss of generality restrict attention to (D, u) such that $D(\theta_0) \in L^2$ and $\mathbb{E}[U(\theta_0)] < \infty$ since otherwise $\mathcal{S}(D, u) = -\infty$. In what follows, I derive an upper bound on $\mathcal{S}$ and construct an allocation that attains the bound.

Step 2: construct the participation multiplier. Define

$$M(\theta) := \begin{cases} 0, & \theta < -x^*, \\ F(\theta) + \theta f(\theta), & -x^* \leq \theta < x^*, \\ 1, & \theta \geq x^*. \end{cases}$$

This measure will play the role of a multiplier on the participation constraint and is supported on the interval where the constraint binds.

On $0 \leq \theta \leq x^*$, write

$$M(\theta) = F(\theta) + \theta f(\theta) = 1 + f(\theta)v_+(\theta), \quad v_+(\theta) = \theta - \frac{1 - F(\theta)}{f(\theta)}.$$

For $0 \leq \theta < \theta' \leq x^*$, the first representation implies $M(\theta') \geq M(\theta)$ if $f(\theta') \geq f(\theta)$. If the density decreases between the two points, the second representation gives the same inequality because $v_+ \leq 0$ and v_+ is nondecreasing. Symmetry gives $M(-\theta) = 1 - M(\theta)$, so M is nondecreasing also on the negative half of the interval. Since $v_+(x^*) = 0$, M is continuous at the endpoints and is a probability distribution supported on $[-x^*, x^*]$.

By construction,

$$\frac{M(\theta) - F(\theta)}{f(\theta)} = \bar{\Lambda}(\theta). \quad (75)$$

Regularity and symmetry also imply

$$|\theta - \bar{\Lambda}(\theta)| \leq |\theta|, \quad |\bar{\Lambda}(\theta)| \leq |\theta|. \quad (76)$$

Consequently,

$$\int_{\Theta} |D(l)| |M(l) - F(l)| dl = \mathbb{E}[|\bar{\Lambda}(\theta_0) D(\theta_0)|] < \infty$$

by Cauchy–Schwarz. Moreover, $\mathbf{1}\{l \geq x^\star\} - M(l)$ is supported on $[-x^\star, x^\star]$, where D is integrable. Thus Fubini’s theorem gives

$$\begin{aligned}\mathbb{E}[U(\theta_0)] &= u + \int_{\Theta} D(l)(\mathbf{1}\{l \geq x^\star\} - F(l)) \, dl, \\ \int_{\Theta} U \, dM &= u + \int_{\Theta} D(l)(\mathbf{1}\{l \geq x^\star\} - M(l)) \, dl.\end{aligned}$$

Subtracting these identities and using (75) yields

$$-\mathbb{E}[U(\theta_0)] + \int_{\Theta} U \, dM = -\mathbb{E}[\bar{\Lambda}(\theta_0)D(\theta_0)]. \quad (77)$$

Step 3: derive an upper bound and show it is attained. Since $U \geq 0$ and M is a probability measure, (77) gives

$$\begin{aligned}\mathcal{S}(D, u) &\leq \mathcal{S}(D, u) + \int_{\Theta} U \, dM \\ &= -\frac{\rho}{2} \mathbb{E} \left[\left(D - \frac{\theta_0}{\rho} \right)^2 \right] - \mathbb{E}[\bar{\Lambda}D] \\ &= -\frac{\rho}{2} \mathbb{E} \left[(D - D^{\text{FOA}})^2 \right] + \frac{1}{2\rho} \mathbb{E} \left[(\theta_0 - \bar{\Lambda}(\theta_0))^2 - \theta_0^2 \right] \\ &\leq \frac{1}{2\rho} \mathbb{E} \left[(\theta_0 - \bar{\Lambda}(\theta_0))^2 - \theta_0^2 \right],\end{aligned} \quad (78)$$

where

$$D^{\text{FOA}}(\theta) = \frac{\theta - \bar{\Lambda}(\theta)}{\rho}.$$

Regularity implies that D^{FOA} is nondecreasing. Below $-x^\star$, $D^{\text{FOA}}(\theta) = (\theta + F(\theta)/f(\theta))/\rho$ and increases to zero. It is zero on $[-x^\star, x^\star]$. Above $x^\star$, $D^{\text{FOA}} = v_+/\rho$ and increases from zero.

Take $D = D^{\text{FOA}}$ and $u = 0$. Symmetry gives

$$U(\theta_0) = \int_{x^\star}^{|\theta_0|} D^{\text{FOA}}(l) \, dl \geq 0,$$

with equality on the support of M . Hence $\int U \, dM = 0$, so both inequalities in (78) bind. Moreover, (76) gives

$$0 \leq U(\theta_0) \leq \frac{\theta_0^2}{2\rho},$$

so U is integrable.

It remains to show that an allocation delivers this bound. For the allocation in (32), stochastic Fubini gives

$$Z = D^{\text{FOA}}(\theta_0) + \int_0^\infty \frac{e^{-(r+\lambda)s}}{\rho} dB_s. \quad (79)$$

Thus $Z \in L^2$, $\mathbb{E}[Z \mid \theta_0] = D^{\text{FOA}}(\theta_0)$, and the current-value loading is $\psi \equiv 1/\rho$. In addition,

$$\tilde{D}_0 = -\frac{\bar{\Lambda}(\theta_0)}{\rho}, \quad \zeta_s = -\bar{\Lambda}(\theta_0)e^{-\lambda s} = \rho\tilde{D}_0e^{-\lambda s},$$

so (73) binds. Combining (74) and (78) proves

$$\Pi^{\text{FOA}} = \frac{1}{2\rho} \left(\frac{1}{r} + \mathbb{E} \left[(\theta_0 - \bar{\Lambda}(\theta_0))^2 \right] \right).$$

Finally, equality in (78) requires $D = D^{\text{FOA}}$ up to F -null sets. For this rent, $U = u$ on the support of M . Participation gives $u \geq 0$, while attainment requires $\int U dM = 0$, so $u = 0$. Equality in (73) then pins down the allocation up to $dt \otimes d\mathbb{P}$ -null sets. $\square$

H.2 Proof of Proposition 4

Proof. The tracking identity. For any finite-tracking-cost continuation, the definition of the rent gives

$$\tilde{D}_t = \mathbb{E}_t \int_t^\infty e^{-(r+\lambda)(s-t)} \zeta_s ds.$$

Conditional Cauchy–Schwarz implies

$$\rho e^{-rt} \mathbb{E}[\tilde{D}_t^2] \leq \mathbb{E} \int_t^\infty e^{-rs} \zeta_s^2 ds \longrightarrow 0.$$

Apply Itô's formula to $e^{-rt} \rho \tilde{D}_t^2 / 2$, localize the martingale, and complete the square. The conditional-expectation representation gives the local square-integrability needed to remove the localization; the preceding estimate removes the terminal term. Monotone convergence for the nonnegative integrals then proves (34), including finiteness of its right-hand side. A continuation with infinite tracking cost has surplus $-\infty$ and cannot improve the value.

The value bound and attainment. Fix an initial state $(\hat{\theta}, D, \psi_0)$. For every admissible continuation strategy that is feasible in either problem, $e^{-\rho t}\psi_t$ is a nonnegative supermartingale by the control equation (25) and nonnegativity. Hence

$$0 \leq \mathbb{E}[\psi_t] \leq e^{\rho t}\psi_0.$$

Jensen's inequality therefore gives

$$\mathbb{E} \left[\left(\psi_t - \frac{1}{\rho} \right)^2 \right] \geq \left(\mathbb{E}[\psi_t] - \frac{1}{\rho} \right)^2 \geq \left[\left(\frac{1}{\rho} - e^{\rho t}\psi_0 \right)^+ \right]^2. \quad (80)$$

Integrating yields

$$\mathbb{E} \int_0^\infty e^{-rt} \left(\psi_t - \frac{1}{\rho} \right)^2 dt \geq P(\psi_0).$$

For $0 \leq \psi_0 \leq 1/\rho$, equality holds for the deterministic path

$$\psi_t = \min\{e^{\rho t}\psi_0, 1/\rho\}. \quad (81)$$

Write $\tilde{D} := D - \hat{\theta}/\rho$. We first show that every continuation from $(\hat{\theta}, D, \psi_0)$ has surplus at most

$$V(\hat{\theta}, D, \psi_0) = \frac{\hat{\theta}^2}{2\rho} + \frac{1}{2r\rho} - \frac{\rho}{2}\tilde{D}^2 - \frac{\rho}{2}P(\psi_0).$$

Since first-best surplus is finite, a continuation with infinite tracking loss has surplus $-\infty$ and satisfies this inequality. We may therefore assume that its tracking loss is finite. Combining (72) with the tracking identity gives

$$\begin{aligned} \mathbb{E} \int_0^\infty e^{-rt} \left(\hat{\theta}_t X_t - \frac{1}{2} X_t^2 \right) dt &= \frac{\hat{\theta}^2}{2\rho} + \frac{1}{2r\rho} - \frac{\rho}{2}\tilde{D}^2 \\ &\quad - \frac{1}{2}\mathbb{E} \int_0^\infty e^{-rt} (\zeta_t - \rho\tilde{D}_t)^2 dt \\ &\quad - \frac{\rho}{2}\mathbb{E} \int_0^\infty e^{-rt} \left(\psi_t - \frac{1}{\rho} \right)^2 dt. \end{aligned} \quad (82)$$

The first integral after the constant terms is the loss from choosing $\zeta_t \neq \rho\tilde{D}_t$; it is nonnegative and vanishes when $\zeta_t = \rho\tilde{D}_t$. The last integral is bounded below by $P(\psi_0)$. Thus (82) proves the upper bound for every $\psi_0 \geq 0$.

Now suppose $0 \leq \psi_0 \leq 1/\rho$. Choose the volatility path in (81) and define

$$K_t := \int_0^t \rho \psi_s \mathbf{1}\{\psi_s = 1/\rho\} ds.$$

Then $d\psi_t = \rho\psi_t dt - dK_t$. Let

$$d\tilde{D}_t = -\lambda\tilde{D}_t dt + \left(\psi_t - \frac{1}{\rho}\right) d\hat{B}_t, \quad D_t = \tilde{D}_t + \frac{\hat{\theta}_t}{\rho}, \quad X_t = \hat{\theta}_t + \rho\tilde{D}_t = \rho D_t.$$

Direct substitution gives

$$dD_t = [(r + \lambda)D_t - X_t]dt + \psi_t d\hat{B}_t.$$

Since ψ is bounded,

$$e^{-(r+\lambda)t} D_t + \int_0^t e^{-(r+\lambda)s} X_s ds$$

is a square-integrable martingale. The explicit linear solution also gives

$$e^{-(r+\lambda)T} D_T \longrightarrow 0 \quad \text{in } L^1.$$

Consequently,

$$D_t = \mathbb{E}_t \int_t^\infty e^{-(r+\lambda)(s-t)} X_s ds.$$

Thus D_t is the continuation rent generated by X , and its current-value volatility is ψ_t . The volatility is bounded, deterministic, and of finite variation, so the construction satisfies the restrictions of both continuation problems.

It remains to verify admissibility. A direct calculation gives

$$\text{vol}(\widehat{W}_t^X) = D_t + \hat{\theta}_t \psi_t, \quad H_t = -\hat{\theta}_t \psi_t,$$

where H is the transfer integrand in Proposition 2. Under a report distortion a ,

$$d\hat{\theta}_t = -\lambda\hat{\theta}_t dt + dB_t + a_t dt, \quad dD_t = -\lambda D_t dt + \psi_t (dB_t + a_t dt).$$

The changes in these states caused by a therefore have the form

$$\int_0^t e^{-\lambda(t-s)} b_s a_s ds,$$

where b is bounded and deterministic. Weighted Young's inequality gives

$$\mathbb{E} \int_0^\infty e^{-rt} \left| \int_0^t e^{-\lambda(t-s)} b_s a_s ds \right|^2 dt \leq \frac{\|b\|_\infty^2}{(\lambda + r/2)^2} \mathbb{E} \int_0^\infty e^{-rt} a_t^2 dt.$$

The right-hand side is finite for every admissible report. Standard estimates for the two linear equations governing $\hat{\theta}$ and D therefore give the required local L^2 bounds, discounted integrability, and terminal limits for $\hat{\theta}$, D , X , and H . Applied to the transfer formula in Proposition 2, the same estimates, the L^2 isometry, Fubini's theorem, and Cauchy–Schwarz give finite discounted expected absolute transfers. Because ψ is deterministic, its diffusion coefficient is zero, so the additional admissibility conditions involving that coefficient hold automatically. Proposition 2, applied to the effective transfer, therefore constructs an admissible $\tilde{T}$. Since X is discounted square-integrable, $T = \tilde{T} - X^2/2$ is an admissible raw transfer.

Finally,

$$\zeta_t = X_t - \hat{\theta}_t = \rho \tilde{D}_t,$$

so the first integral in (82) vanishes. The chosen volatility path makes (80) bind. Hence the continuation attains $V(\hat{\theta}, D, \psi_0)$ in both classes.

Now suppose $\psi_0 > 1/\rho$. Because $K_0 = 0$, the volatility cannot jump at date zero. Let

$$\delta := \frac{\psi_0 - 1/\rho}{2}, \quad \tau := \inf\{t > 0 : |\psi_t - \psi_0| \geq \delta\} \wedge 1.$$

Right continuity gives $\tau > 0$ almost surely, and $|\psi_t - 1/\rho| \geq \delta$ on $[0, \tau)$. Hence

$$\mathbb{E} \int_0^\infty e^{-rt} \left(\psi_t - \frac{1}{\rho} \right)^2 dt \geq \mathbb{E} \int_0^\tau e^{-rt} \delta^2 dt > 0.$$

Since $P(\psi_0) = 0$, no continuation from this state attains the upper bound. It can nevertheless be approached by lowering ψ linearly from ψ_0 to $1/\rho$ on $[0, \varepsilon]$ and then holding it fixed. Set $dK_t = (\rho\psi_t - \dot{\psi}_t)dt$ and retain $X_t = \rho D_t$. This bounded, absolutely continuous volatility is feasible, and its additional loss is at most

$$\frac{\rho\varepsilon}{2} \left(\psi_0 - \frac{1}{\rho} \right)^2 \longrightarrow 0.$$

Thus $V(\hat{\theta}, D, \psi_0)$ is the supremum of the continuation surplus for both the finite-variation and local programs. Since $P(\psi) = 0$ for every $\psi \geq 1/\rho$, replacing an initial volatility above $1/\rho$ by $1/\rho$ preserves continuation value and weakly relaxes either date-zero incentive constraint.

The variational inequalities. For $0 < \psi < 1/\rho$, differentiation of P gives

$$rP(\psi) = \left(\psi - \frac{1}{\rho}\right)^2 + \rho\psi P'(\psi), \quad P''(\psi) = 2 \int_0^{\rho^{-1} \log(1/(\rho\psi))} e^{(2\rho-r)t} dt > 0. \quad (83)$$

The zero extension above $1/\rho$ is C^2 on $(0, \infty)$, with $P' = P'' = 0$ at $1/\rho$. For the value in (35), direct substitution into the generator gives

$$\begin{aligned} rV - \left(\hat{\theta}x - \frac{x^2}{2} + \mathcal{L}^{x,v}V\right) &= \frac{1}{2}(x - \rho D)^2 \\ &\quad + \frac{\rho}{2} \left[\left(\psi - \frac{1}{\rho}\right)^2 - rP(\psi) + \rho\psi P'(\psi) \right] \\ &\quad + \frac{\rho}{4} P''(\psi) v^2. \end{aligned}$$

Below $1/\rho$, the second line's bracket is zero and the maximum is attained at $x = \rho D, v = 0$. At and above $1/\rho$, the bracket is nonnegative and $V_\psi = 0$. Thus the variational inequality (27) holds in both classes. At $\psi = 0$, $P(0) = 1/(r\rho^2)$; applying the generator only to the $(\hat{\theta}, D)$ variables gives equality in (56) at $x = \rho D$. $\square$

H.3 Proof of Proposition 5

Suppose F is uniform on $[-1, 1]$. Then

$$\Lambda(\theta) = \text{sgn}(\theta)(1 - |\theta|), \quad x^\star = \frac{1}{2}.$$

On $(0, 1/\rho)$, P is strictly decreasing and strictly convex, with $P'(0^+) = -\infty$. At first best,

$$P(1/\rho) = P'(1/\rho) = P''(1/\rho) = 0, \quad P'''(1/\rho^-) = -2.$$

The part of either date-zero objective that depends on (D, ψ_0) is

$$\mathcal{J}(D, \psi_0) = \mathbb{E} \left[-\frac{\rho}{2} \left(D(\theta_0) - \frac{\theta_0}{\rho} \right)^2 - \frac{\rho}{2} P(\psi_0(\theta_0)) - \Lambda(\theta_0) D(\theta_0) \right]. \quad (84)$$

Total revenue adds $\mathbb{E}[\theta_0^2]/(2\rho) + 1/(2r\rho)$. Reflection and averaging weakly improve any feasible triple (D, ψ_0, U) , so we may take D odd, ψ_0 even, and U even. Truthful utility is then minimized at zero, where participation binds,

and we set $D(0) = U(0) = 0$. A possible jump of an odd nondecreasing D at zero is allowed in competitors: dD below includes its right jump as an atom at zero. Thus both programs maximize (84) over nondecreasing D and measurable $\psi_0 \in [0, 1/\rho]$. The finite-variation program imposes (30); the local program imposes $\psi_0 d\theta \leq dD$. By symmetry, it is enough to work on $[0, 1]$.

Under the conditions on D below, the mirror report $m = -\hat{\theta}_0$ produces the smallest upper bound $D(\hat{\theta}_0)/\hat{\theta}_0$ on initial volatility for every nonzero type.

Lemma 7 (The date-zero volatility cap). *Let D be odd, nondecreasing, and locally absolutely continuous, with $D(0) = 0$, and suppose that $D(\theta)/\theta$ is nondecreasing on $(0, 1]$. Define*

$$c(\theta) := \begin{cases} D(\theta)/\theta, & \theta \neq 0, \\ \lim_{x \downarrow 0} D(x)/x, & \theta = 0. \end{cases}$$

Then, for every $m \neq \hat{\theta}_0$,

$$\frac{2\mathcal{B}(m, \hat{\theta}_0)}{(m - \hat{\theta}_0)^2} \geq c(\hat{\theta}_0).$$

Hence $\psi_0 = c$ satisfies (30) and is the largest feasible initial volatility at each type. For $\hat{\theta}_0 \neq 0$, equality holds exactly when c is constant between $|m|$ and $|\hat{\theta}_0|$; the mirror report $m = -\hat{\theta}_0$ always binds. At $\hat{\theta}_0 = 0$, equality for $m \neq 0$ holds exactly when c is constant on $[0, |m|]$.

Proof. By symmetry take $\hat{\theta}_0 > 0$, set $c^* = c(\hat{\theta}_0)$, and define

$$\Phi(m) := 2\mathcal{B}(m, \hat{\theta}_0) - c^*(m - \hat{\theta}_0)^2.$$

Oddness gives

$$\Phi'(m) = 2[D(m) - c^*m] = 2m[c(|m|) - c^*].$$

Monotonicity of c makes Φ attain its minimum on the positive half-line at $m = \hat{\theta}_0$, and on the negative half-line at $m = -\hat{\theta}_0$. Both minima are zero. This proves the inequality and its equality statement. At $\hat{\theta}_0 = 0$,

$$\frac{2\mathcal{B}(m, 0)}{m^2} = \frac{2}{m^2} \int_0^{|m|} lc(l) dl \geq c(0),$$

with the asserted equality condition. □

Proposition 9 (Solution of the finite-variation program, uniform case). *Let $s \in (0, 1/\rho)$ be the unique solution of*

$$P'(s) = -\frac{3}{8\rho(2 - \rho s)}, \quad (85)$$

and set $b = 3/[4(2 - \rho s)] \in (3/8, 3/4)$. Define D^{FV} on $[0, 1]$ by

$$D^{\text{FV}}(\theta) = \begin{cases} s\theta, & 0 \leq \theta \leq b, \\ \theta\varsigma(\theta), & b < \theta \leq 1, \end{cases}$$

where $\varsigma(\theta)$ is the unique solution of

$$2\rho\theta^2\varsigma + \rho P'(\varsigma) = 2\theta(2\theta - 1), \quad (86)$$

in $(0, 1/\rho]$. Extend D^{FV} oddly, and let

$$\psi_0^{\text{FV}}(\theta) = \begin{cases} D^{\text{FV}}(\theta)/\theta, & \theta \neq 0, \\ s, & \theta = 0. \end{cases}$$

Extend ψ_0^{FV} evenly to negative types. This pair uniquely solves the finite-variation program up to null sets. Moreover, ς increases strictly from s at b to $1/\rho$ at 1. Thus volatility is below first best at every interior type and reaches it only at the endpoints. Every date-zero constraint between two distinct types in $[-b, b]$ binds; outside this interval, only the mirror-type constraint binds.

Proof. Let $\mathcal{P}$ be the finite-variation program and let $\mathcal{R}$ retain only the local and mirror-type implications of (30). On $[0, 1]$, these are

$$0 \leq \psi_0 \leq \frac{1}{\rho}, \quad \psi_0 d\theta \leq dD, \quad \theta\psi_0 \leq D. \quad (87)$$

Here dD is the Lebesgue–Stieltjes measure of the nondecreasing function D . Every $\mathcal{P}$ -feasible pair is feasible for $\mathcal{R}$.

The left side of (85), after moving all terms to one side, is strictly increasing from $-\infty$ to a positive value, so s is unique. For each $\theta \in [b, 1]$, the left side of (86) is strictly increasing in ς , with derivative $\rho(2\theta^2 + P''(\varsigma)) > 0$. It crosses the right side once, below $1/\rho$ when $\theta < 1$ and at $1/\rho$ when $\theta = 1$. Equation (85) is equivalent to

$$-\frac{\rho}{2}P'(s) = \frac{b}{4},$$

which gives $\varsigma(b) = s$. Implicit differentiation gives

$$\varsigma'(\theta) = \frac{2[1 - (\rho/\theta)|P'(\varsigma(\theta))|]}{\rho[2\theta^2 + P''(\varsigma(\theta))]}.$$
 (88)

The numerator is positive at b . If it first vanished at $T > b$, then monotonicity up to T would imply

$$\frac{\rho}{T}|P'(\varsigma(T))| < \frac{\rho}{b}|P'(s)| = \frac{1}{2},$$

a contradiction. Thus ς is strictly increasing. Since $D^{\text{FV}}(\theta)/\theta = s$ on $(0, b]$ and equals $\varsigma(\theta)$ on $(b, 1]$, with $\varsigma(b) = s$, the ratio $D^{\text{FV}}(\theta)/\theta$ is nondecreasing on $(0, 1]$. Hence Lemma 7 applies to D^{FV} .

It remains to certify optimality in $\mathcal{R}$. Let $\eta \geq 0$ and $\nu \geq 0$ be the multipliers on the second and third constraints in (87). Stationarity requires

$$-\frac{\rho}{2}P'(\psi_0) = \eta + \theta\nu, \quad 2\theta - 1 - \rho D = \eta' - \nu.$$
 (89)

On $[b, 1]$, set

$$\eta = 0, \quad \nu = -\frac{\rho P'(\varsigma)}{2\theta}.$$

Equation (86) gives (89). On $[0, b]$, set

$$\nu(\theta) = \frac{1}{2} - \frac{(2 - \rho s)\theta}{3}, \quad \eta(\theta) = \frac{b}{4} - \frac{\theta}{2} + \frac{(2 - \rho s)\theta^2}{3}.$$

These functions also satisfy (89). They are nonnegative and continuous at b : $\nu(b) = 1/4$, while η is a convex quadratic with minimum $\eta(b) = 0$. Both constraints bind on $[0, b]$; only the mirror constraint binds on $(b, 1)$. Thus complementary slackness holds.

For any feasible (D, ψ_0) in $\mathcal{R}$, concavity of (84), (89), and Lebesgue–Stieltjes integration by parts, using complementary slackness at the candidate, give

$$\begin{aligned} \mathcal{J}(D, \psi_0) - \mathcal{J}(D^{\text{FV}}, \psi_0^{\text{FV}}) &\leq \int_{[0,1]} \eta(\psi_0 \, d\theta - dD) \\ &\quad + \int_0^1 \nu(\theta\psi_0 - D) \, d\theta \leq 0. \end{aligned}$$

The boundary terms vanish because $D(0) = D^{\text{FV}}(0) = 0$ and $\eta(1) = 0$. Hence the candidate solves $\mathcal{R}$. By Lemma 7, it is also feasible for the full family (30); therefore it solves $\mathcal{P}$. Strict concavity gives uniqueness of the optimal (D, ψ_0) . Since $D^{\text{FV}}(\theta)/\theta$ is constant on $(0, b]$ and strictly increasing on $(b, 1]$, Lemma 7 gives the binding pattern stated in the proposition. $\square$

Proposition 10 (Solution of the local program, uniform case). *Define*

$$g(w) := \rho^2(wP'(w) - P(w)) - 2\rho P'(w).$$

This function decreases strictly from $+\infty$ to zero on $(0, 1/\rho]$. Let w_0 be the unique solution of $g(w_0) = 1$, and define

$$\vartheta(w) := \int_{w_0}^w \frac{\rho P''(u)}{2\sqrt{g(u)}} du, \quad a := \vartheta(1/\rho) \in (1/2, 1).$$

Up to null sets, the local program is uniquely solved on $[0, a]$ by

$$(\theta, D^{\text{loc}}, \psi_0^{\text{loc}}) = \left(\vartheta(w), \frac{2\vartheta(w) - 1 + \sqrt{g(w)}}{\rho}, w \right), \quad w \in [w_0, 1/\rho],$$

and on $(a, 1]$ by

$$D^{\text{loc}}(\theta) = \frac{2\theta - 1}{\rho}, \quad \psi_0^{\text{loc}}(\theta) = \frac{1}{\rho},$$

with the odd/even extension to negative types. The local constraint binds on $(-a, a)$, and the first-order schedule with first-best volatility applies outside. The rent is continuous, with $D'(a^-) = 1/\rho$ and $D'(a^+) = 2/\rho$.

Proof. On $[0, 1]$, it is without loss to restrict the initial volatility to

$$0 \leq \psi_0 \leq \frac{1}{\rho}.$$

The local incentive constraint is

$$\psi_0 d\theta \leq dD.$$

Let $\eta \geq 0$ be its multiplier. Stationarity requires

$$-\frac{\rho}{2}P'(\psi_0) = \eta, \quad \eta' = 2\theta - 1 - \rho D, \quad \eta(1) = 0. \quad (90)$$

Where the constraint is slack, $\eta = 0$, so

$$\psi_0 = 1/\rho, \quad D = (2\theta - 1)/\rho.$$

Where it binds, write $\psi_0 = D' =: w$ and $m := -\rho P'(w)/2 = \eta$. Then

$$m' = 2\theta - 1 - \rho D, \quad m'' = 2 - \rho w.$$

Multiplying the second equation by m' and integrating gives

$$(m')^2 = g(w). \quad (91)$$

The integration constant is zero because, at $\theta = a$, $\psi_0 = 1/\rho$ and $D = (2a-1)/\rho$, so $m = m' = 0$. At $\theta = 0$, oddness gives $D(0) = 0$ and hence $m'(0) = -1$, so $g(w_0) = 1$. Since

$$g'(w) = \rho P''(w)(\rho w - 2) < 0,$$

this root is unique.

Since m' increases from -1 at $\theta = 0$ to 0 at $\theta = a$, (91) gives

$$m' = -\sqrt{g(w)}, \quad \rho D - (2\theta - 1) = \sqrt{g(w)}.$$

Differentiation gives

$$\frac{d\theta}{dw} = \frac{\rho P''(w)}{2\sqrt{g(w)}},$$

which yields the stated parametrization and $D' = w$. The integral defining a is finite because, as $w \uparrow 1/\rho$,

$$P''(w) \sim 2(1/\rho - w), \quad g(w) \sim \rho(1/\rho - w)^2.$$

Moreover,

$$1 = m'(a) - m'(0) = \int_0^a (2 - \rho w) d\theta.$$

Since $0 < w < 1/\rho$ before the junction, $1/2 < a < 1$.

The multiplier $\eta = -\rho P'(w)/2$ is positive on $[0, a)$ and vanishes at a , so stationarity and complementary slackness hold. For any feasible pair, concavity and Lebesgue–Stieltjes integration by parts therefore give

$$\mathcal{J}(D, \psi_0) - \mathcal{J}(D^{\text{loc}}, \psi_0^{\text{loc}}) \leq \int_{[0,1]} \eta(\psi_0 d\theta - dD) \leq 0.$$

Thus $(D^{\text{loc}}, \psi_0^{\text{loc}})$ solves the local program. Strict concavity makes this maximizing pair unique up to null sets.

At $\theta = a$, the solution changes from the region in which the local constraint binds to

$$D^{\text{loc}}(\theta) = \frac{2\theta - 1}{\rho}, \quad \psi_0^{\text{loc}}(\theta) = \frac{1}{\rho}.$$

Consequently,

$$(D^{\text{loc}})'(a^-) = \frac{1}{\rho}, \quad (D^{\text{loc}})'(a^+) = \frac{2}{\rho}, \quad \psi_0^{\text{loc}}(a) = \frac{1}{\rho}.$$

Thus both one-sided date-zero conditions hold despite the kink. Finally, strict convexity of D on $(0, a)$, followed by the upward slope jump, makes $D(\theta)/\theta$ strictly increasing. Lemma 7 therefore shows that the finite-variation cap is $D(\theta)/\theta$, which is strictly below ψ_0 at every $0 < \theta < 1$. This proves the failure of the global date-zero condition stated in the main text. $\square$

H.4 Admissibility in the Exchange Setting

The initial report precedes the first allocation. Subsequently, trader i observes his own types, reports, and allocations. Let ω^{-i} denote the other traders' initial types and entire Brownian paths; these primitives are independent of trader i 's.

Definition 5 (Admissible exchange reports). *A reporting rule consists of an initial report y^i and a nonanticipating drift rule*

$$a_t^i = \alpha_t^i((\theta_s^i, \widehat{\theta}_s^i, X_s^i)_{0 \leq s \leq t}).$$

The resulting report and allocation equations must have a unique adapted solution. For almost every fixed opponents' primitive history, the induced own-history rule must be admissible under Definition 2. For every $x \in \Theta$ and $T < \infty$, also require, conditional on $\theta_0^i = x$,

$$\text{Law}_x(\widehat{\theta}_{[0,T]}^i, \omega^{-i}) \ll \text{Law}_{y^i}(\theta_{[0,T]}^i) \otimes \text{Law}(\omega^{-i}), \quad \mathbb{E}_x \int_0^\infty e^{-rt} |a_t^i|^2 dt < \infty.$$

Fixing the opponents' histories gives an admissible dealer report based on trader i 's own history. Reports that ignore allocations include the original dealer reports.

Definition 6 (Admissible exchange mechanisms). *An exchange mechanism assigns report-adapted allocations and transfers and satisfies, under truth and every admissible unilateral report,*

$$\mathbb{E}_x \int_0^\infty e^{-rt} ((X_t^j)^2 + |T_t^j|) dt < \infty, \quad j = 1, \dots, N.$$

Truthful ex ante integrals satisfy the same condition. For every i , its reduced dealer mechanism is also admissible under Definition 4. It is defined, for each own report path, by

$$\begin{aligned} Y_t^i(\widehat{\theta}^i) &:= \mathbb{E}_{-i}[X_t^i(\widehat{\theta}^i, \theta^{-i})], \\ \widetilde{T}_t^{i,\text{red}}(\widehat{\theta}^i) &:= \mathbb{E}_{-i} \left[T_t^i(\widehat{\theta}^i, \theta^{-i}) + \frac{1}{2} X_t^i(\widehat{\theta}^i, \theta^{-i})^2 \right], \\ T_t^{i,\text{red}} &:= \widetilde{T}_t^{i,\text{red}} - \frac{1}{2} (Y_t^i)^2. \end{aligned}$$

Definition 4 applies to $(Y^i, \tilde{T}^{i,\text{red}})$; $T^{i,\text{red}}$ is the actual payment. In class FV, this reduced mechanism satisfies the finite-variation continuation restrictions and (30). In class L, it satisfies the continuation and date-zero restrictions of the local program.

Membership in $\mathcal{E}_C$ additionally requires market clearing after every report profile, interim participation, and the selected incentive condition against every admissible unilateral report. For G and FV, the incentive condition is global incentive compatibility. For L, it is Definition 3 with allocation-dependent continuation directions.

H.5 Proof of Corollary 2.2

In class G, Corollary 2.2 requires optimal dealer implementations satisfying the following condition under every admissible allocation-dependent unilateral report in the construction below:

$$\mathbb{E}_x \int_0^\infty e^{-rt} \left[(X_t^{i*}(\hat{\theta}^i))^2 + |\tilde{T}_t^{i*}(\hat{\theta}^i)| \right] dt < \infty \quad \forall i, x. \quad (92)$$

The expectation integrates over the opponents' independent histories. The estimates below verify this condition for the optimal finite-variation implementations used in FV and L.

Proof. Fix a class and suppress its subscript. The case $N = 1$ is the zero exchange, so assume $N \geq 2$. Let $\mathcal{H}$ be the discounted L^2 space on the product of the traders' primitive spaces, with

$$\langle A, B \rangle = \mathbb{E} \int_0^\infty e^{-rt} A_t B_t dt, \quad \|A\|^2 = \langle A, A \rangle.$$

Let $\mathcal{H}_i$ be the closed subspace adapted to trader i 's own type filtration, and let $\mathcal{P}_i$ be its orthogonal projection. Denote by $\mathcal{V}_i \subseteq \mathcal{H}_i$ the dealer allocations implementable in the selected class.

Dealer optimization. Effective transfers make payoffs linear in allocations and transfers. Addition and nonnegative scaling therefore preserve participation and global or local incentive inequalities, as well as the finite-variation and date-zero restrictions. The corresponding raw transfers are

$$T^\oplus = T + T' - XX', \quad T^c = cT + \frac{c - c^2}{2} X^2.$$

Discounted square-integrability and Cauchy–Schwarz preserve transfer integrability. The remaining admissibility conditions are preserved by linearity and the triangle inequality. Thus $\mathcal{V}_i$ is a convex cone.

For an allocation Y , write

$$D_Y(x) = \mathbb{E}_x \int_0^\infty e^{-(r+\lambda)t} Y_t dt, \quad J_i(Y) = \ell_i(Y) - \frac{1}{2} \|Y\|^2, \\ \ell_i(Y) = \langle \theta^i, Y \rangle - \mathbb{E}[\Lambda(\theta_0^i) D_{\mathcal{P}_i Y}(\theta_0^i)].$$

The rent-pricing integral is the finite integral in (22); all uses below concern feasible reduced allocations or their finite linear combinations. The revenue of a dealer mechanism with allocation Y is $J_i(Y) - U(0)$.

Reflect the initial type, Brownian path, report, and allocation simultaneously. This maps $\mathcal{V}_i$ into itself, preserves ℓ_i , and preserves the norm. Averaging a mechanism with its reflection makes its rent odd and nondecreasing, so utility is minimized at zero. Its utility at zero can then be normalized to zero. Consequently, every $Y \in \mathcal{V}_i$ has a symmetric feasible counterpart whose dealer revenue is at least $J_i(Y)$. It follows that a dealer optimum maximizes J_i on $\mathcal{V}_i$. Strict concavity makes its allocation unique and reflection-symmetric. In particular,

$$\mathbb{E}[X_t^{i*}] = 0 \quad \text{for almost every } t, \quad U^{i*}(0) = 0, \quad \Pi_i^* = J_i(X^{i*}). \quad (93)$$

The variational inequality and its radial equality are

$$\ell_i(Y - X^{i*}) - \langle X^{i*}, Y - X^{i*} \rangle \leq 0 \quad (Y \in \mathcal{V}_i), \\ \ell_i(X^{i*}) = \|X^{i*}\|^2. \quad (94)$$

The second relation follows by taking $Y = 0$ and $Y = 2X^{i*}$.

The exchange upper bound. Consider any feasible exchange mechanism. A report that ignores allocations has the same expected payoff in that mechanism and in its reduced dealer mechanism: substitute the definitions in Definition 6 and apply conditional Fubini. The reduced mechanism therefore inherits participation and the relevant incentive condition. Under truth,

$$Y^i = \mathcal{P}_i X^i \in \mathcal{V}_i.$$

Since $\ell_i(X^i) = \ell_i(Y^i)$, orthogonality and (94) imply

$$J_i(X^i) \leq J_i(X^{i*}) - \frac{1}{2} \|X^i - X^{i*}\|^2.$$

The envelope and participation bound exchange revenue by $\sum_i J_i(X^i)$. Write $\bar{X}^* = N^{-1} \sum_j X^{j*}$. Market clearing gives

$$\sum_i \|X^i - X^{i*}\|^2 = N\|\bar{X}^*\|^2 + \sum_i \|X^i - (X^{i*} - \bar{X}^*)\|^2.$$

Hence

$$\Pi^{\text{ex}} \leq \sum_i \Pi_i^* - \frac{N}{2} \|\bar{X}^*\|^2. \quad (95)$$

Allocations and transfers attaining the bound. Set

$$c = 1 - \frac{1}{N}, \quad q^i = \frac{1}{N} \sum_{j \neq i} X^{j*}, \quad \hat{X}^i = cX^{i*} - q^i,$$

and, using dealer implementations normalized by $U^{i*}(0) = 0$, define actual exchange payments by

$$\hat{T}^i = c\tilde{T}^{i*} - \frac{1}{2}(\hat{X}^i)^2. \quad (96)$$

Allocations clear after every report profile. The payoff identity is

$$\theta^i \hat{X}^i - \frac{1}{2}(\hat{X}^i)^2 - \hat{T}^i = c(\theta^i X^{i*} - \tilde{T}^{i*}) - \theta^i q^i. \quad (97)$$

When opponents are truthful, q^i is independent of trader i 's primitives, has mean zero, and is unaffected by his report. Thus interim truthful utility is cU^{i*} , and the reduced rent is cD^{i*} . Participation holds and binds at zero. Independence, mean zero, and the radial equality in (94) give

$$\ell_i(\bar{X}^*) = \langle X^{i*}, \bar{X}^* \rangle = \frac{1}{N} \|X^{i*}\|^2.$$

Therefore

$$\sum_i J_i(\hat{X}^i) = \sum_i \Pi_i^* - \frac{N}{2} \|\bar{X}^*\|^2.$$

The base utilities are zero, so this is the candidate's revenue.

Reports that use observed allocations. Under the candidate, trader i can recover $q^i = cX^{i*} - \hat{X}^i$ from his own report and allocation. Condition on the opponents' primitive histories. The process q^i is then fixed, and every admissible report induces a dealer report with the properties of Definition 2. The final term in (97) is independent of that report. In class G, conditional deviation gains are nonpositive at every initial type. Joint payoff integrability and Fubini

permit averaging over opponents, proving global incentive compatibility at every initial type.

For both FV and L, select the deterministic continuation of Proposition 4 at the optimal date-zero choices. Its value equals the optimal continuation value, and strict concavity of the dealer objective makes its truthful allocation equal to X^{i*} . Its volatility lies in $[0, 1/\rho]$ and has finite variation along every report. In class FV, the dealer mechanism is globally incentive compatible by Theorems 1a and 1b; the preceding conditional argument proves the same property in the exchange.

In class L, (55) supplies the uniform finite-deviation bound

$$U_*^x(y, b) - U^*(x) \leq -\mathcal{B}(x, y) + \frac{1}{2}\psi_0^*(y)(x - y)^2 \quad (98)$$

for every continuation report b . Its right-hand side does not depend on that report or the opponents' histories. The local restrictions say that D^* is nondecreasing and $dD^* \geq \psi_0^*(l)dl$; therefore

$$\mathcal{B}(x, y) \geq \frac{1}{2}(x - y)^2 \inf_{l \in [x \wedge y, x \vee y]} \psi_0^*(l).$$

This follows by integrating the measure inequality first between y and l , then between y and x , with the appropriate orientation. It applies also at a kink or a jump of D^* , taking its monotone representative. Continuity of ψ_0^* , required by admissibility, bounds the right-hand side of (98) above by $o((x - y)^2)$. For $y = x + \varepsilon h$, average the finite-deviation inequality over the opponents before dividing by ε^2 and taking the limsup. Equation (97) then proves the local incentive condition for every allocation-dependent direction. When $h = 0$, the finite-deviation bound is already nonpositive.

Admissibility of the construction. For the deterministic implementations in FV and L, conditional on a fixed initial report y ,

$$d\hat{\theta}_t = -\lambda\hat{\theta}_t dt + dB_t + a_t dt, \quad dD_t = -\lambda D_t dt + \psi_t(y)(dB_t + a_t dt).$$

Here $0 \leq \psi_t(y) \leq 1/\rho$ is deterministic. The weighted convolution estimate in Appendix H.2 applies directly on the product probability space. It bounds the allocation and transfer integrals by a finite constant depending on $(x, y, D(y))$ times $1 + \mathbb{E}_x \int_0^\infty e^{-rt} a_t^2 dt$. Thus (92) holds in these two classes. In G, it holds by the corollary's hypothesis.

For each component, Cauchy–Schwarz gives

$$|\hat{X}^k|^2 \leq 2c^2 |X^{k*}|^2 + \frac{2(N-1)}{N^2} \sum_{j \neq k} |X^{j*}|^2, \quad |\hat{T}^k| \leq c|\tilde{T}^{k*}| + \frac{1}{2}|\hat{X}^k|^2.$$

These estimates establish all joint allocation and transfer integrals in Definition 6. Averaging over truthful opponents gives the reduced allocation cX^{i*} and effective transfer $c\tilde{T}^{i*}$. Nonnegative scaling preserves dealer admissibility and the selected class restrictions. The candidate is therefore feasible and attains (95). $\square$

H.6 Optimal Static Mechanism

Proposition 11 (The static benchmark). *Under the assumptions of Proposition 3, the optimal static mechanism has*

$$X_t = (\theta_0 - \bar{\Lambda}(\theta_0))e^{-\lambda t} = \mathbb{E}[X_t^{\text{FOA}} \mid \theta_0]$$

and the rent in (31). Its revenue is

$$\Pi_{\text{stat}} = \frac{\mathbb{E}[(\theta_0 - \bar{\Lambda}(\theta_0))^2]}{2\rho} = \Pi^{\text{FOA}} - \frac{1}{2r\rho}.$$

Proof. Recall from the proof of Proposition 3 that $\rho := r + 2\lambda$ and

$$\mathcal{S}(D, u) := -\frac{\rho}{2}\mathbb{E}\left[\left(D(\theta_0) - \frac{\theta_0}{\rho}\right)^2\right] - \mathbb{E}[U(\theta_0)], \quad U(\theta) = u + \int_{x^*}^{\theta} D(l) \, dl.$$

If report m selects the allocation path $x(m)$, define

$$Z(m) := \int_0^\infty e^{-(r+\lambda)t} x_t(m) \, dt, \quad C(m) := \frac{1}{2} \int_0^\infty e^{-rt} x_t(m)^2 \, dt.$$

Since $\mathbb{E}[\theta_t \mid \theta_0] = \theta_0 e^{-\lambda t}$, the expected gross utility of type θ_0 from this path is $\theta_0 Z(m) - C(m)$. Static incentive compatibility is equivalent to convexity of U and $Z(m) \in \partial U(m)$. Thus Z is nondecreasing and $U' = Z$ almost everywhere. Conversely, the usual payment identity sets the discounted payment following report m equal to $mZ(m) - C(m) - U(m)$, so these conditions are also sufficient. Writing $D := Z$, expected revenue is therefore

$$\mathbb{E}[\theta_0 D(\theta_0) - C(\theta_0) - U(\theta_0)].$$

For a fixed D , Cauchy–Schwarz gives

$$D^2 \leq \frac{1}{\rho} \int_0^\infty e^{-rt} x_t^2 \, dt,$$

with equality exactly when $x_t = \rho D e^{-\lambda t}$ almost everywhere. This path therefore minimizes the allocation cost required to deliver D . Substituting its cost gives

$$\theta_0 D - \frac{\rho}{2} D^2 = \frac{\theta_0^2}{2\rho} - \frac{\rho}{2} \left(D - \frac{\theta_0}{\rho} \right)^2.$$

After optimizing over allocation paths, the static problem is

$$\Pi_{\text{stat}} = \frac{\mathbb{E}[\theta_0^2]}{2\rho} + \sup_{D,u} \mathcal{S}(D, u),$$

where the supremum is over nondecreasing D satisfying the envelope condition and participation.

Equation (78) in the proof of Proposition 3 was derived using only the envelope condition and participation, without imposing monotonicity on D . It therefore gives, for every feasible static pair,

$$\mathcal{S}(D, u) \leq \frac{\mathbb{E}[(\theta_0 - \bar{\Lambda}(\theta_0))^2] - \mathbb{E}[\theta_0^2]}{2\rho}.$$

The proof of Proposition 3 shows that equality holds at $u = 0$ and

$$D^{\text{FOA}}(\theta) = \frac{\theta - \bar{\Lambda}(\theta)}{\rho}.$$

Regularity makes D^{FOA} nondecreasing, so this pair is feasible for the static problem and attains its upper bound. Its least-cost allocation is

$$X_t = \rho D^{\text{FOA}}(\theta_0) e^{-\lambda t} = (\theta_0 - \bar{\Lambda}(\theta_0)) e^{-\lambda t} = \mathbb{E}[X_t^{\text{FOA}} \mid \theta_0],$$

where the last equality follows from (32). Finally,

$$\Pi_{\text{stat}} = \frac{\mathbb{E}[(\theta_0 - \bar{\Lambda}(\theta_0))^2]}{2\rho} = \Pi^{\text{FOA}} - \frac{1}{2r\rho},$$

by (33). □

H.7 Numerical Details

The uniform entries in Table 1 evaluate the characterizations in Appendix H.3. The Gaussian entries solve the corresponding date-zero programs on a symmetric grid. This subsection records the objective, constraints, and numerical normalizations.

For evaluating the continuation penalty, put $z = \rho\psi$ and $\alpha = r/\rho$. Direct integration gives

$$P(\psi) = \begin{cases} \frac{1 - z^2 + 2z \log z}{r^3}, & \lambda = 0, \\ \frac{1}{\rho^2} \left[\frac{1 - z^\alpha}{r} - \frac{z^\alpha - z}{\lambda} + \frac{z^\alpha - z^2}{r + 4\lambda} \right], & \lambda > 0, \end{cases} \quad 0 < z \leq 1. \quad (99)$$

The continuous endpoint value is $P(0) = 1/(r\rho^2)$, and $P(\psi) = 0$ for $\psi \geq 1/\rho$.

Let $\theta_1 < \dots < \theta_n$ be an equally spaced grid containing zero, and let w_i be density-weighted trapezoidal weights, normalized to sum to one. The uniform grid covers $[-1, 1]$; the standard normal grid covers $[-4, 4]$, which omits probability approximately 6.3×10^{-5} . For the Gaussian density, the inverse-hazard weight can be evaluated stably as

$$\Lambda(\theta) = \text{sgn}(\theta) \sqrt{\frac{\pi}{2}} \text{erfcx} \left(\frac{|\theta|}{\sqrt{2}} \right),$$

where $\text{erfcx}(z) = e^{z^2} \text{erfc}(z)$.

The variables (D_i, ψ_i) maximize

$$\sum_i w_i \left[-\frac{\rho}{2} \left(D_i - \frac{\theta_i}{\rho} \right)^2 - \frac{\rho}{2} P(\psi_i) - \Lambda(\theta_i) D_i \right].$$

The implementation imposes $D(0) = 0$, monotonicity of D_i , and $0 \leq \psi_i \leq 1/\rho$. Symmetry makes utility minimized at zero, where participation binds. If A_i is the trapezoidal antiderivative of D at θ_i , the finite-variation constraints are

$$A_i - A_j - D_j(\theta_i - \theta_j) \geq \frac{1}{2} \psi_j (\theta_i - \theta_j)^2, \quad i \neq j.$$

There are $n(n-1)$ such linear constraints. The local program replaces them by

$$\psi_i \leq \frac{D_{i+1} - D_{i-1}}{2\Delta\theta}, \quad i = 2, \dots, n-1.$$

Both are concave maximization problems with linear constraints.

The companion computations use a trust-region constrained solver, analytic gradients and Hessians, and a gradient tolerance of 10^{-10} . The recorded refinements use $n = 81, 161$, with $n = 321$ also used for the local program. The uniform local values are 0.54363, 0.54356, 0.54354 at these three resolutions; their second-order extrapolation is 0.54353. The reported Gaussian local values converge similarly to the levels shown in the table.

To recover total revenue, add

$$\frac{\mathbb{E}[\theta_0^2]}{2\rho} + \frac{1}{2r\rho}, \quad \mathbb{E}[\theta_0^2] = \begin{cases} 1/3, & F \text{ uniform on } [-1, 1], \\ 1, & F \text{ standard normal.} \end{cases}$$

The static and first-order rows are evaluated from Appendix H.6 and Proposition 3. For uniform types,

$$\Pi_{\text{stat}} = \frac{1}{12\rho}, \quad \Pi^{\text{FOA}} = \frac{1}{12\rho} + \frac{1}{2r\rho}.$$

For the standard normal, numerical quadrature gives

$$x^* = 0.7517915\dots, \quad \Pi_{\text{stat}} = \frac{0.2061911\dots}{\rho},$$

$$\Pi^{\text{FOA}} = \frac{0.2061911\dots}{\rho} + \frac{1}{2r\rho}.$$

At $r = 1$, these values and the reported local levels give the static and first-order percentages in Table 1.

I Geometric Brownian Motion

This appendix supplies the admissibility conditions and proofs for Section 7.1. The relative reporting gap replaces the OU gap, and the relevant supermartingale is $A_t = e^{-rt}\hat{\theta}_t\psi_t$. I also give the discrete-time calculation and the control formulation with allocation cost C .

I.1 Admissible Reports

Use the probability space, information, and payoff conventions of Section 4, with

$$d\theta_t = \mu\theta_t dt + \gamma\theta_t dB_t, \quad \Theta \subseteq (0, \infty), \quad r > \mu.$$

A report (y, a) starts at $y \in \Theta$ and satisfies

$$d\hat{\theta}_t = \mu\hat{\theta}_t dt + \gamma\hat{\theta}_t d\hat{B}_t, \quad d\hat{B}_t = dB_t + a_t dt.$$

For actual initial type x , define

$$\begin{aligned}
z_t &:= \log \frac{\hat{\theta}_t}{\theta_t} = \log \frac{y}{x} + \gamma \int_0^t a_s \, ds, \\
\delta_t &:= \hat{\theta}_t - \theta_t, & \phi_t &:= \frac{\delta_t}{\hat{\theta}_t} = 1 - e^{-z_t}, \\
H(z) &:= z - 1 + e^{-z}, & \Psi_t &:= H(z_t) = -\phi_t - \log(1 - \phi_t).
\end{aligned} \tag{100}$$

The nonnegative process Ψ has continuous finite variation, and

$$d\phi_t = \gamma(1 - \phi_t)a_t \, dt, \quad d\Psi_t = \gamma\phi_t a_t \, dt. \tag{101}$$

Fix $q > 1$ and $\eta > 0$, independently of the mechanism.

Definition 7 (Admissible GBM reports). *A reporting strategy (y, a) is admissible if a is progressively measurable and, for every $x \in \Theta$:*

(G1) *On every finite horizon, its induced report law under $\mathbb{P}_x$ is absolutely continuous with respect to the truthful GBM law starting from y .*

(G2) *With z as in (100),*

$$\mathbb{E}_x \left[\sup_{t \geq 0} e^{-\eta t} (e^{qz_t} + e^{-qz_t}) + \int_0^\infty e^{-\eta t} (1 + a_t^2) (e^{qz_t} + e^{-qz_t}) \, dt \right] < \infty. \tag{102}$$

These conditions depend only on reports. They admit truthful reporting and every bounded distortion with bounded time support, including all the short corrections used below. Feasible directions and local incentive compatibility have the meaning in Definition 3, using this reporting domain.

I.2 Admissible Mechanisms

The functionals D , W , and $\psi = \text{vol}(D)$ are defined under truthful continuation at each report history and evaluated along the induced report under a deviation. Write $g_t := \text{vol}(W_t)$. When $A = e^{-rt}\hat{\theta}\psi$ has a semimartingale representative, write $\nu := \text{vol}(A)$ for its Brownian diffusion coefficient, evaluated along the same report.

Definition 8 (Admissible GBM mechanisms). *A mechanism $M = (X, T)$ is admissible if:*

(GM1) *Under truthful reporting,*

$$\int_0^\infty e^{-rt} \theta_t X_t dt \in L^2, \quad \frac{1}{\theta_0} \int_0^\infty e^{-rt} \theta_t X_t dt \in L^2.$$

(GM2) *For every $x \in \Theta$ and admissible report (y, a) ,*

$$\mathbb{E}_x \int_0^\infty e^{-rt} (|\theta_t \hat{X}_t| + |\hat{T}_t|) dt < \infty.$$

(GM3) *Under truthful continuation from every $y \in \Theta$, A has a predictable representative satisfying*

$$\begin{aligned} \mathbb{E}_y \int_0^T A_t^2 dt &< \infty \quad (T < \infty), \\ A_t &\longrightarrow A_0(y) = y\psi_0(y) \quad \text{in } L^1 \quad (t \downarrow 0), \end{aligned}$$

where $A_0(y)$ is finite and deterministic. Under every admissible report, the process

$$\int_0^\cdot e^{-rt} (g_t - \gamma D_t \delta_t - \psi_t \delta_t) dB_t$$

is a true martingale on every finite horizon. Whenever A has a semimartingale representative, the same is true of

$$\int_0^\cdot \Psi_t \nu_t dB_t.$$

(GM4) *For every $x \in \Theta$ and admissible report (y, a) , $W_T - D_T \delta_T$ is integrable for every $T < \infty$, and*

$$\mathbb{E}_x [e^{-rT} (W_T - D_T \delta_T)] \longrightarrow 0, \quad \mathbb{E}_x \int_0^\infty |A_t \phi_t a_t| dt < \infty.$$

Whenever A has a semimartingale representative, also require

$$\mathbb{E}_x \int_0^\infty \Psi_t |\nu_t a_t| dt < \infty.$$

(GM5) *Couple truthful continuations from different initial reports using the same Brownian innovations, and denote them by $A^{(y,0)}$. For every $x \in \Theta$ and $T < \infty$,*

$$\begin{aligned} A_0(y) &\longrightarrow A_0(x), \\ \mathbb{E}_x \int_0^T |A_t^{(y,0)} - A_t^{(x,0)}|^2 dt &\longrightarrow 0 \quad (y \rightarrow x \text{ within } \Theta). \end{aligned}$$

Whenever A has a semimartingale representative, every feasible direction (h, a) at x also satisfies

$$\mathbb{E}_x \int_0^\infty \Psi_t^{(\varepsilon, h, a)} |\nu_t^{(\varepsilon, h, a)} \varepsilon a_t| dt = o(\varepsilon^2), \quad (103)$$

where the superscripts denote evaluation along $(x + \varepsilon h, \varepsilon a)$.

These are the GBM versions of (M1)–(M5). The two martingale conditions justify expectations in the payoff and integration-by-parts identities. The infinite-horizon conditions control their tails. Continuity and (103) justify the local calculations. If A has finite variation, $\nu = 0$, so its additional martingale, integrability, and small-deviation conditions hold automatically. Multiplication by e^{rt} gives equivalent conditions in the state $p_t = e^{rt} A_t$ used in the control problem.

I.3 Discrete-Time Preview: Linear Mechanisms

The GBM calculation uses linearity of gross continuation utility. Let $\beta = e^{-r\Delta t}$, $m = 1 + \mu\Delta t > 0$, and

$$\theta_{t+1} = \theta_t(m + \gamma\varepsilon_{t+1}), \quad \mathbb{E}\varepsilon_{t+1} = 0, \quad \mathbb{E}\varepsilon_{t+1}^2 = \Delta t.$$

Take independent innovations with $m + \gamma\varepsilon_{t+1} > 0$ almost surely. This preserves the positive type space. Define

$$D_t = \mathbb{E}_t \sum_{s=t}^\infty \beta^{s-t} \frac{\theta_s}{\theta_t} X_s \Delta t, \quad \hat{\varepsilon}_{t+1} = \frac{\hat{\theta}_{t+1} - m\hat{\theta}_t}{\gamma\hat{\theta}_t}.$$

The counterpart of Definition 1 is a loading b_t , measurable with respect to reports through period t , such that, along every report path,

$$\hat{\theta}_{t+1} D_{t+1} = \frac{\hat{\theta}_t D_t - \hat{\theta}_t X_t \Delta t}{\beta} + b_t \hat{\varepsilon}_{t+1}. \quad (104)$$

Write $q_t := b_t/(\gamma\hat{\theta}_t)$. The loading b_t belongs to gross continuation utility; it is not the marginal-rent volatility ψ_t .

Fix a truthful history through $t - 1$, a current report y , and a possible true type l . In this paragraph $D_t(y)$, $X_t(y)$, and $q_t(y)$ are evaluated at that reported history. The innovations inferred after reporting y and then reporting truthfully are

$$\hat{\varepsilon}_t = \varepsilon_t + \frac{y - l}{\gamma\theta_{t-1}}, \quad \hat{\varepsilon}_{t+1} = \frac{l}{y}\varepsilon_{t+1} + \frac{m(l - y)}{\gamma y}.$$

Equation (104), first at $t - 1$ and then at t , gives

$$lD_t(l) - yD_t(y) = q_{t-1}(l - y),$$

and

$$\begin{aligned} D_t(y, l) &= X_t(y)\Delta t + \frac{\beta}{l}\mathbb{E}_l[\theta_{t+1}D_{t+1} \mid \hat{\theta}_t = y] \\ &= \frac{y}{l}D_t(y) + \frac{l - y}{l}[X_t(y)\Delta t + \beta m_{q_t}(y)]. \end{aligned}$$

Consequently,

$$D_t(l, l) - D_t(y, l) = [q_{t-1} - \beta m_{q_t}(y) - X_t(y)\Delta t] \frac{l - y}{l}. \quad (105)$$

Because

$$J(y, x) := \int_y^x \frac{l - y}{l} dl = x - y - y \log \frac{x}{y} \geq 0,$$

with strict inequality when $x \neq y$, integral monotonicity at $t \geq 1$ is equivalent to

$$q_{t-1} \geq \beta m_{q_t}(y) + X_t(y)\Delta t. \quad (106)$$

At date zero, the same conditional expectation gives

$$D_0(y, l) = D(y) + [X_0(y)\Delta t + \beta m_{q_0}(y) - D(y)] \frac{l - y}{l}.$$

Thus date-zero integral monotonicity requires

$$\int_y^x \left[D(l) - D(y) - (X_0(y)\Delta t + \beta m_{q_0}(y) - D(y)) \frac{l - y}{l} \right] dl \geq 0. \quad (107)$$

To compare with continuous time, put $Y_t = \beta^{t+1}q_t$. Then (106) is

$$Y_{t-1} \geq (1 + \mu\Delta t)Y_t + \beta^t X_t \Delta t.$$

Its finite-variation limit is $dY_t + \mu Y_t dt \leq -e^{-rt} X_t dt$. The continuous-time identity below is $Y_t = e^{-rt}(D_t + \psi_t/\gamma)$; when Y has finite variation, the product rule gives

$$dA_t = \gamma \hat{\theta}_t [dY_t + (\mu Y_t + e^{-rt} X_t) dt] \leq 0.$$

Also $q_0 - D \rightarrow \psi_0/\gamma$, so (107) becomes (38). These calculations identify the continuous-time restrictions when the stated finite-difference limits exist.

I.4 Marginal Rent and Its Volatility

The impulse response is θ_s/θ_t , giving (36). At a truthful reported history, gross continuation utility is therefore exactly

$$W_t^X = \hat{\theta}_t D_t = \mathbb{E}_t \int_t^\infty e^{-r(s-t)} \hat{\theta}_s X_s ds. \quad (108)$$

Hence

$$e^{-rt} \hat{\theta}_t D_t + \int_0^t e^{-rs} \hat{\theta}_s X_s ds$$

is a martingale. Its Brownian representation and the product rule give

$$dD_t = [(r - \mu)D_t - X_t - \gamma\psi_t]dt + \psi_t d\hat{B}_t, \quad (109)$$

$$d(e^{-rt} \hat{\theta}_t D_t) = -e^{-rt} \hat{\theta}_t X_t dt + e^{-rt} \hat{\theta}_t (\gamma D_t + \psi_t) d\hat{B}_t. \quad (110)$$

In particular, $\text{vol}(W_t^X) = \hat{\theta}_t(\gamma D_t + \psi_t)$, and the rescaled gross-rent loading used above is $Y_t = e^{-rt}(D_t + \psi_t/\gamma)$.

The GBM transfer construction is Proposition 2 with the required volatility replaced by $\gamma\hat{\theta}_t D_t$. Explicitly, put

$$H_t^G := \gamma\hat{\theta}_t D_t - \text{vol}(W_t^X) = -\hat{\theta}_t \psi_t.$$

When H^G is square-integrable on every finite horizon under every admissible report and the resulting mechanism is admissible, the transfers

$$T_t = r \left[yD(y) - \int_c^y D(l) dl - U_c \right] - r \int_0^t H_s^G d\hat{B}_s, \quad y = \hat{\theta}_0, \quad (111)$$

satisfy the two transfer conditions of Theorem 3. Indeed, the martingale $N_t = \int_0^t H_s^G d\hat{B}_s$ makes the truthful continuation value of these transfers equal to their constant bracket minus N_t . Thus $\text{vol}(W) = \text{vol}(W^X) + H^G = \gamma\hat{\theta}D$, and expected discounted transfers equal the bracket. Subtracting that bracket from $W_0^X(y) = yD(y)$ gives the envelope formula. As in the OU construction, admissibility supplies the infinite-horizon integrability for this calculation.

I.5 Implementability

Lemma 8 (GBM deviation formula). *For an admissible GBM mechanism satisfying $\text{vol}(W_t) = \gamma\hat{\theta}_t D_t$, every admissible report satisfies*

$$U_x(y, a) = U(y) + (x - y)D(y) - \mathbb{E}_x \int_0^\infty A_t \phi_t a_t dt. \quad (112)$$

Proof. When distortion stops at t , the gap obeys $\delta_s = (\hat{\theta}_s/\hat{\theta}_t)\delta_t$ for $s \geq t$, so the true continuation payoff is $W_t - D_t\delta_t$. Under the deviation,

$$d\delta_t = (\mu\delta_t + \gamma\hat{\theta}_ta_t)dt + \gamma\delta_t dB_t.$$

Apply Itô's formula to

$$G_t = \int_0^t e^{-rs}(\theta_s\hat{X}_s - \hat{T}_s) ds + e^{-rt}(W_t - D_t\delta_t).$$

The covariation in $D\delta$ is $\gamma\psi_t\delta_t dt$. Together with (109), it cancels the terms independent of a , leaving

$$\begin{aligned} dG_t &= e^{-rt}[(g_t - \gamma\hat{\theta}_tD_t)a_t - \psi_t\delta_ta_t]dt \\ &\quad + e^{-rt}(g_t - \gamma D_t\delta_t - \psi_t\delta_t)dB_t. \end{aligned} \tag{113}$$

The first-order term is zero by the transfer condition, and $e^{-rt}\psi_t\delta_t = A_t\phi_t$. Taking expectations using (GM3), then letting $t \rightarrow \infty$ using (GM2)–(GM4), proves (112). $\square$

Lemma 9 (GBM gain decomposition). *Under the preceding hypotheses, suppose A has a nonnegative càdlàg supermartingale representative under truthful continuation from y . Write its localized Doob–Meyer decomposition as*

$$dA_t = -dL_t + \nu_t d\hat{B}_t, \quad L_0 = 0,$$

where L is predictable and nondecreasing. Then $\mathbb{E}_x \int_0^\infty \Psi_t dL_t < \infty$, the limit $\ell = \lim_{T \rightarrow \infty} \mathbb{E}_x[A_T\Psi_T] \geq 0$ exists, and

$$\begin{aligned} U_x(y, a) - U(y) - (x - y)D(y) &= \frac{A_0(y)}{\gamma} H\left(\log \frac{y}{x}\right) - \frac{1}{\gamma} \mathbb{E}_x \int_0^\infty \Psi_t dL_t - \frac{\ell}{\gamma} \\ &\quad + \frac{1}{\gamma} \mathbb{E}_x \int_0^\infty \Psi_t \nu_t a_t dt. \end{aligned} \tag{114}$$

Proof. Absolute continuity of report laws transfers the decomposition and the pathwise inequalities to the induced report. Because Ψ is continuous with finite variation, integration by parts gives

$$\gamma \int_0^T A_t \phi_t a_t dt = A_T \Psi_T - A_0(y) \Psi_0 + \int_0^T \Psi_t dL_t - \int_0^T \Psi_t \nu_t dB_t - \int_0^T \Psi_t \nu_t a_t dt.$$

Localize first if necessary. The two terms on the right involving $A_T\Psi_T$ and $\int \Psi dL$ are nonnegative; (GM3)–(GM4) make their sum integrable and allow

expectations in this identity. The master integral and the last integral converge absolutely in expectation by (GM4). Consequently the sum of the two nonnegative expectations has a finite limit. The compensator term is increasing in T , so both expectations converge separately. Substitute these limits in (112). $\square$

Proof of Theorem 3(i). Suppose first that the supermartingale representative exists. For a truthful initial report, $\Psi_0 = 0$. Discarding the two nonpositive terms in (114) and using (103) gives, for every feasible direction $(0, a)$,

$$\limsup_{\varepsilon \downarrow 0} \frac{U_x(x, \varepsilon a) - U(x)}{\varepsilon^2} \leq 0.$$

If A has finite variation, its continuous local martingale part vanishes, so $\nu = 0$. Equation (114) then gives $U_x(x, a) \leq U(x)$ for every admissible a .

For necessity, initially treat A as the predictable equivalence class in (GM3). Let $\mathcal{L}$ be its right $L^2(\Omega)$ -Lebesgue points, a set of full Lebesgue measure. Fix $s < t$ in $\mathcal{L}$, $E \in \mathcal{F}_s$, and $0 < \tau < (t - s)/2$. Let b rise linearly from zero to one on $(s, s + \tau]$, remain one until t , and fall linearly to zero on $(t, t + \tau]$. Use the bounded direction

$$a_u = \frac{\mathbf{1}_E}{\gamma\tau} (\mathbf{1}_{(s, s+\tau]}(u) - \mathbf{1}_{(t, t+\tau]}(u)).$$

The report is truthful through s , so E is also an event of the reported history through s . The scaled report has $z_u = \varepsilon \mathbf{1}_E b_u$. The master formula is therefore

$$\frac{U_x(x, \varepsilon a) - U(x)}{\varepsilon^2} = -\frac{1}{\gamma} \mathbb{E}_x \int_0^{t+\tau} A_u^{(x, \varepsilon a)} d \left[\frac{H(\varepsilon \mathbf{1}_E b_u)}{\varepsilon^2} \right].$$

Uniformly for $0 \leq b \leq 1$, $H(\varepsilon b)/\varepsilon^2 \rightarrow b^2/2$, together with its derivative in b . The proof of Lemma 4 applies to GBM without change: for this bounded distortion, the density of the report law has L^2 distance $o(1)$ from one. Condition (GM3) and Cauchy–Schwarz thus give the limit

$$-\frac{1}{2\gamma} \mathbb{E}_x \left[\mathbf{1}_E \left(\frac{2}{\tau^2} \int_s^{s+\tau} (u - s) A_u du - \frac{2}{\tau^2} \int_t^{t+\tau} (t + \tau - u) A_u du \right) \right].$$

Local incentive compatibility makes this expression nonpositive. Letting $\tau \downarrow 0$ at the two Lebesgue points gives

$$\mathbb{E}_x[\mathbf{1}_E(A_t - A_s)] \leq 0.$$

Using just the first ramp, with the log gap left constant afterward, instead gives $\mathbb{E}_x[\mathbf{1}_E A_s] \geq 0$. Since E was arbitrary, $A_s \geq 0$ almost surely and $\mathbb{E}_x[A_t | \mathcal{F}_s] \leq$

A_s for $s < t$ in $\mathcal{L}$. Lemma 5 supplies the required càdlàg representative, and the L^1 limit in (GM3) supplies its initial value. Global incentive compatibility implies the local test, completing the finite-variation equivalence. $\square$

Proof of Theorem 3(ii). Fix x and a feasible direction (h, a) , and put $y_\varepsilon = x + \varepsilon h$. The envelope condition gives

$$U(y_\varepsilon) + (x - y_\varepsilon)D(y_\varepsilon) - U(x) = -\frac{\varepsilon^2 h^2}{2} D'(x) + o(\varepsilon^2). \quad (115)$$

Moreover,

$$\frac{1}{\varepsilon^2} H\left(\log \frac{y_\varepsilon}{x}\right) \longrightarrow \frac{h^2}{2x^2}, \quad A_0(y_\varepsilon) \longrightarrow A_0(x).$$

Apply (114) from the report y_ε , discard its nonpositive terms, and use (GM5). Then

$$\limsup_{\varepsilon \downarrow 0} \frac{U_x(y_\varepsilon, \varepsilon a) - U(x)}{\varepsilon^2} \leq \frac{h^2}{2} \left[\frac{A_0(x)}{\gamma x^2} - D'(x) \right]. \quad (116)$$

Since $A_0(x) = x\psi_0(x)$, condition (37) proves local sufficiency.

For local necessity, fix a nonzero feasible h and, for $\tau > 0$, use

$$a_t^\tau = -\frac{h}{\gamma x \tau} \mathbf{1}_{(0, \tau]}(t).$$

On $[0, \tau]$, the scaled report satisfies

$$z_t^\varepsilon = \log \frac{x + \varepsilon h}{x} - \frac{\varepsilon h t}{x \tau}, \quad \frac{\phi_t^\varepsilon}{\varepsilon} \longrightarrow \frac{h}{x} \left(1 - \frac{t}{\tau}\right)$$

uniformly. The residual log gap at τ is $O(\varepsilon^2)$; it adds no master-integral term because the distortion has stopped. For fixed τ , (GM3)–(GM5) imply

$$\mathbb{E}_x \int_0^\tau w_t A_t^{(y_\varepsilon, \varepsilon a^\tau)} dt \longrightarrow \mathbb{E}_x \int_0^\tau w_t A_t^{(x, 0)} dt \quad (w \in L^2[0, \tau]).$$

To verify this limit, change from the distorted report law to the truthful law from y_ε . The density has L^2 distance $o(1)$ from one, uniformly in the initial report. Cauchy–Schwarz and (GM5) bound the weighted truthful integrals in L^2 and make them continuous in the initial report. The same estimate handles the uniformly convergent weight $\phi^\varepsilon/\varepsilon$. Using the master formula and (115), the directional gain therefore converges to

$$-\frac{h^2}{2} D'(x) + \frac{h^2}{\gamma x^2 \tau} \mathbb{E}_x \int_0^\tau A_t^{(x, 0)} \left(1 - \frac{t}{\tau}\right) dt.$$

The L^1 limit at zero in (GM3), followed by $\tau \downarrow 0$, makes this expression $\frac{h^2}{2}[A_0(x)/(\gamma x^2) - D'(x)]$. Local incentive compatibility requires it to be non-positive. At an endpoint, choose h pointing into Θ ; the same calculation uses the one-sided derivative.

For global sufficiency, assume that A has finite variation after every truthful initial report. Then $\nu = 0$, and (114) implies

$$\begin{aligned} U_x(y, a) - U(x) &\leq - \int_y^x [D(l) - D(y)] dl + \frac{A_0(y)}{\gamma} H\left(\log \frac{y}{x}\right) \\ &= - \int_y^x \left[D(l) - D(y) - \frac{\psi_0(y)}{\gamma} \frac{l - y}{l} \right] dl. \end{aligned}$$

The last equality uses $yH(\log(y/x)) = x - y - y \log(x/y)$. Condition (38) makes the bound nonpositive. $\square$

Necessity of the global date-zero condition. Under the regularity in Remark 1, with A in place of the OU volatility, fix x, y and correct the initial log gap with

$$a_t^\tau = - \frac{\log(y/x)}{\gamma \tau} \mathbf{1}_{(0, \tau]}(t).$$

Then $z_t = \log(y/x)(1 - t/\tau)$ until τ and is zero thereafter. The regularity means that $\sup_{t \leq \tau} |A_t^{(y, a^\tau)} - A_0(y)| \rightarrow 0$ in probability along these admissible corrections. Finite variation and the supermartingale condition make $0 \leq A_t^{(y, a^\tau)} \leq A_0(y)$, so this convergence also holds in L^1 . Since Ψ decreases from $H(\log(y/x))$ to zero,

$$-\mathbb{E}_x \int_0^\tau A_t^{(y, a^\tau)} \phi_t a_t^\tau dt \longrightarrow \frac{A_0(y)}{\gamma} H\left(\log \frac{y}{x}\right).$$

The corrective deviation therefore attains the preceding upper bound in the limit. Global incentive compatibility implies (38).

I.6 Optimal Control

The allocation cost changes continuation surplus but not information rents or incentives. Use the current-value state

$$p_t := e^{rt} A_t = \hat{\theta}_t \psi_t, \quad \Xi_t := (\hat{\theta}_t, D_t, p_t) \in (0, \infty) \times \mathbb{R} \times [0, \infty).$$

The state equations are

$$\begin{aligned} d\hat{\theta}_t &= \mu\hat{\theta}_t dt + \gamma\hat{\theta}_t d\hat{B}_t, \\ dD_t &= \left[(r - \mu)D_t - X_t - \gamma\frac{p_{t-}}{\hat{\theta}_t} \right] dt + \frac{p_{t-}}{\hat{\theta}_t} d\hat{B}_t, \\ dp_t &= rp_t dt + v_t d\hat{B}_t - dK_t, \quad p_t \geq 0. \end{aligned} \quad (117)$$

Here K is predictable, càdlàg, nondecreasing, and $K_0 = 0$; viability requires $0 \leq \Delta K_t \leq p_{t-}$. Thus $dA_t = e^{-rt}v_t d\hat{B}_t - e^{-rt}dK_t$. The finite-variation program sets $v = 0$; the local program allows $v \in \mathbb{R}$. In both programs zero is absorbing, with $v = 0$ and K constant thereafter. These are the control sets $\mathcal{A}_P$ of Section 5.3.

An admissible continuation is a truthful continuation of an admissible GBM mechanism solving (117) in the specified class. A constructed continuation must in particular deliver the promised rent:

$$\hat{\theta}_t D_t = \mathbb{E}_t \int_t^\infty e^{-r(s-t)} \hat{\theta}_s X_s ds \quad \forall t \geq 0 \text{ outside one null set.} \quad (118)$$

Integrating (110) and taking conditional expectations gives (118) if its stochastic integral is a true martingale on finite horizons and $\mathbb{E}[e^{-rT}|\hat{\theta}_T D_T|] \rightarrow 0$.

For every such continuation from (ϑ, d, p) ,

$$\mathbb{E} \int_0^\infty e^{-rt} [\hat{\theta}_t X_t - C(X_t)] dt = \vartheta d - \mathbb{E} \int_0^\infty e^{-rt} C(X_t) dt. \quad (119)$$

Consequently $V(\vartheta, d, p) = \vartheta d$ is the continuation value when $C = 0$ and a rent-delivering continuation exists. With a nonzero cost, the remaining control problem minimizes expected discounted allocation cost subject to rent delivery and incentives.

The variational inequality and initial schedules. For $p > 0$, the generator with K held fixed is

$$\begin{aligned} \mathcal{L}_G^{x,v} V &:= \mu\vartheta V_\vartheta + \left[(r - \mu)d - x - \gamma\frac{p}{\vartheta} \right] V_D + rpV_p \\ &\quad + \frac{\gamma^2\vartheta^2}{2} V_{\vartheta\vartheta} + \gamma p V_{\vartheta D} + \frac{p^2}{2\vartheta^2} V_{DD} + \frac{v^2}{2} V_{pp} + v \left(\gamma\vartheta V_{\vartheta p} + \frac{p}{\vartheta} V_{Dp} \right). \end{aligned} \quad (120)$$

For $P \in \{\text{FV}, \text{loc}\}$, the candidate $V^{G,P}$ solves

$$\min \left\{ rV - \sup_{(x,v) \in \mathcal{A}_P} [\vartheta x - C(x) + \mathcal{L}_G^{x,v} V], V_p \right\} = 0 \quad (p > 0). \quad (121)$$

At $p = 0$, its C^2 restriction to (ϑ, d) obeys

$$rV(\vartheta, d, 0) - \sup_{x \in \mathcal{X}} [\vartheta x - C(x) + \mathcal{L}_G^{x,0} V(\vartheta, d, 0)] \geq 0, \quad (122)$$

where terms involving p , v , or derivatives in p are omitted. Pointwise allocation selectors maximize $x(\vartheta - V_D) - C(x)$. Existence and measurability of an attaining selector are verification hypotheses. In the local program, the volatility supremum is finite exactly when $V_{pp} < 0$, or when $V_{pp} = 0$ and $\gamma\vartheta V_{\vartheta p} + (p/\vartheta)V_{Dp} = 0$. In the first case its maximizer is

$$v^* = -\frac{\gamma\vartheta V_{\vartheta p} + (p/\vartheta)V_{Dp}}{V_{pp}};$$

in the second, any admissible measurable choice attains it.

Use h_c and the participation condition (23) from Section 5.3. The date-zero programs are

$$\Pi_P^G := \sup_{D(\cdot), p_0(\cdot), U_c} \left\{ \int_{\Theta} [V^{G,P}(\theta, D(\theta), p_0(\theta)) - h_c(\theta)D(\theta)] dF(\theta) - U_c \right\}, \quad (123)$$

subject to (23) and

$$\begin{aligned} \text{FV : } & p_0 \geq 0, \quad \int_y^x \left[D(l) - D(y) - \frac{p_0(y)}{\gamma y} \frac{l-y}{l} \right] dl \geq 0 \quad (x, y \in \Theta), \\ \text{loc : } & D \text{ nondecreasing, } \quad 0 \leq p_0(\theta) \leq \gamma\theta^2 D'(\theta) \quad F\text{-almost everywhere.} \end{aligned} \quad (124)$$

Schedules are measurable, $D \in L^2(F)$, $p_0 \in L^1(F)$, $h_c D \in L^1(F)$, and $\theta \mapsto V^{G,P}(\theta, D(\theta), p_0(\theta))$ is integrable under F . Date-zero comparisons below use these same schedule requirements and the aggregate condition

$$\int_{\Theta} \mathbb{E}_{\theta} \int_0^{\infty} e^{-rt} (|\theta_t X_t| + |T_t| + C(X_t)) dt dF(\theta) < \infty. \quad (125)$$

These conditions justify applying Fubini's theorem separately to gross surplus and information rents.

Verification conditions. The GBM version of Condition 1 uses the state Ξ , generator $\mathcal{L}_G$, and rent delivery (118). More explicitly, domination requires integrability of $V(\Xi_T)$, a true martingale on every finite horizon

$$M_t^{G,V} := \int_0^t e^{-rs} \left(\gamma \hat{\theta}_s V_{\vartheta} + \frac{p_{s-}}{\hat{\theta}_s} V_D + v_s V_p \right) (\Xi_{s-}) d\hat{B}_s, \quad (126)$$

and existence of a nonnegative limit $\lim_{T \rightarrow \infty} \mathbb{E}[e^{-rT} V(\Xi_T)]$, along every admissible continuation in class P . At $p = 0$, the martingale integrand is $e^{-rs} \gamma \hat{\theta}_s V_{\vartheta}(\hat{\theta}_s, D_s, 0)$.

Attainment additionally requires an admissible adapted solution from the selected initial state, measurable attaining controls, and

$$\begin{aligned} rV(\Xi_{t-}) &= \hat{\theta}_t X_t - C(X_t) + \mathcal{L}_G^{X_t, v_t} V(\Xi_{t-}) & dt \otimes d\mathbb{P}\text{-a.e.}, \\ V_p(\Xi_{t-}) dK_t^c &= 0 & \text{on } \{p_{t-} > 0\}, \\ V(\hat{\theta}_t, D_t, p_t) &= V(\hat{\theta}_t, D_t, p_{t-}) & \text{at every jump of } K. \end{aligned}$$

The Bellman equality uses the boundary convention when $p = 0$. The last two equalities hold outside one null set, with the continuous equality interpreted as equality of random measures. Require (118),

$$\mathbb{E} \int_0^\infty e^{-rt} (|\hat{\theta}_t X_t| + C(X_t)) dt < \infty, \quad \mathbb{E}[e^{-rT} V(\Xi_T)] \longrightarrow 0,$$

and the corresponding flow-integrability condition after integrating the selected continuations over F . For date-zero attainment, the assembled mechanism must also satisfy (125). No initial adjustment through K is allowed because $K_0 = 0$.

Proposition 12 (GBM verification). *Fix P and let $V = V^{G,P}$ have the regularity in Condition 1, with $(\hat{\theta}, D, \psi)$ replaced by (ϑ, d, p) and $\vartheta > 0$. Suppose it solves (121)–(122) and satisfies the GBM domination conditions. Then V bounds the net continuation surplus of every admissible continuation in class P . A continuation satisfying the GBM attainment conditions attains this bound.*

Suppose (123) has a maximizer (D^, p_0^*, U_c^*) , and attaining continuations from $(\theta, D^*(\theta), p_0^*(\theta))$ can be selected measurably and combined with the GBM envelope transfers into an admissible mechanism. In the local program, also suppose D^* is absolutely continuous and differentiable at every type (one-sided at endpoints) and $p_0^*(\theta) \leq \gamma \theta^2 D^*(\theta)$ at every type. Then the mechanism is individually rational and attains Π_P^G . For $P = \text{FV}$, it is globally incentive compatible and optimal among admissible, incentive-compatible, individually rational finite-variation mechanisms satisfying the integral constraint in (124). For $P = \text{loc}$, it is locally incentive compatible and optimal among admissible locally incentive-compatible mechanisms with an absolutely continuous initial rent, the envelope transfers, and individual rationality. All comparisons use the integrability requirements stated above.*

Proof. For a continuation in class P , put

$$\begin{aligned} R_t &:= rV(\Xi_{t-}) - [\hat{\theta}_t X_t - C(X_t) + \mathcal{L}_G^{X_t, v_t} V(\Xi_{t-})], \\ J_t &:= V(\hat{\theta}_t, D_t, p_{t-}) - V(\hat{\theta}_t, D_t, p_t). \end{aligned}$$

The variational inequality and boundary condition give $R_t \geq 0$. They also give $V_p \geq 0$ in the interior, so downward jumps of p satisfy $J_t \geq 0$, including jumps to zero by continuity. The covariance terms are

$$d\langle \hat{\theta}, D \rangle_t = \gamma p_{t-} dt, \quad d\langle \hat{\theta}, p \rangle_t = \gamma \hat{\theta}_t v_t dt, \quad d\langle D, p \rangle_t = \frac{p_{t-}}{\hat{\theta}_t} v_t dt.$$

Itô's formula, including the exact value change at each jump, therefore gives

$$\begin{aligned} V(\Xi_0) &= \int_0^T e^{-rt} [\hat{\theta}_t X_t - C(X_t)] dt + e^{-rT} V(\Xi_T) + \int_0^T e^{-rt} R_t dt \\ &\quad + \int_0^T e^{-rt} V_p(\Xi_{t-}) dK_t^c + \sum_{0 < t \leq T} e^{-rt} J_t - M_T^{G,V}. \end{aligned} \quad (127)$$

The continuous adjustment integral is restricted to $\{p_{t-} > 0\}$. Localize on compact subsets of $\{p > 0\}$; if p reaches zero, continue with the C^2 boundary restriction. The nonnegative residual and adjustment terms and the true-martingale hypothesis justify the limiting expectation inequality. Taking expectations, discarding those nonnegative terms, and using the terminal condition yields

$$\mathbb{E} \int_0^\infty e^{-rt} [\hat{\theta}_t X_t - C(X_t)] dt \leq V(\Xi_0).$$

Under attainment, each residual and adjustment term vanishes, and the terminal expectation tends to zero. Equality follows.

At date zero, $h_c D \in L^1(F)$ gives

$$\int_{\Theta} \int_{\min\{c, \theta\}}^{\max\{c, \theta\}} |D(l)| dl dF(\theta) = \int_{\Theta} |h_c(\theta) D(\theta)| dF(\theta) < \infty.$$

The envelope formula, Fubini, and (125) therefore give

$$\begin{aligned} \Pi(M) &= \int_{\Theta} [S_C(\theta) - h_c(\theta) D(\theta)] dF(\theta) - U_c, \\ S_C(\theta) &:= \mathbb{E}_{\theta} \int_0^\infty e^{-rt} [\theta_t X_t - C(X_t)] dt. \end{aligned}$$

For the constructed mechanism, continuation attainment replaces $S_C(\theta)$ by $V(\theta, D^*(\theta), p_0^*(\theta))$. Theorem 3 and (124) give the asserted incentive property; (23) gives participation. Its payoff is Π_P^G . A comparator in the finite-variation class is feasible for its initial program. For a locally incentive-compatible

comparator with absolutely continuous rent, the pointwise necessity argument above gives $D'(\theta) \geq p_0(\theta)/(\gamma\theta^2) \geq 0$ at almost every differentiability point. Absolute continuity makes D nondecreasing, so its initial schedules are feasible for the local program. The continuation bound and the same revenue identity prove both optimality comparisons. $\square$

The programs also give the GBM bracket. If a mechanism is globally incentive compatible, (112) with $a = 0$ gives

$$U(x) \geq U(y) + (x - y)D(y) \quad \forall x, y \in \Theta.$$

Applying the inequality in both directions gives $(x - y)(D(x) - D(y)) \geq 0$, so D is nondecreasing. Its derivative exists almost everywhere, and the pointwise necessity calculation gives $0 \leq p_0(\theta) \leq \gamma\theta^2 D'(\theta)$ there. Thus, when the local candidate satisfies domination, every globally incentive-compatible, individually rational comparator with the stated integrability satisfies $\Pi(M) \leq \Pi_{\text{loc}}^G$. If the finite-variation program is attained as above, its mechanism is globally incentive compatible, and

$$\Pi_{\text{FV}}^G \leq \sup_M \Pi(M) \leq \Pi_{\text{loc}}^G,$$

where the supremum uses this comparison class.

J Contracting without Transfers

This section proves Proposition 6. The incentive arguments require the envelope properties of a contract; they do not require transfers to be available as a choice variable.

Proof of Proposition 6. When $T \equiv 0$, the continuation value is

$$W_t = \mathbb{E}_t \int_t^\infty e^{-r(s-t)} \hat{\theta}_s X_s \, ds.$$

The first-order property is precisely $\text{vol}(W_t) = \gamma D_t$, the hypothesis on transfers used in Lemma 1. In the gains calculation, the drift of accumulated payoff plus $e^{-rt}(W_t - D_t \delta_t)$ is therefore $-e^{-rt} \psi_t \delta_t a_t$. Conditions (M3)–(M4) give the same master deviation formula. The proof of Theorem 1a uses this formula, the admissibility conditions, and the reporting technology, so all its conclusions apply.

If the date-zero envelope condition also holds, the initial-report gain is the same expression $U(y) + (x - y)D(y) - U(x)$ used in the proof of Theorem 1b. Its local characterization and finite-variation sufficiency result follow under exactly the hypotheses stated there. $\square$

K Beyond Finite Variation

This section gives the finite expansion underlying (39) and states conditions for its infinite limit. It then proves Proposition 7 directly, without assuming convergence of that expansion.

K.1 The finite expansion and its limit

Fix a truthful initial report x and an admissible distortion a . Write

$$\mathcal{G}(a) := U_x(x, a) - U(x) = -\mathbb{E}_x \int_0^\infty e^{-rt} \psi_t \delta_t a_t dt, \quad \delta_0 = 0.$$

For an integer $K \geq 2$, set $f_0 = \psi$ and suppose the successive volatility coefficients admit the Itô decompositions

$$df_{j,t} = b_{j,t} dt + f_{j+1,t} d\hat{B}_t, \quad j = 0, \dots, K-2,$$

along the report. Thus $f_j = \text{vol}^j(\psi)$, and b_j is its drift under truthful continuation, evaluated along that report. Under $\mathbb{P}_x$, its drift is $b_j + f_{j+1}a$.

For each $k = 2, \dots, K$, assume

$$\mathbb{E}_x \int_0^\infty e^{-rt} |\delta_t|^k [(r + k\lambda)|f_{k-2,t}| + |b_{k-2,t}| + |f_{k-1,t}a_t|] dt < \infty, \quad (128)$$

$$e^{-rT} \delta_T^k f_{k-2,T} \longrightarrow 0 \quad \text{in } L^1 \text{ as } T \rightarrow \infty, \quad (129)$$

and assume that $\int_0^t e^{-rs} \delta_s^k f_{k-1,s} dB_s$ is a true martingale on every finite horizon. All these conditions concern the fixed deviation a .

Under these conditions, the finite expansion is

$$\begin{aligned} \mathcal{G}(a) = \sum_{k=2}^K \frac{(-1)^{k-1}}{k! \gamma^{k-1}} \mathbb{E}_x \int_0^\infty e^{-rt} \delta_t^k \\ \times [(r + k\lambda)f_{k-2,t} - b_{k-2,t}] dt + \mathcal{R}_K(a), \end{aligned} \quad (130)$$

where the remainder is

$$\mathcal{R}_K(a) := \frac{(-1)^K}{K! \gamma^{K-1}} \mathbb{E}_x \int_0^\infty e^{-rt} \delta_t^K f_{K-1,t} a_t dt. \quad (131)$$

To verify the expansion, use $d\delta_t = (-\lambda\delta_t + \gamma a_t)dt$. Itô's formula and the martingale condition give

$$\begin{aligned} \mathbb{E}_x [e^{-rT} \delta_T^k f_{k-2,T}] = \mathbb{E}_x \int_0^T e^{-rt} \delta_t^k [b_{k-2,t} - (r + k\lambda)f_{k-2,t} + f_{k-1,t}a_t] dt \\ + k\gamma \mathbb{E}_x \int_0^T e^{-rt} \delta_t^{k-1} f_{k-2,t} a_t dt. \end{aligned}$$

Letting $T \rightarrow \infty$ expresses the last integral in terms of the k -th drift term and the next volatility integral. Starting with $\mathcal{G}(a)$ and applying this identity for $k = 2, \dots, K$ gives (130).

If these assumptions hold at every order and $\mathcal{R}_K(a) \rightarrow 0$, the partial sums in (130) converge to $\mathcal{G}(a)$, yielding (39). Interchanging the series with expectation and time integration is justified, for example, by the further condition

$$\sum_{k=2}^{\infty} \frac{1}{k! \gamma^{k-1}} \mathbb{E}_x \int_0^{\infty} e^{-rt} |\delta_t|^k |(r + k\lambda) f_{k-2,t} - b_{k-2,t}| dt < \infty. \quad (132)$$

For an Itô process of finite variation, $f_1 = 0$, so the remainder vanishes already at $K = 2$. Singular finite-variation terms are handled by the measure-valued integration by parts in the proof of Theorem 1a.

K.2 Proportional volatility

If $d\psi = m dt + \alpha\psi d\hat{B}$, then $f_j = \alpha^j \psi$ and $b_j = \alpha^j m$. Set

$$z_t := \frac{\alpha\delta_t}{\gamma}, \quad h(z) := e^{-z} - 1 + z, \quad g(z) := 1 - (1 + z)e^{-z}.$$

Under the preceding convergence and interchange conditions,

$$\sum_{k=2}^{\infty} \frac{(-1)^k z^k}{k!} = h(z), \quad \sum_{k=2}^{\infty} \frac{k(-1)^k z^k}{k!} = zh'(z) = h(z) + g(z)$$

sum the cascade to

$$\mathcal{G}(a) = \frac{\gamma}{\alpha^2} \mathbb{E}_x \int_0^{\infty} e^{-rt} [(m_t - (r + \lambda)\psi_t)h(z_t) - \lambda\psi_t g(z_t)] dt. \quad (133)$$

Proposition 7 follows directly from Itô's formula, without these series assumptions or a vanishing terminal correction.

Proof of Proposition 7. Fix a truthful initial report x and any admissible distortion a . Absolute continuity of report laws carries the hypothesized dynamics and drift inequality to the deviating report. In particular,

$$d\psi_t = (m_t + \alpha\psi_t a_t) dt + \alpha\psi_t dB_t, \quad dz_t = (-\lambda z_t + \alpha a_t) dt.$$

Define the nonnegative correction

$$F_t := \frac{\gamma}{\alpha^2} e^{-rt} \psi_t h(z_t).$$

Here $h \geq 0$ follows from $e^{-z} \geq 1 - z$, while $g(0) = 0$ and $g'(z) = ze^{-z}$ show that $g \geq 0$ on both half-lines. These facts hold for either sign of α . Using $h + h' = z$ and $zh' - h = g$, Itô's formula gives

$$\begin{aligned} dF_t &= e^{-rt} \psi_t \delta_t a_t dt \\ &\quad + \frac{\gamma}{\alpha^2} e^{-rt} [(m_t - (r + \lambda)\psi_t)h(z_t) - \lambda\psi_t g(z_t)] dt \\ &\quad + \frac{\gamma}{\alpha} e^{-rt} \psi_t h(z_t) dB_t. \end{aligned} \tag{134}$$

Since $\delta_0 = 0$, we have $F_0 = 0$. Thus, on every finite horizon, the exact local identity is

$$\begin{aligned} - \int_0^T e^{-rt} \psi_t \delta_t a_t dt &= -F_T \\ &\quad + \frac{\gamma}{\alpha^2} \int_0^T e^{-rt} [(m_t - (r + \lambda)\psi_t)h(z_t) - \lambda\psi_t g(z_t)] dt \\ &\quad + \frac{\gamma}{\alpha} \int_0^T e^{-rt} \psi_t h(z_t) dB_t. \end{aligned} \tag{135}$$

Choose stopping times $\tau_n \uparrow \infty$ that make the stochastic integral a true martingale and the other stopped terms integrable on each finite horizon. The first two terms on the right of (135) are nonpositive because $F \geq 0$, $\psi \geq 0$, $\lambda \geq 0$, and $m \leq (r + \lambda)\psi$. Taking expectations at $T \wedge \tau_n$ gives

$$-\mathbb{E}_x \int_0^{T \wedge \tau_n} e^{-rt} \psi_t \delta_t a_t dt \leq 0.$$

Condition (M4) bounds the absolute value of the stopped integral by the integrable variable $\int_0^\infty e^{-rt} |\psi_t \delta_t a_t| dt$. Dominated convergence first removes the stopping times and then lets $T \rightarrow \infty$. Lemma 1 identifies the limit with $\mathcal{G}(a)$, so $\mathcal{G}(a) \leq 0$. This holds for every initial report x and every admissible continuation distortion a , proving global incentive compatibility after the initial report. $\square$

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